Full waveform inversion with an auxiliary bump functional

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Pawan Bharadwaj†

† Delft University of Technology

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Notation

- q observed data
- p modeled data
- c estimated velocity model
- c_o initial velocity guess
- J objective function
- $\alpha > 0$ step length
- τ_d is the dominant period

Assumptions

- acoustic medium
- constant density
- known source signature

$q,\ c{=}c_o$

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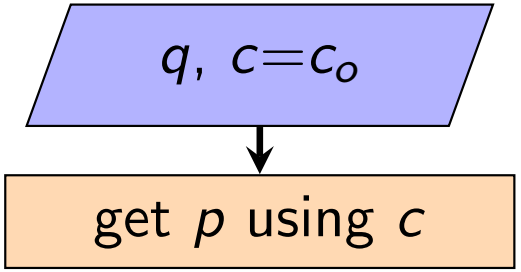

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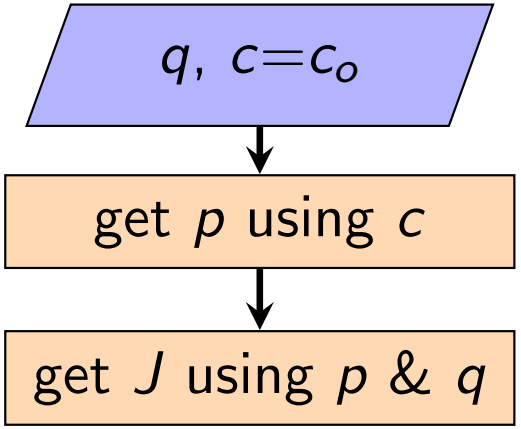

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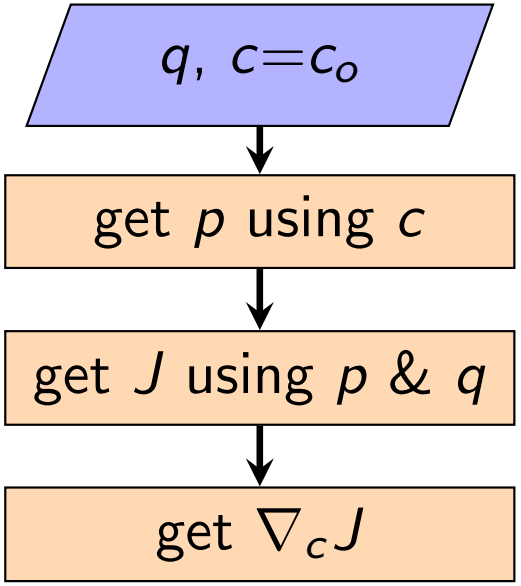

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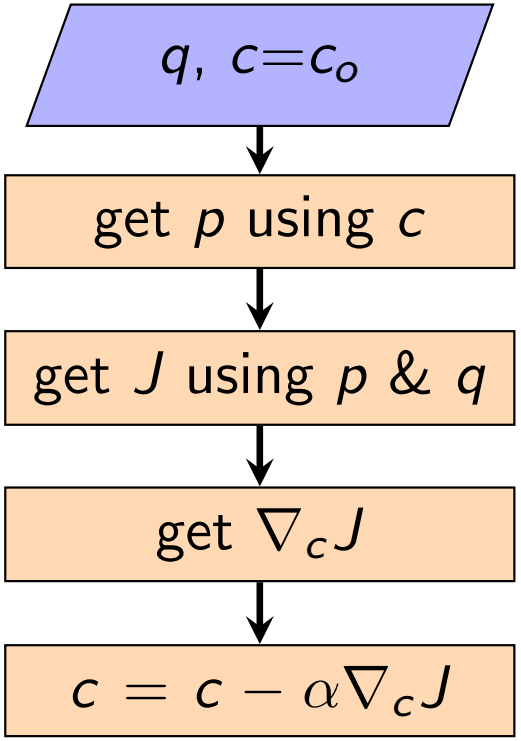

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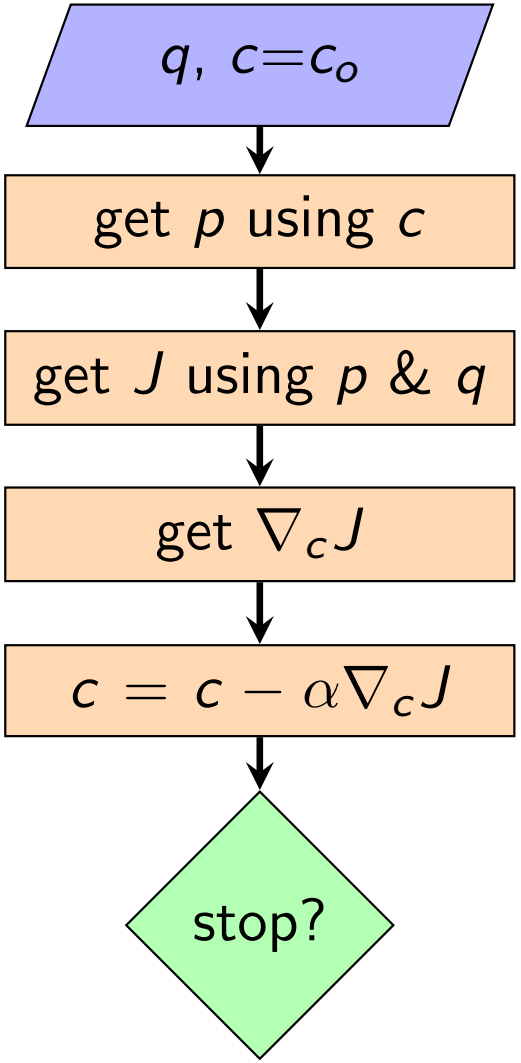

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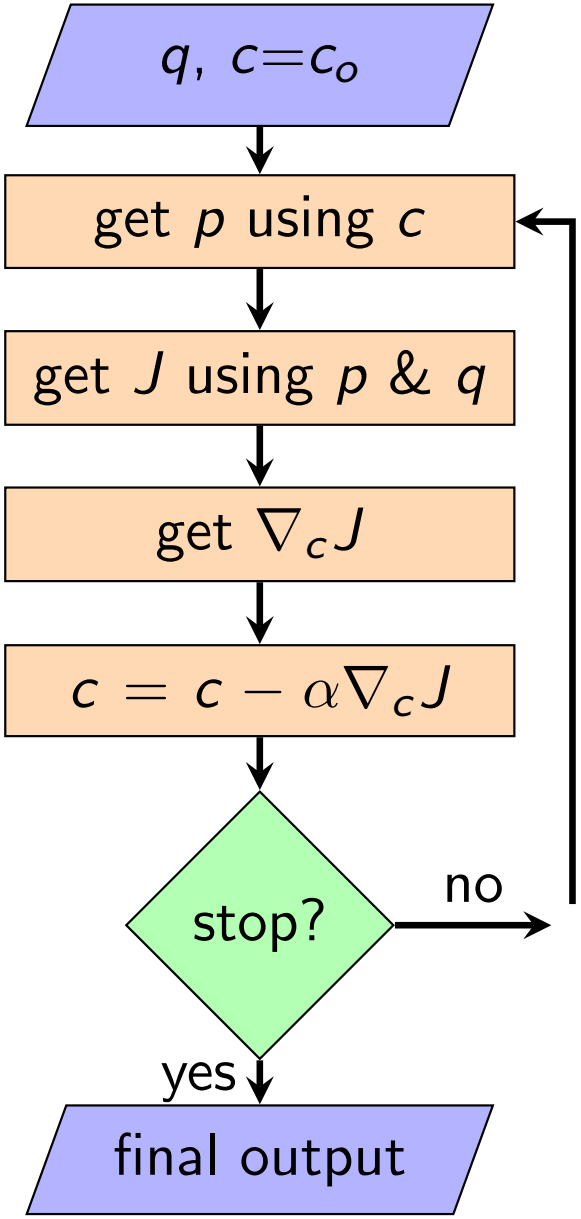

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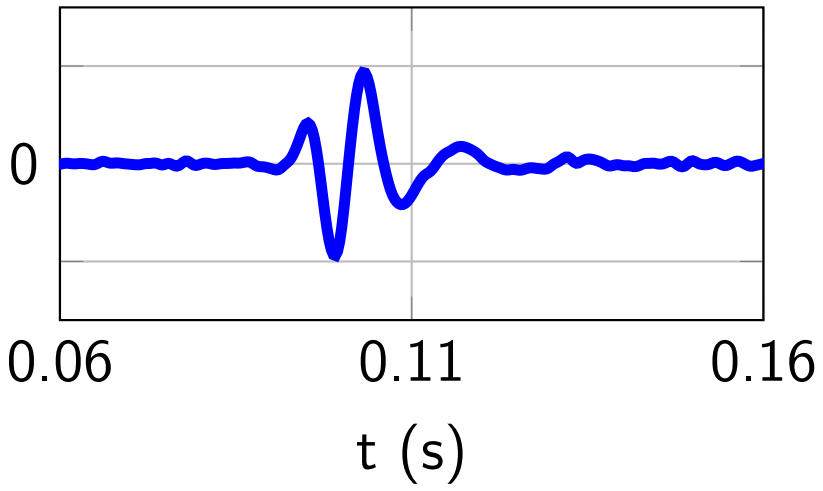
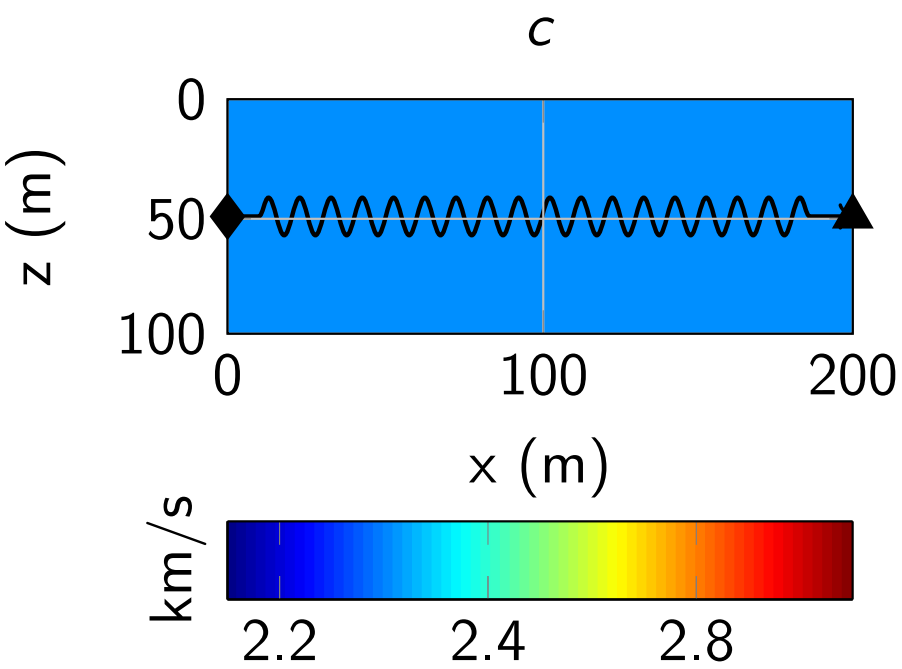
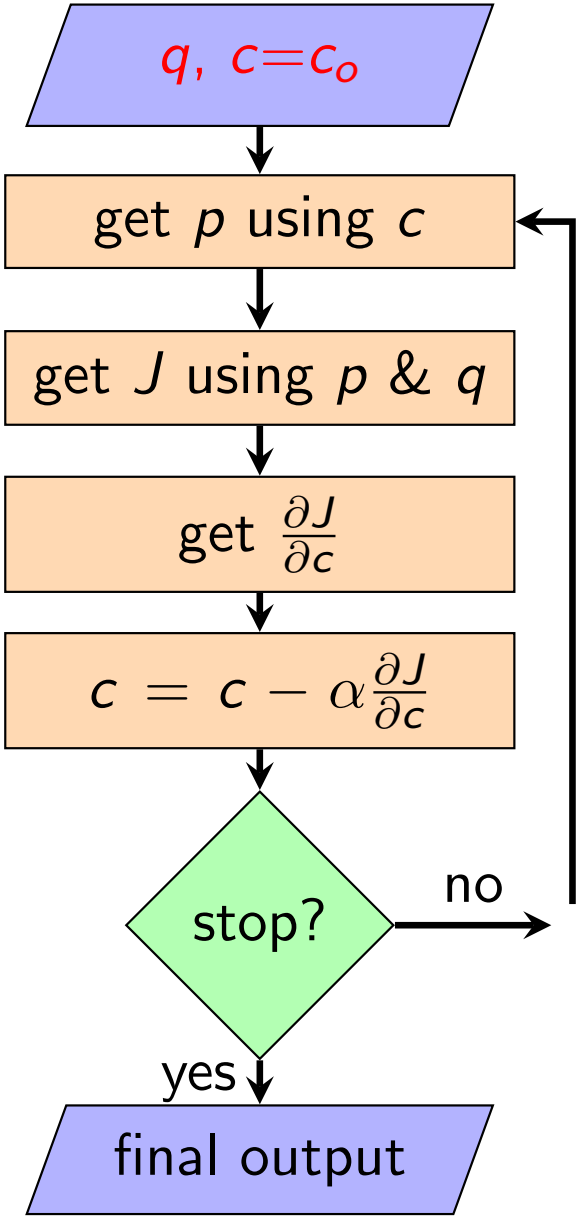
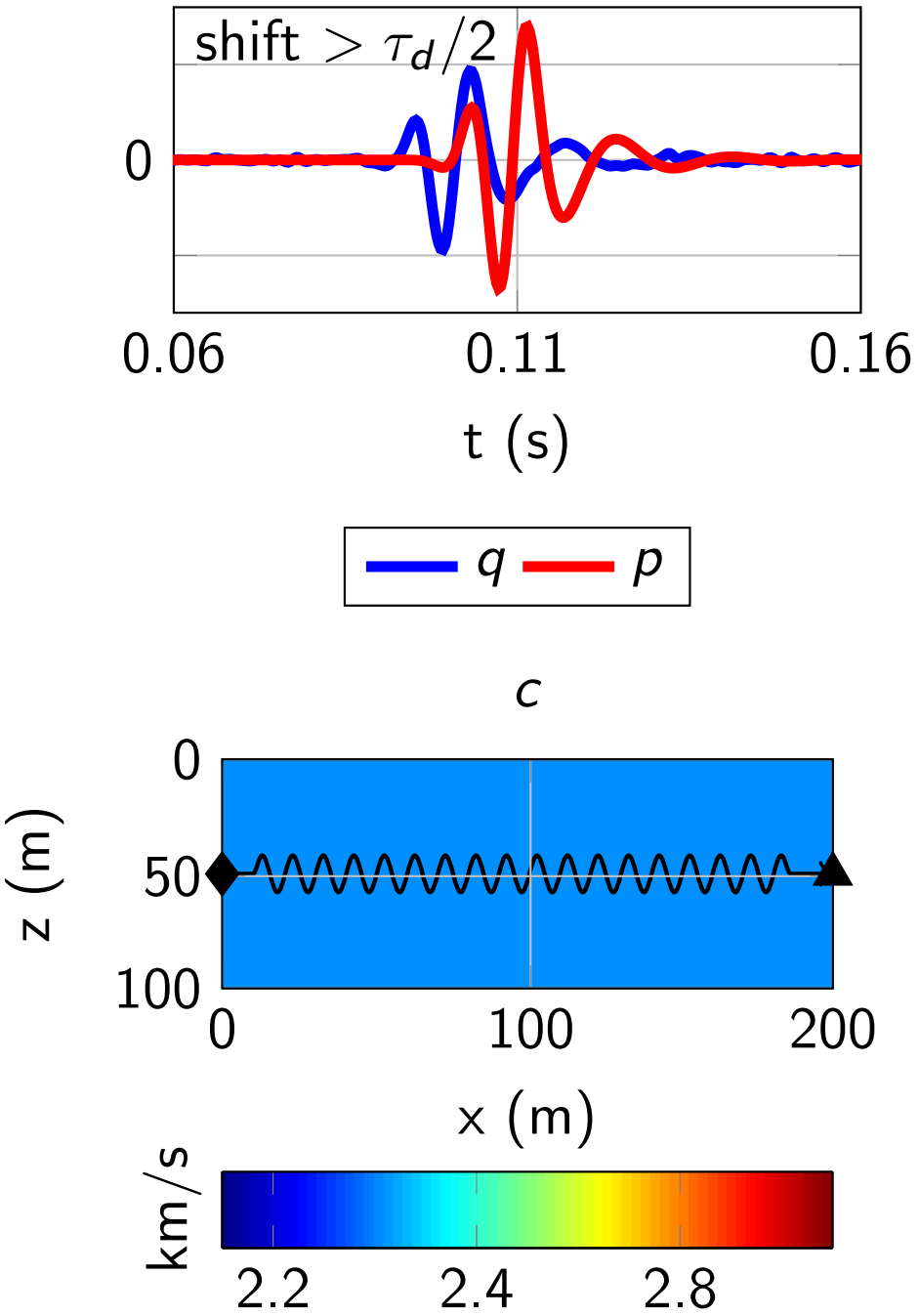

0
0.06
0.11
0.16
t (s)

q


c
z (m)
0
50
100
0
100
200
x (m)
km/s
2.2
2.4
2.8


q, c=c_o
get p using c
get J using p & q
get ∂J/∂c
c = c − α ∂J/∂c
stop?
no
yes
final output


shift > $\tau_d/2$
0
0.06
0.11
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t (s)
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0
50
100
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0
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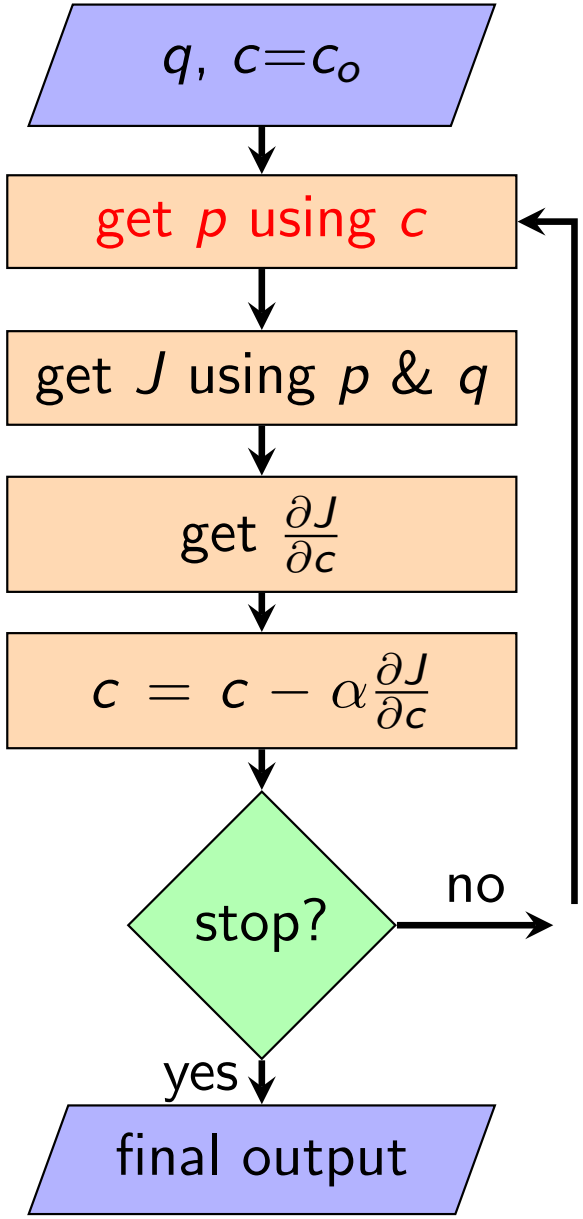
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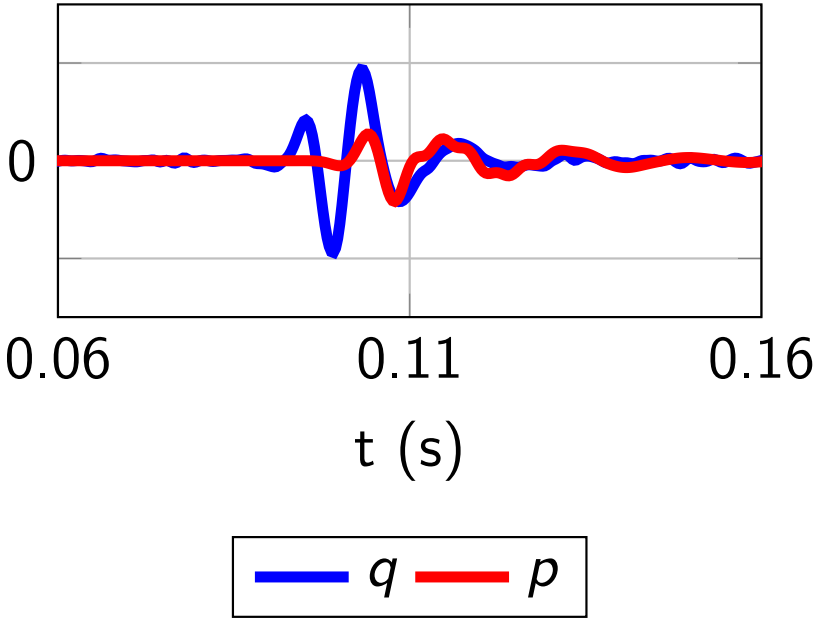
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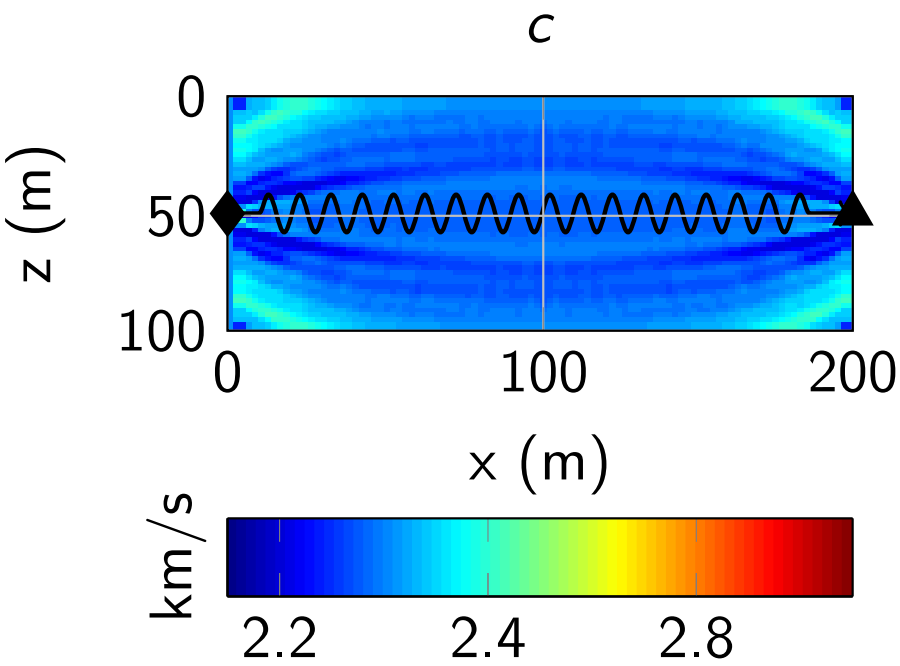
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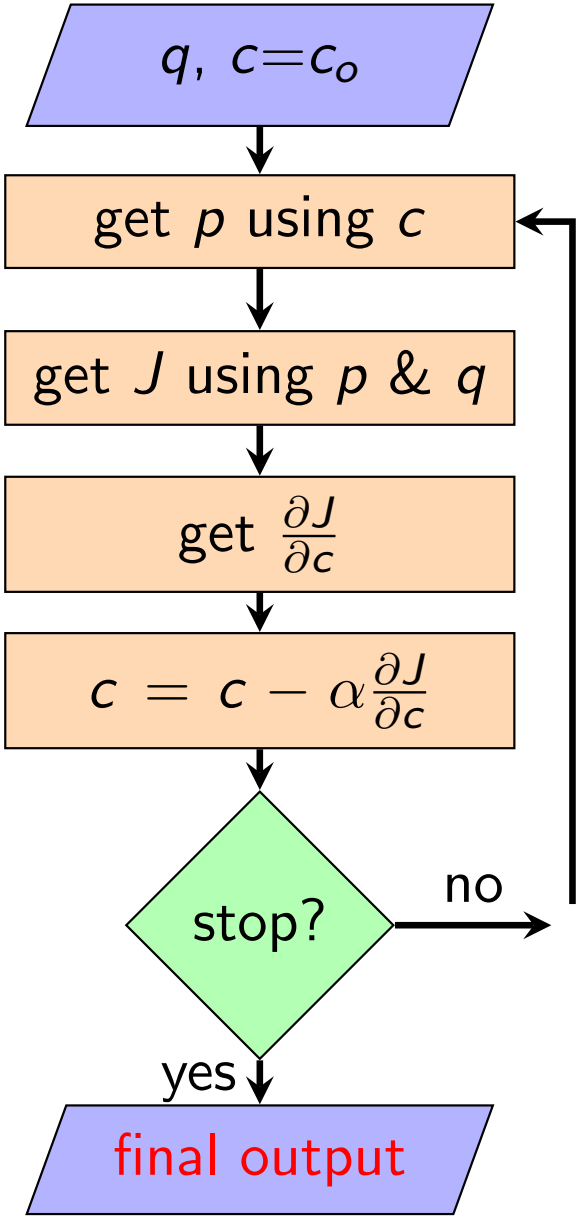
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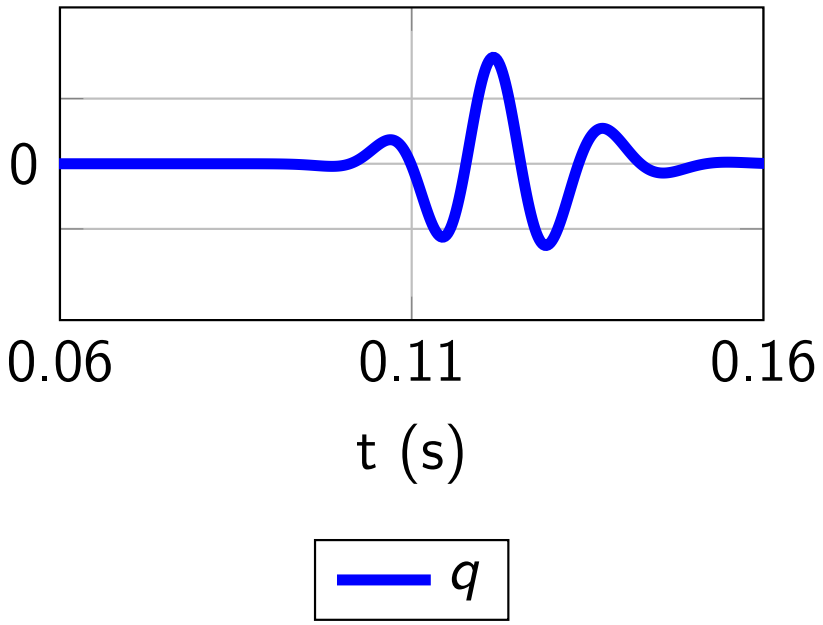
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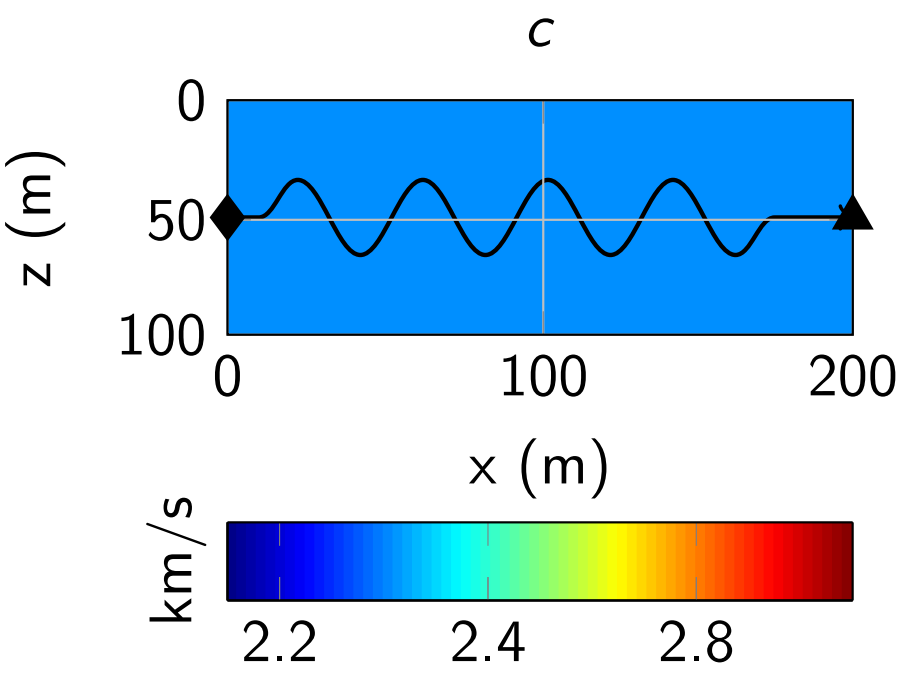
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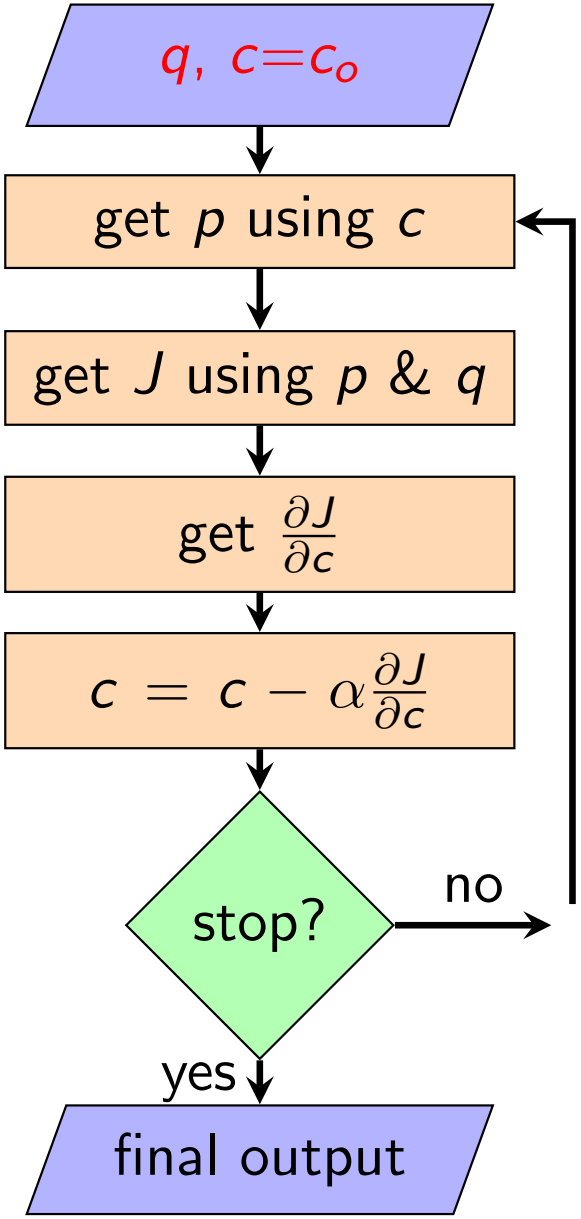
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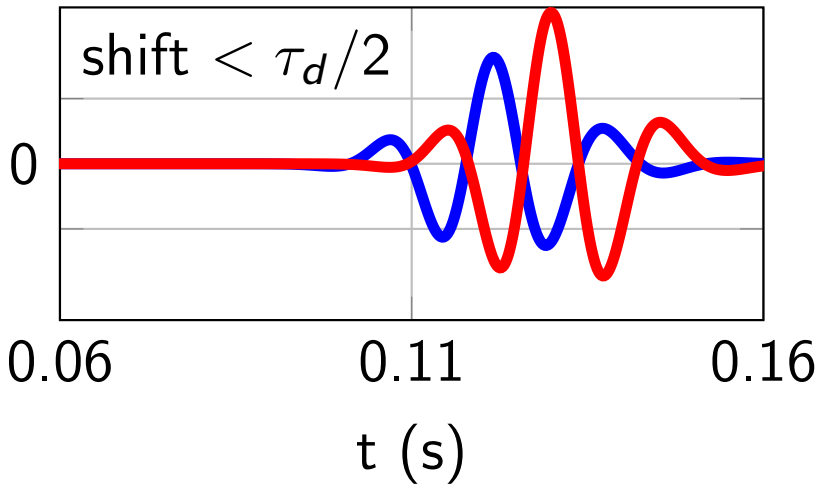
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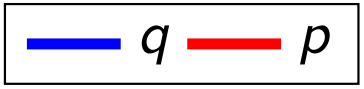
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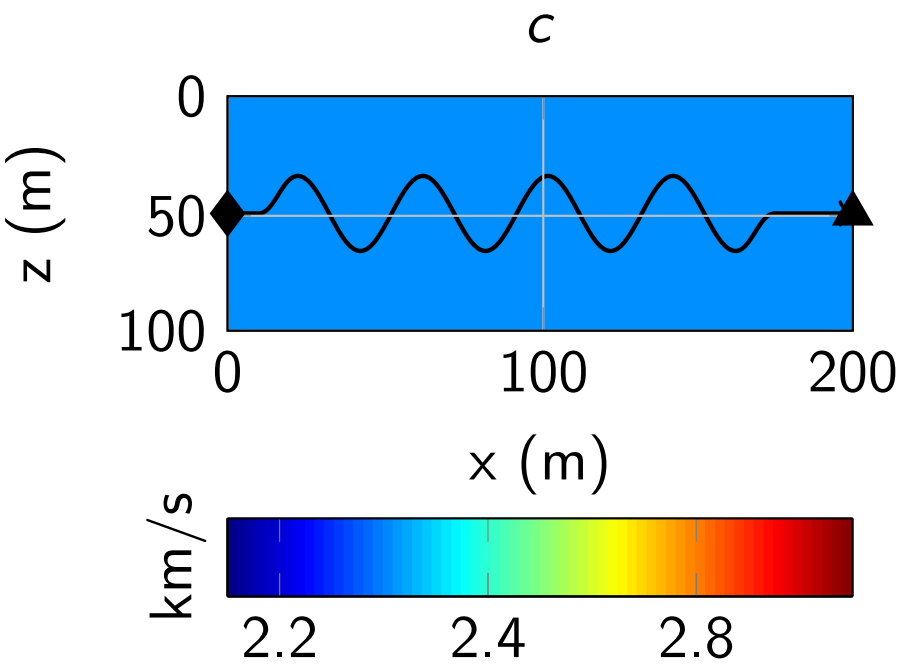
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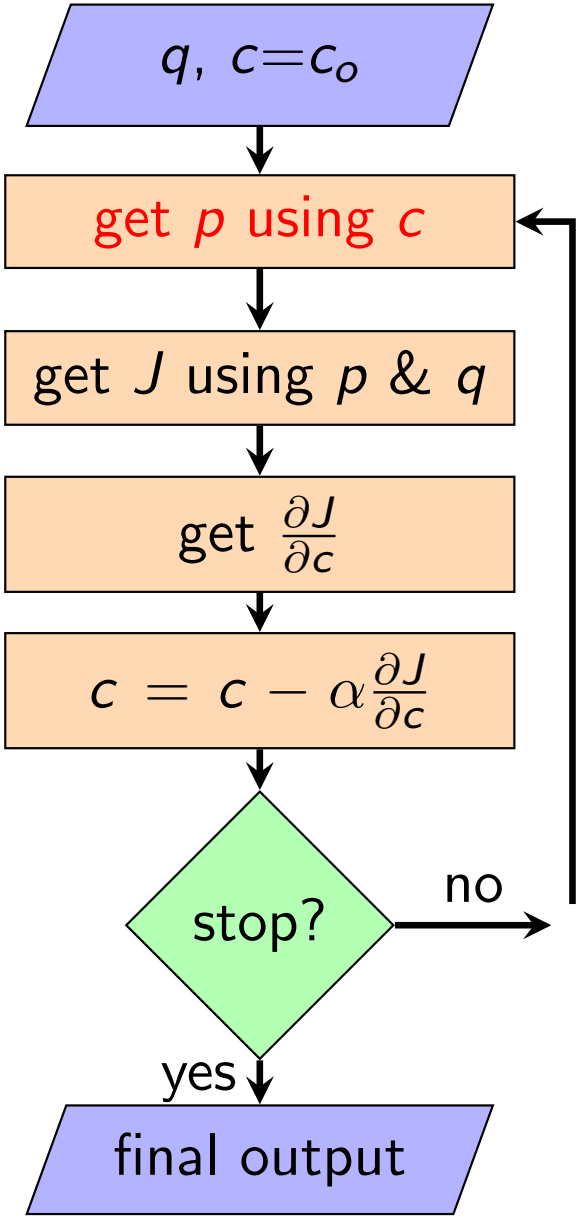
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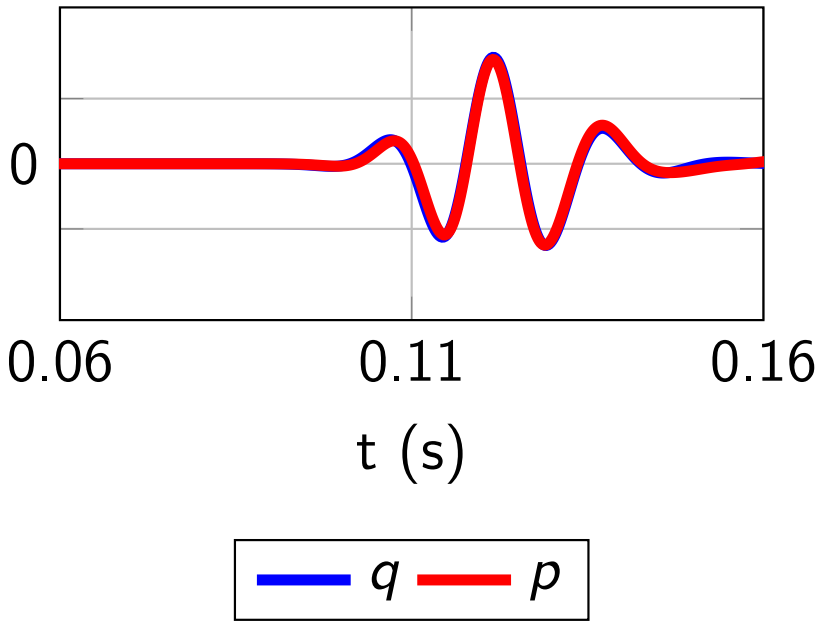
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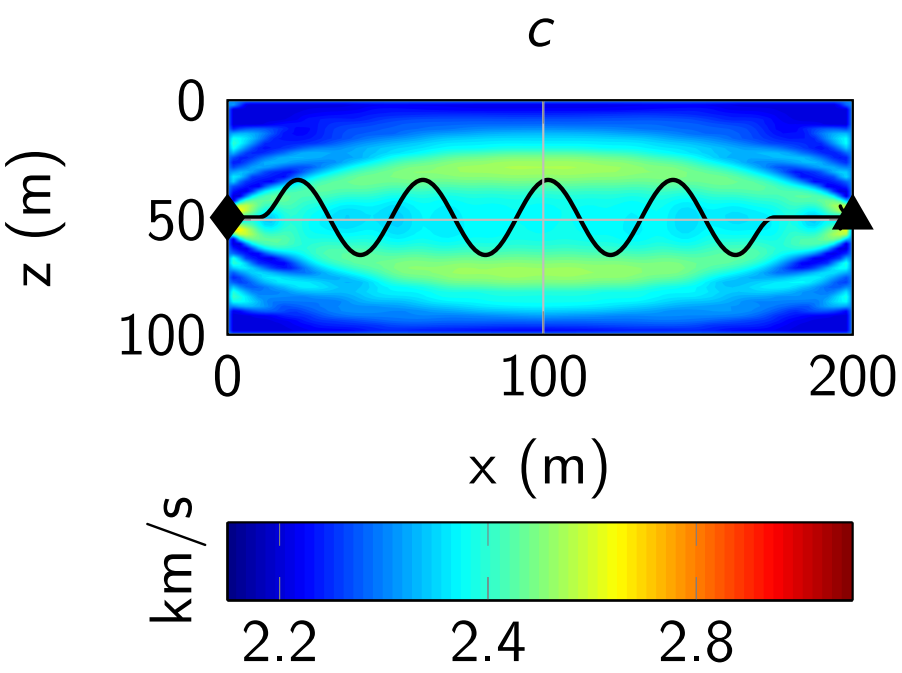
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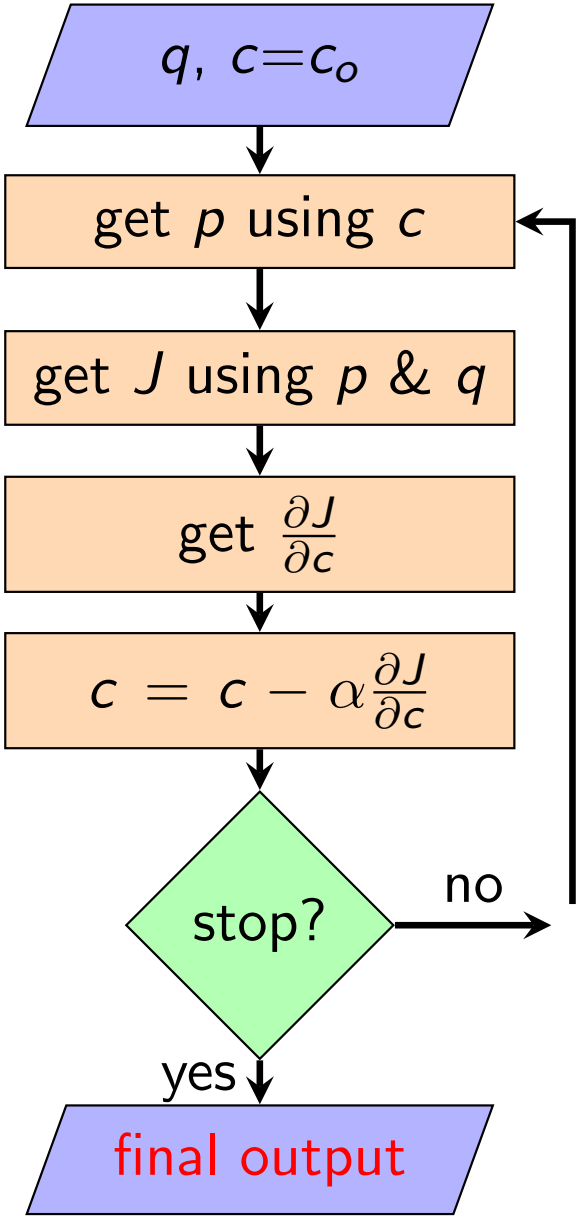
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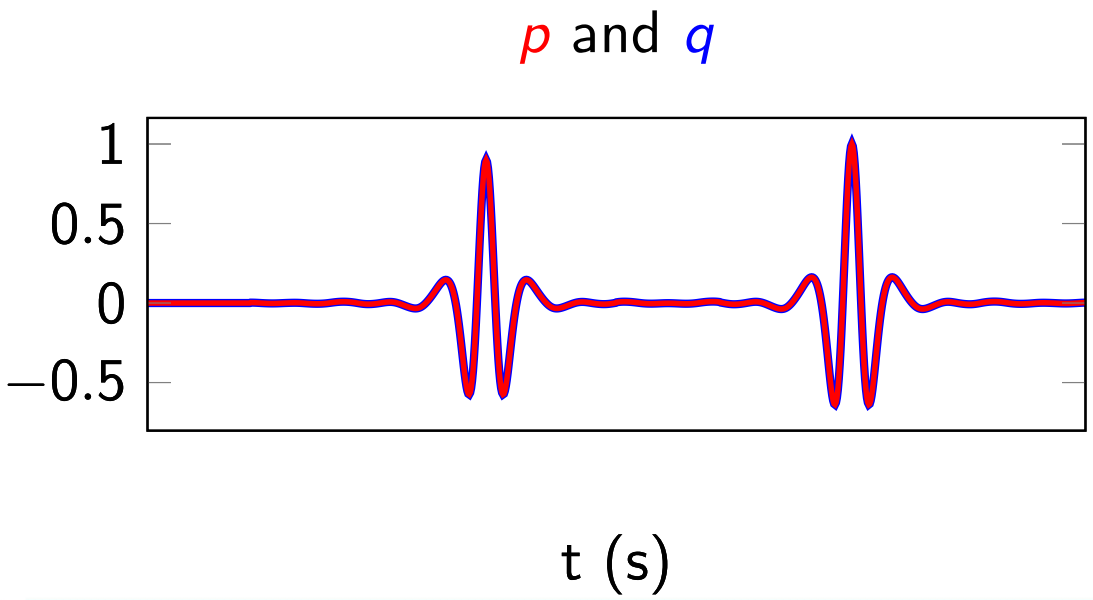
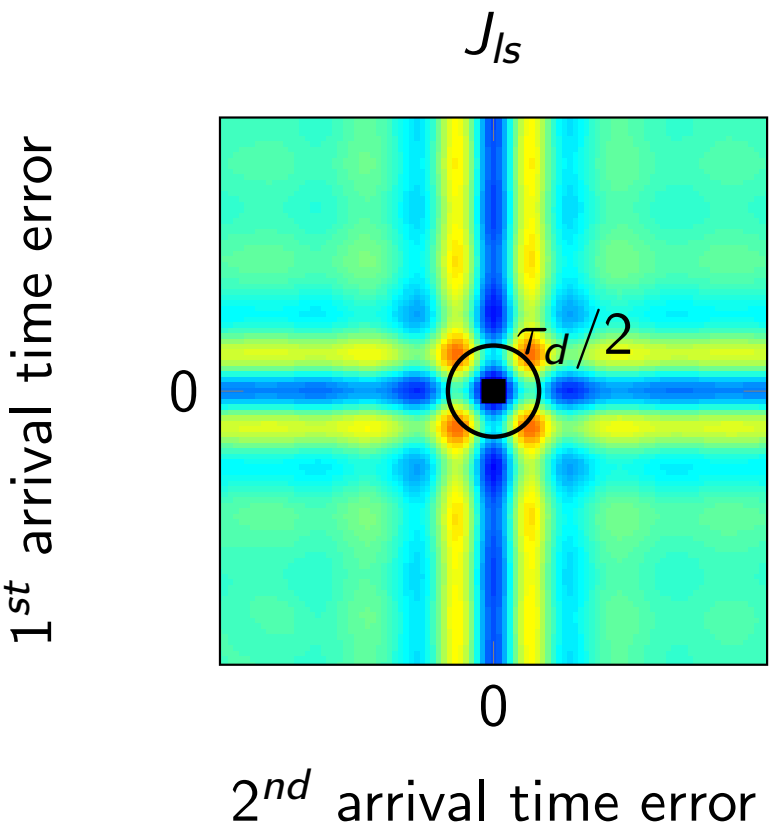
Least-squares Functional

- subtract p and q to measure the error

$$J_{ls} = \frac{1}{2} \sum_t [p(t) - q(t)]^2$$

- no sensitivity if arrival time error $> \tau_d$





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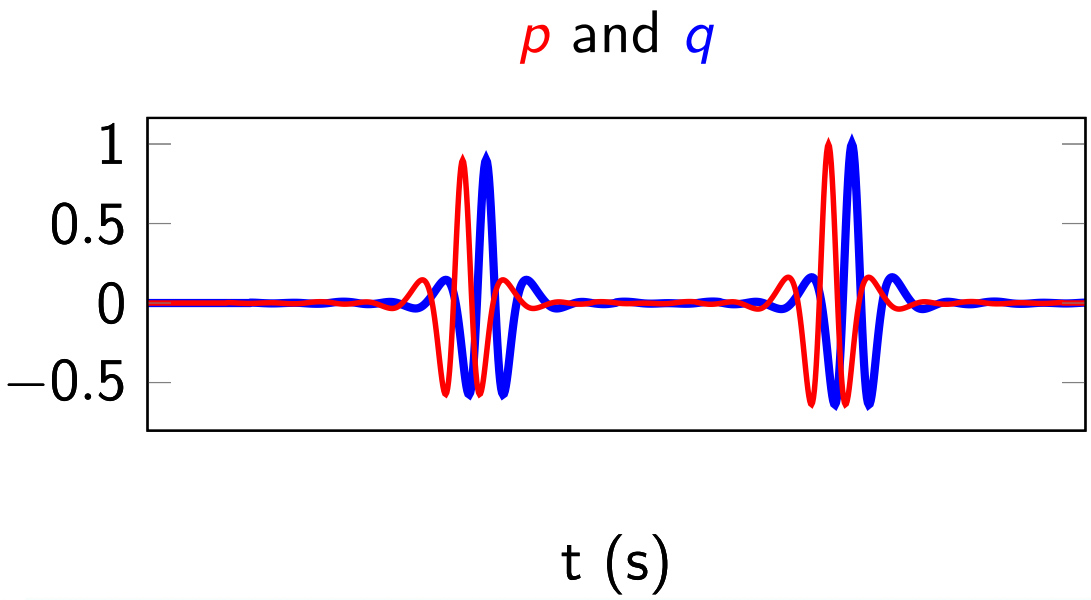

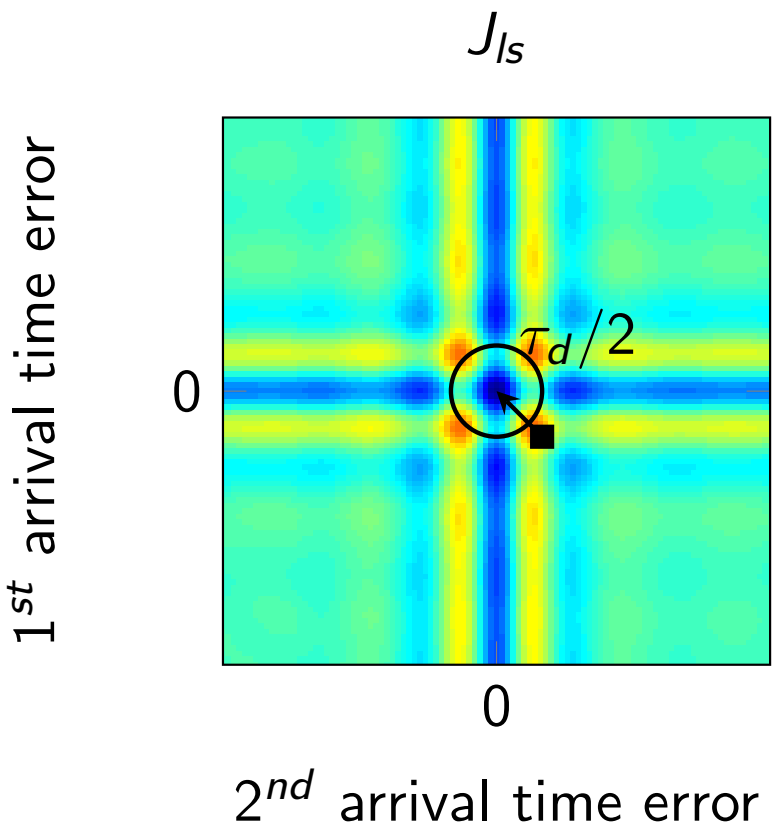

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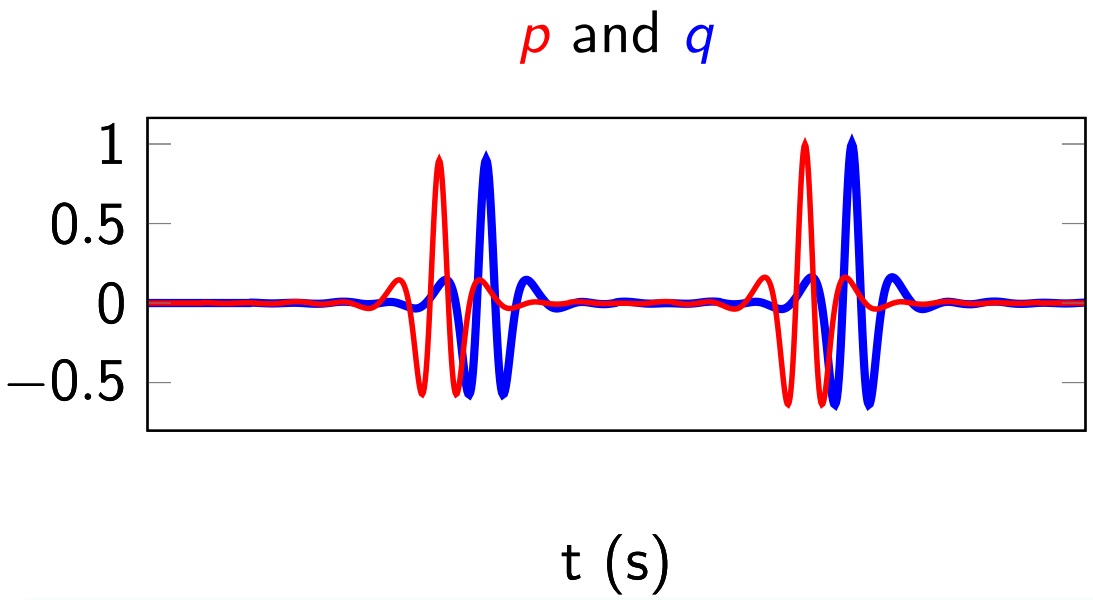

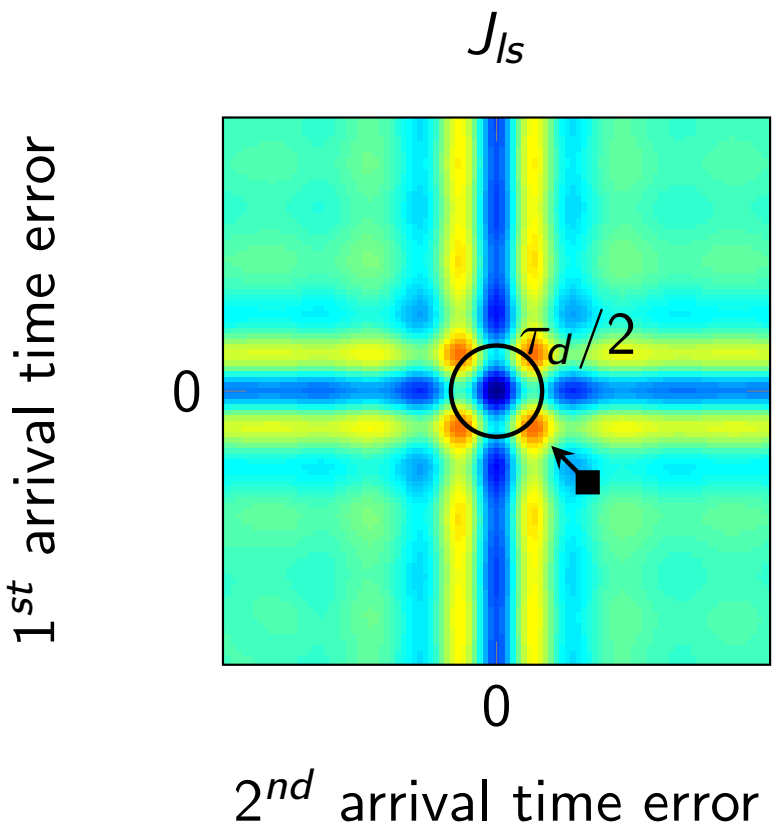

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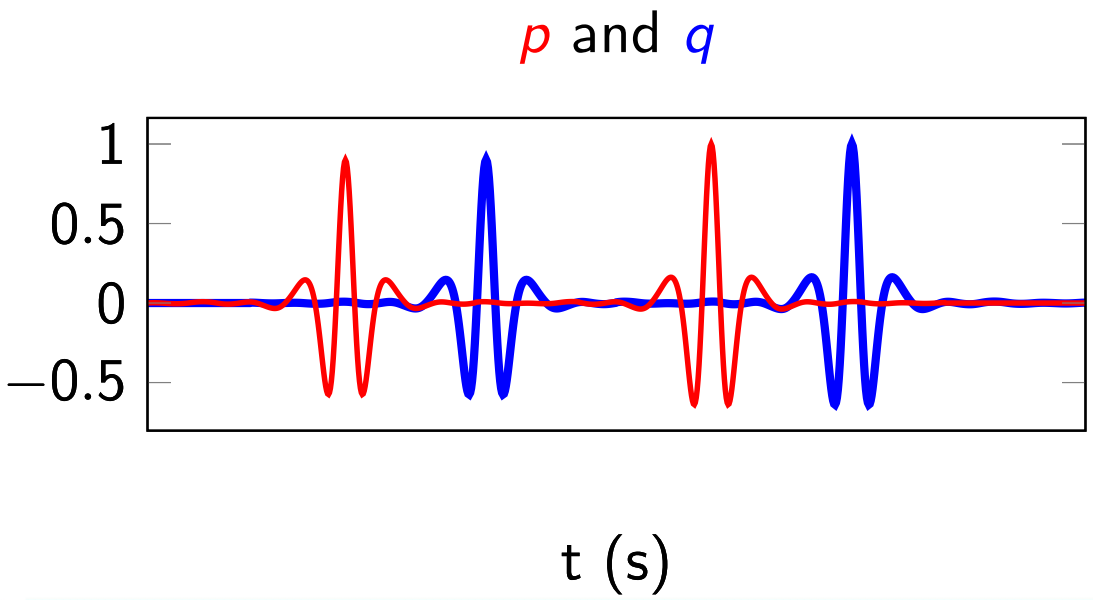

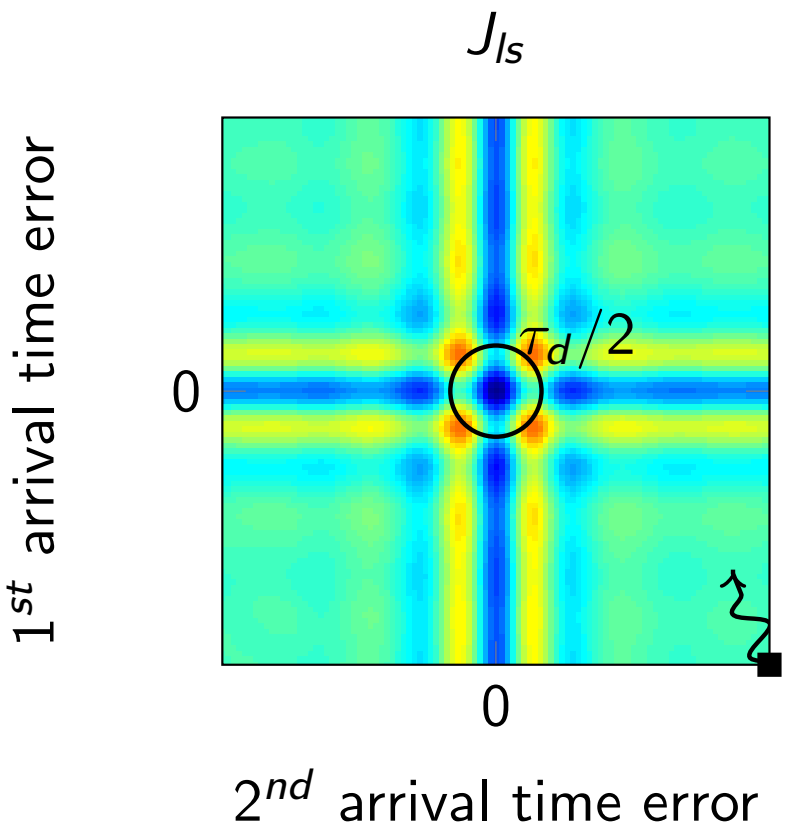

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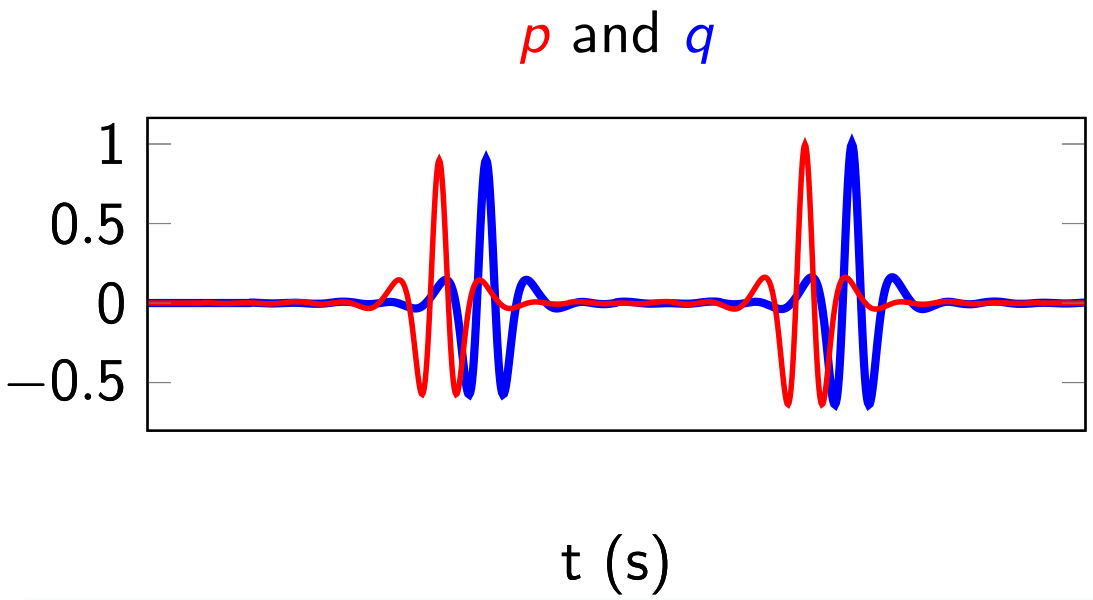
Envelope Functional†

- envelopes $E(p)$ and $E(q)$
- error is difference between them

$$J^E = \frac{1}{2}\sum_t [E(p) - E(q)]^2$$

- no sensitivity if arrival time error $> \tau_d$



† Bozdag, E., Trampert, J., and Tromp, J., 2011. Misfit functions for full waveform inversion based on instantaneous phase and envelope measurements, Geophysical Journal International

† Luo, J. and Wu, R.-S., 2015. Seismic envelope inversion: reduction of local minima and noise resistance, Geophysical Prospecting

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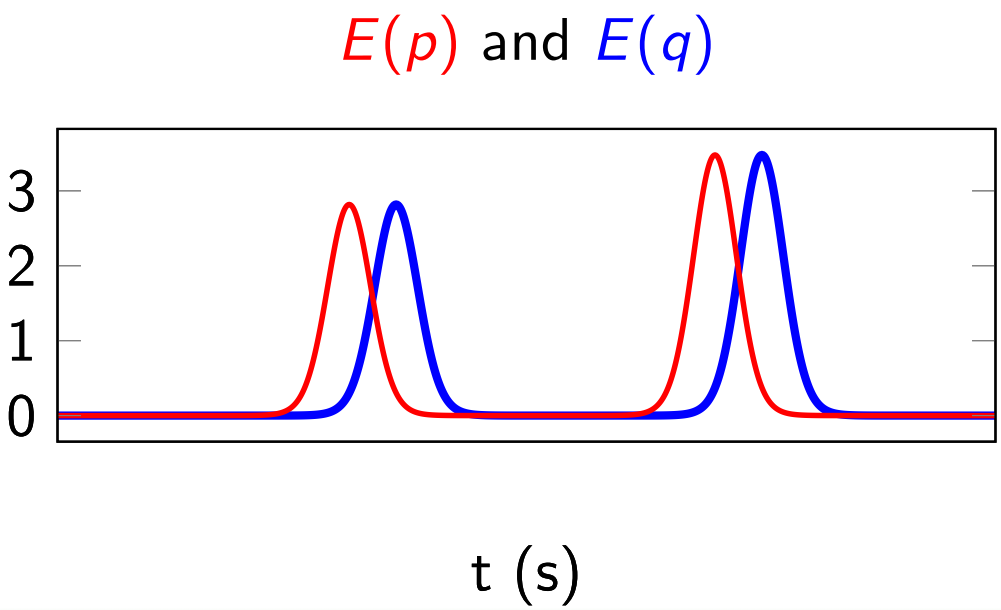

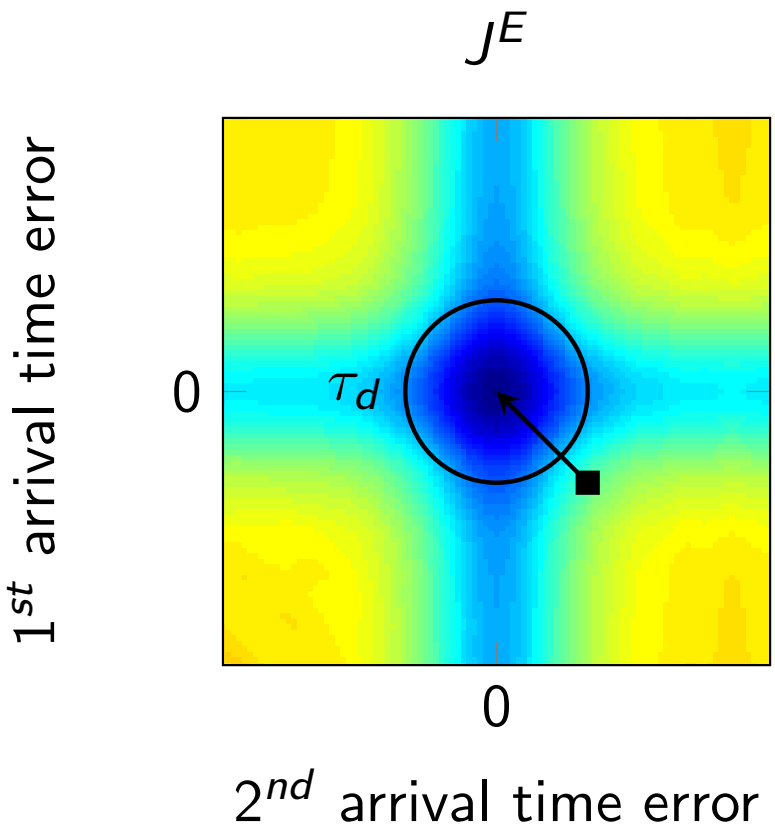

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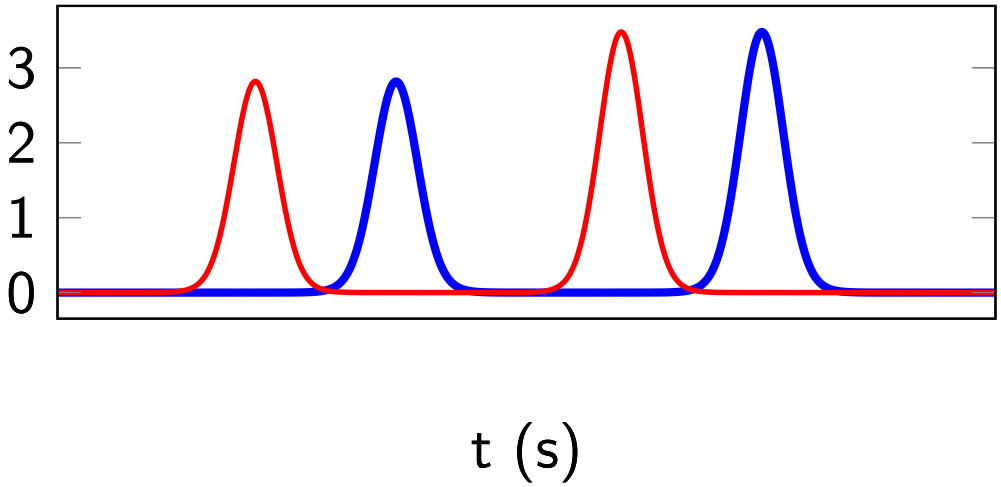
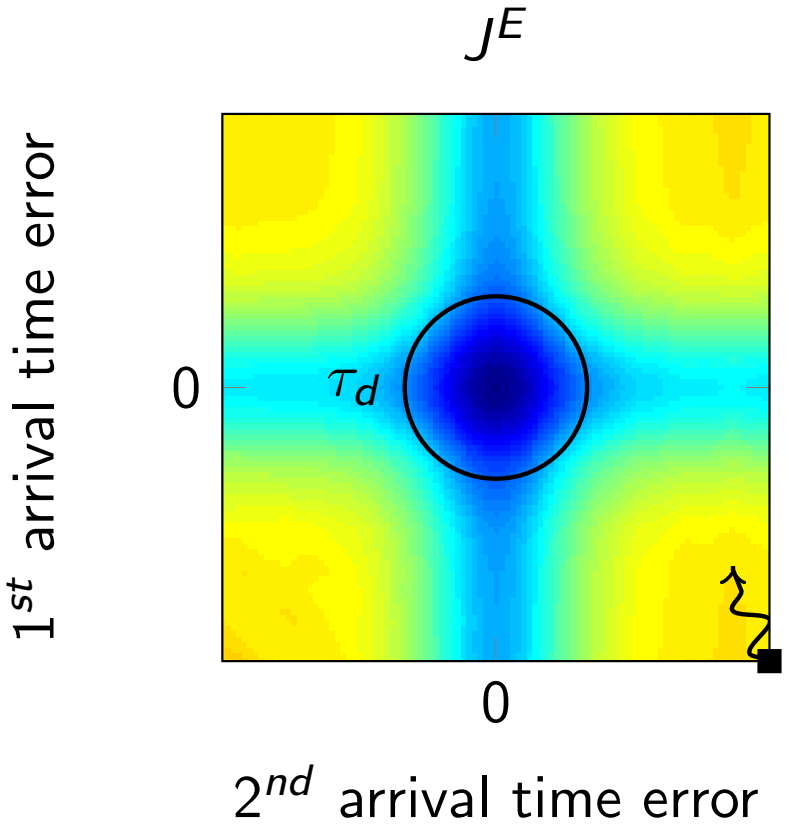
$E(p)$ and $E(q)$





Summary

- Cycle skipping causes local minima when using J_{ls} objective.
- J_{ls} inversion cannot reconstruct velocity anomalies with an arrival time error $> \tau_d/2$.
- The envelope based misfit is insensitive to arrival time errors $> \tau_d$.

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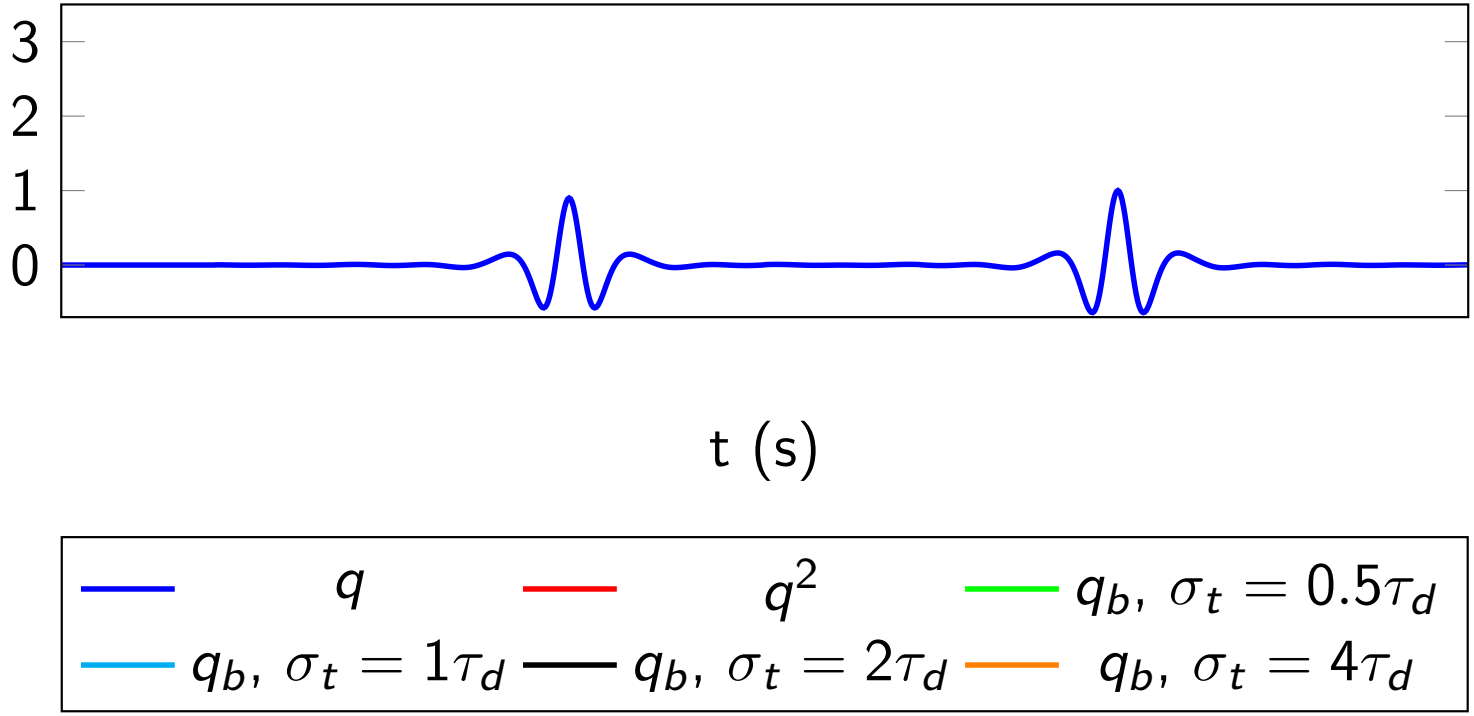
The Bumpy Data (denoted using subscript b)

- squared or absolute-valued data
- Gaussian blurring with $b(t, \sigma_t) = \exp \frac{-t^2}{\sigma_t^2}$

$$q_b = b(t, \sigma_t) *_t (q^2)$$

- σ_t controls the width of the bumps



The Bumpy Data (denoted using subscript b)

- squared or absolute-valued data
- Gaussian blurring with $b(t, \sigma_t) = \exp \frac{-t^2}{\sigma_t^2}$

$$q_b = b(t, \sigma_t) *_t (q^2)$$

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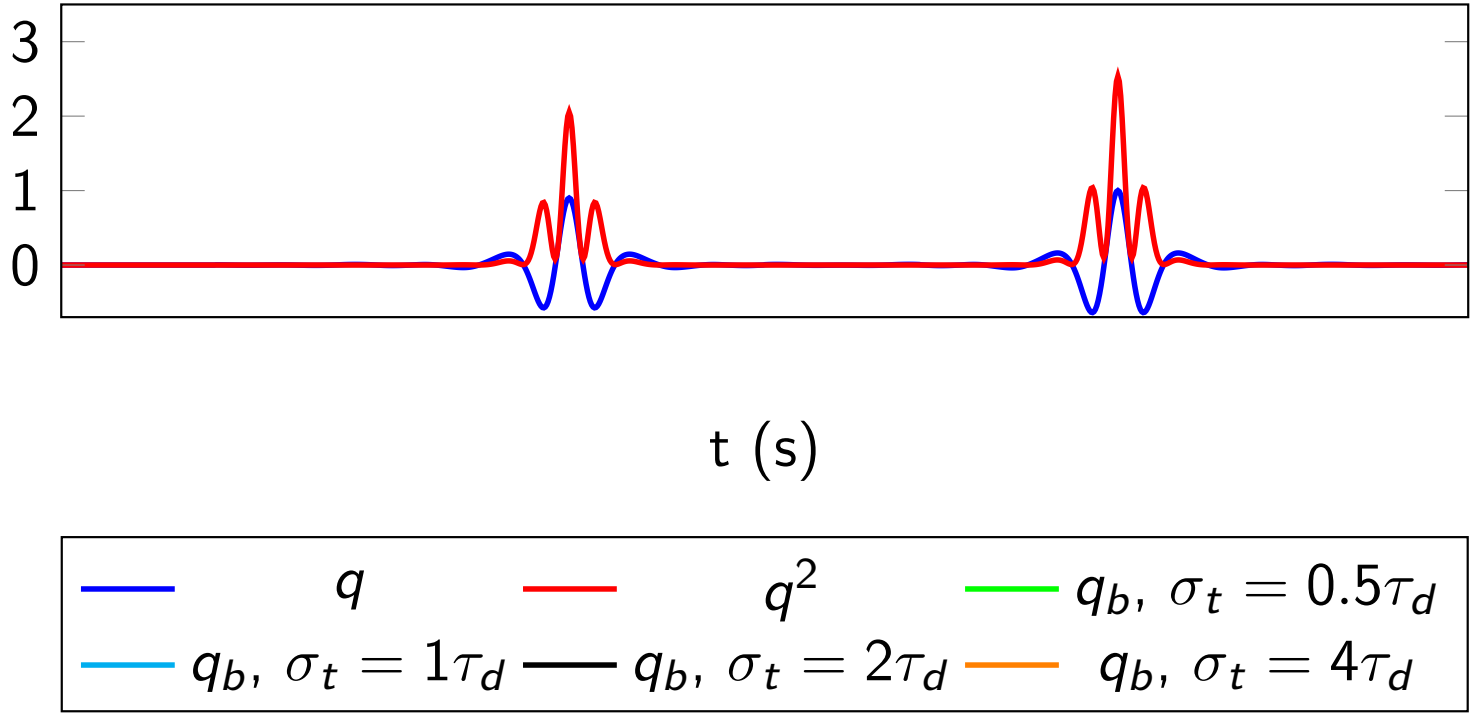

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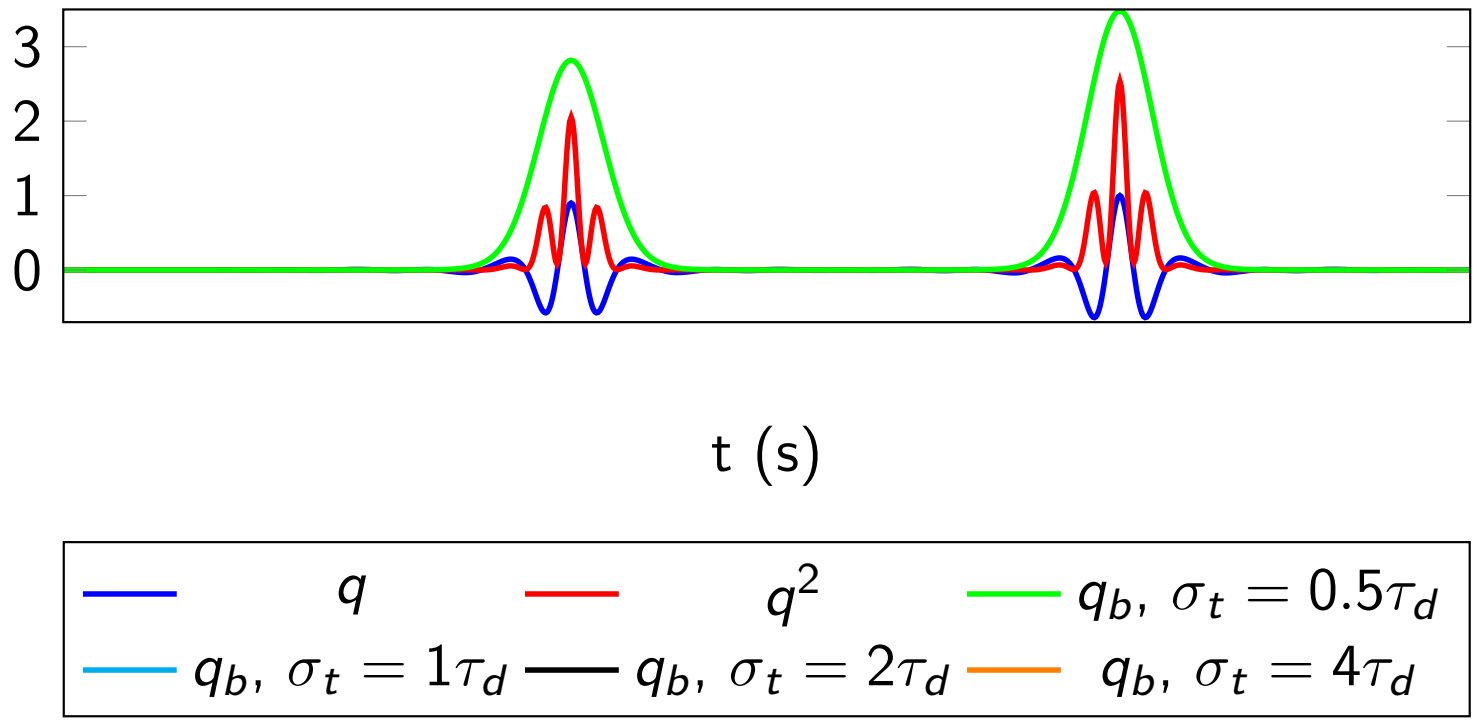

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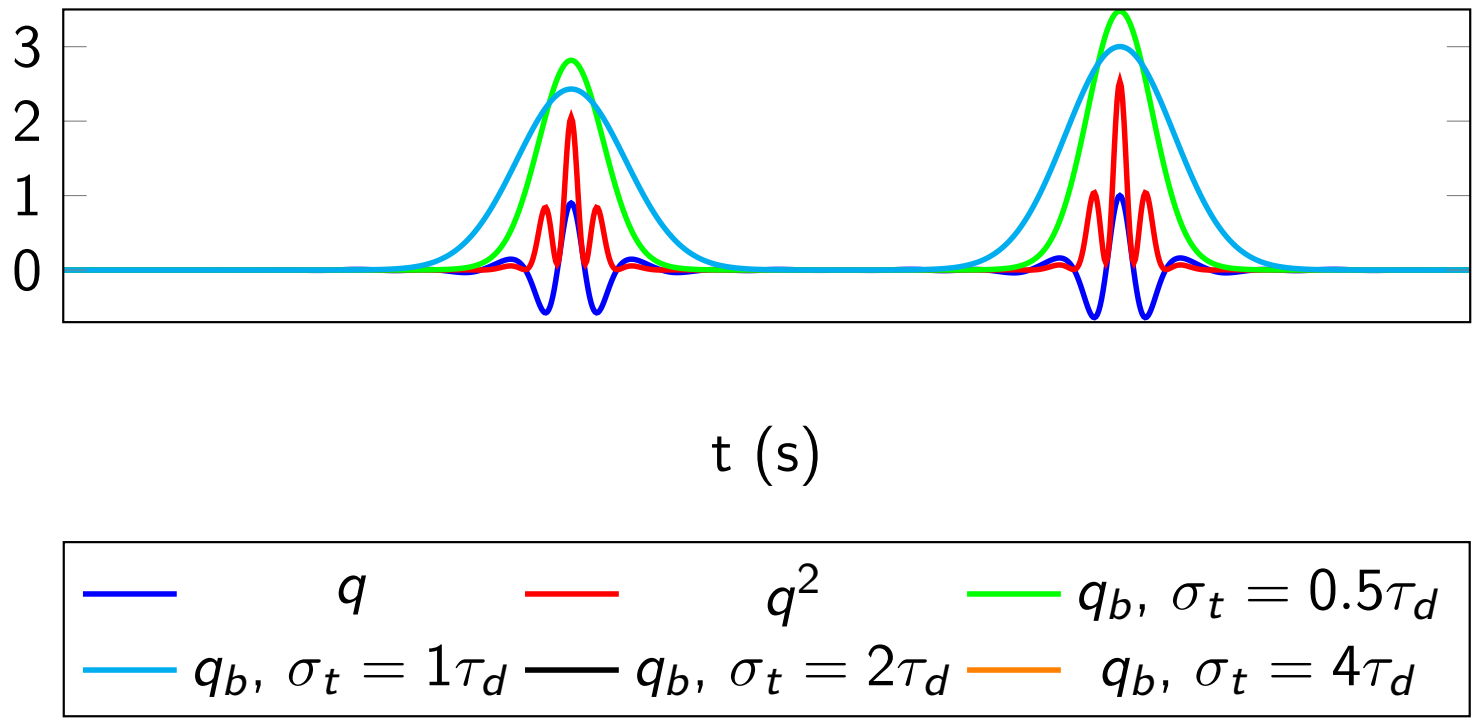

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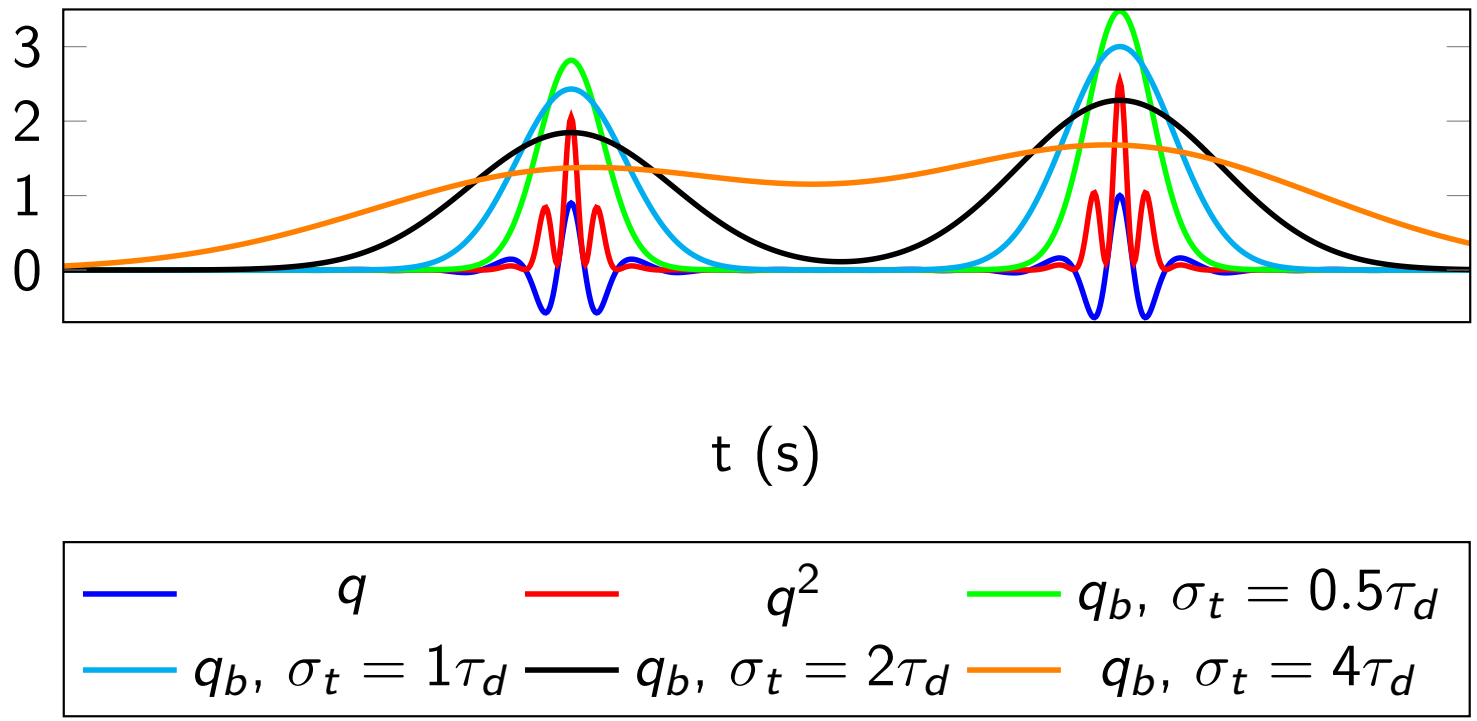
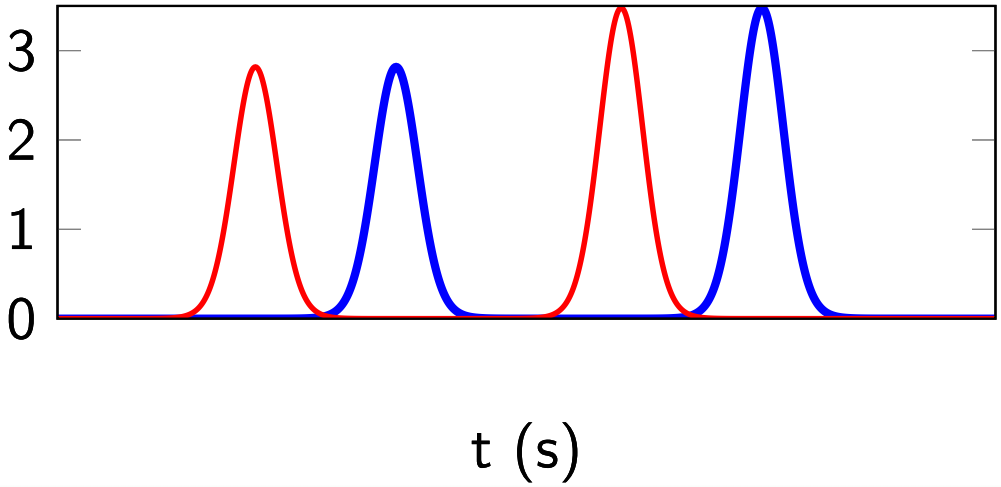


The Bump Functional

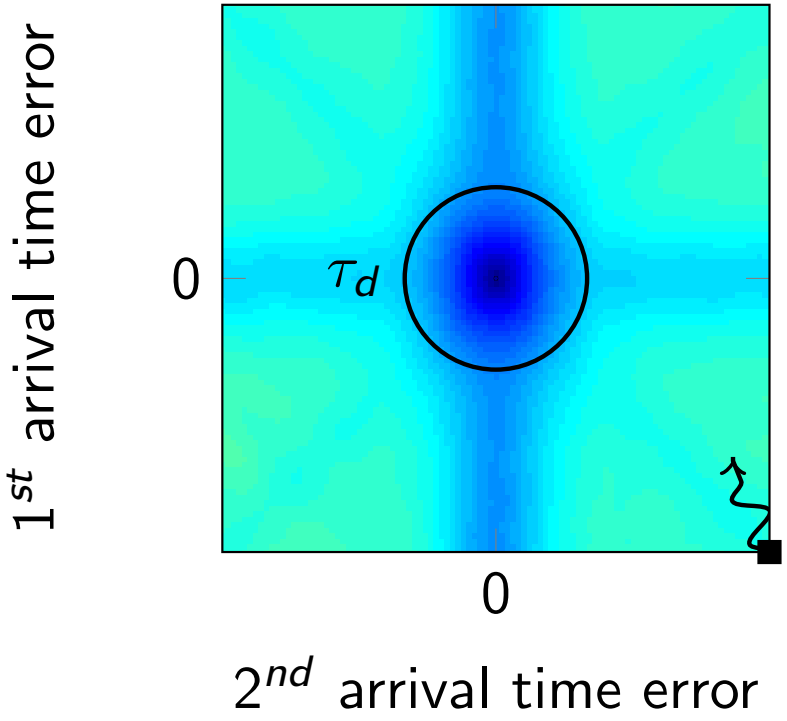
- generate bumpy data
  $p_b = b(t, \sigma_t) *_t (p^2)$
  $q_b = b(t, \sigma_t) *_t (q^2)$
- difference in bumpy data

$$J_b = \frac{1}{2} \sum_t [p_b - q_b]^2$$

p_b and q_b



$J_b\ (\sigma_t = 0.5\tau_d)$



The Bump Functional

- generate bumpy data
  $p_b = b(t, \sigma_t) *_t (p^2)$
  $q_b = b(t, \sigma_t) *_t (q^2)$
- difference in bumpy data

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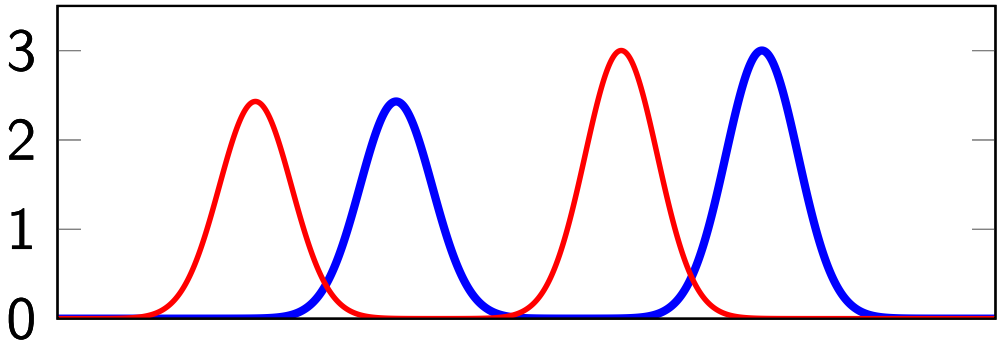
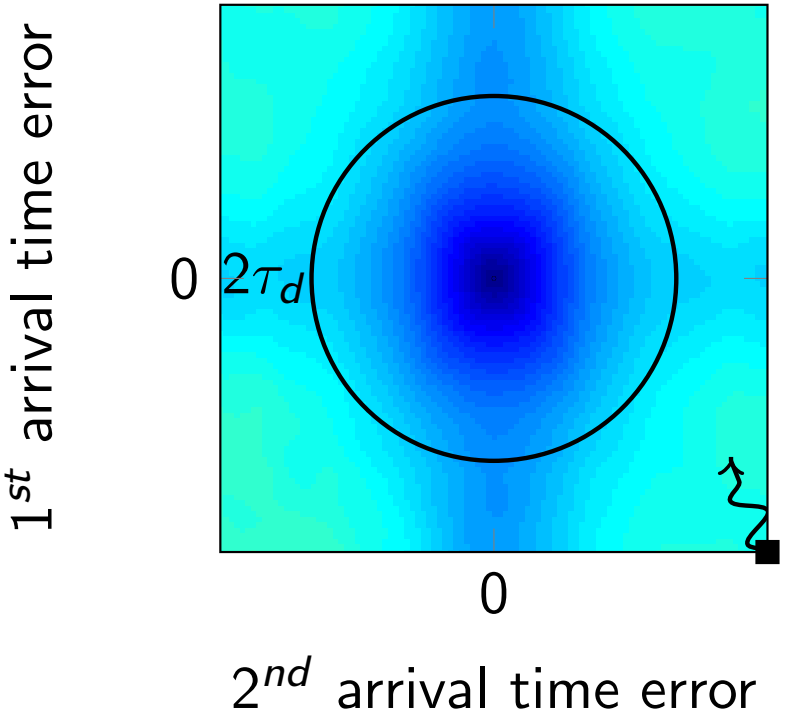
p_b and q_b



t (s)

$J_b\ (\sigma = \tau_d)$



The Bump Functional

- generate bumpy data
  $p_b = b(t, \sigma_t) *_t (p^2)$
  $q_b = b(t, \sigma_t) *_t (q^2)$
- difference in bumpy data

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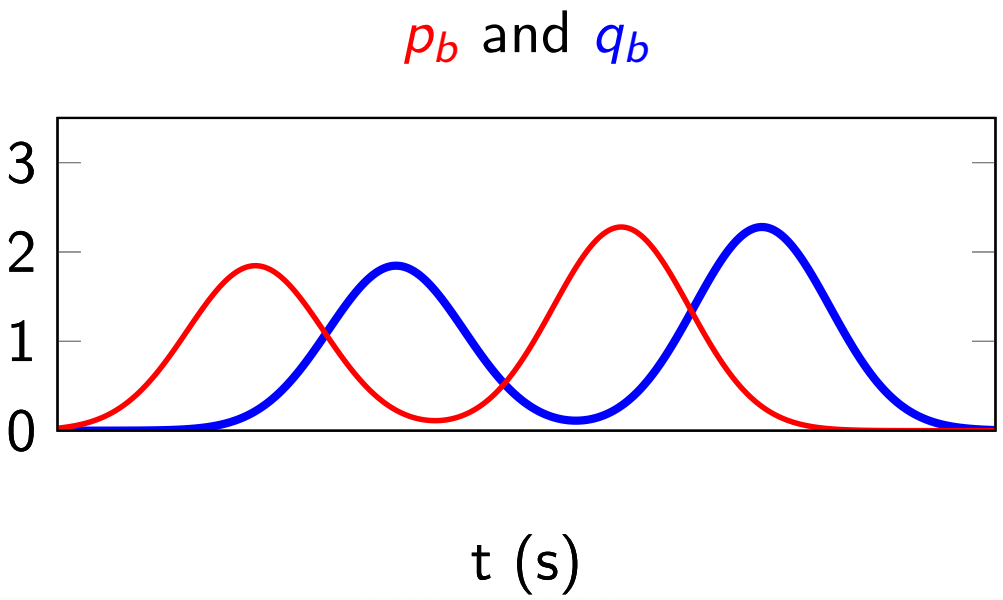

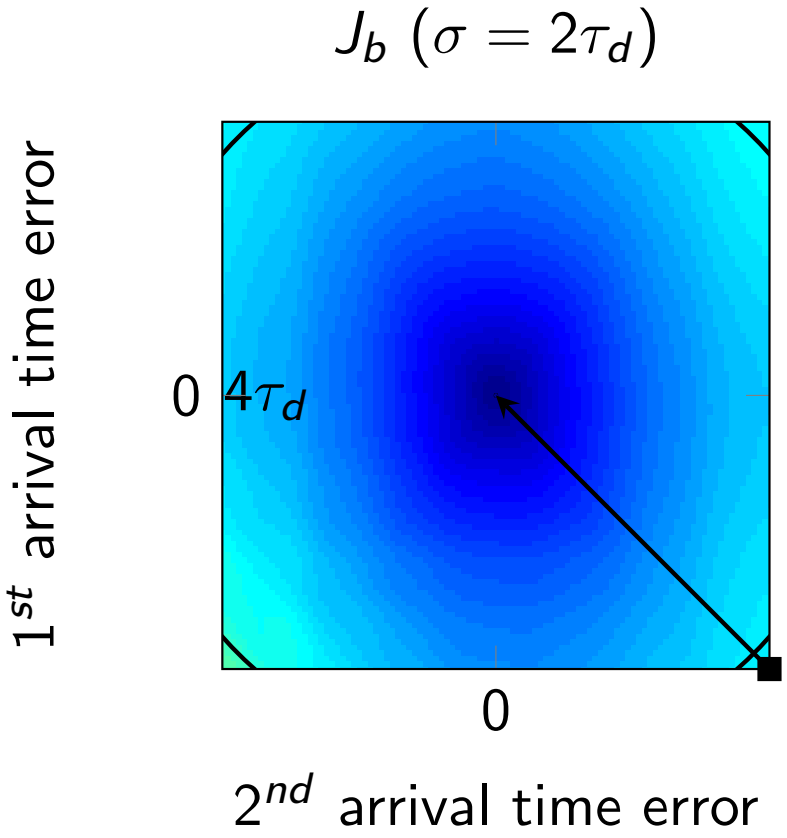

The Bump Functional

- generate bumpy data
  $p_b = b(t, \sigma_t) *_t (p^2)$
  $q_b = b(t, \sigma_t) *_t (q^2)$
- difference in bumpy data

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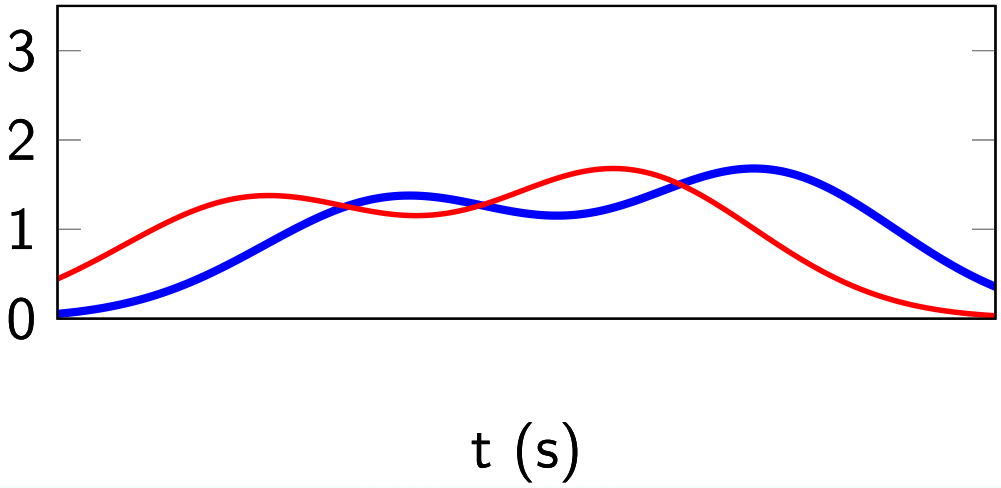
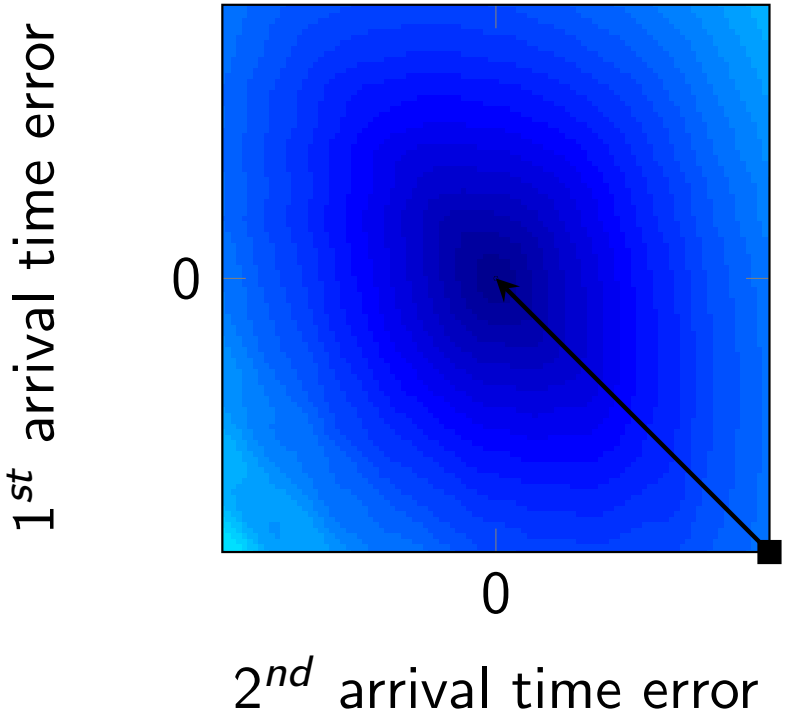
p_b and q_b



$J_b \ (\sigma = 4\tau_d)$

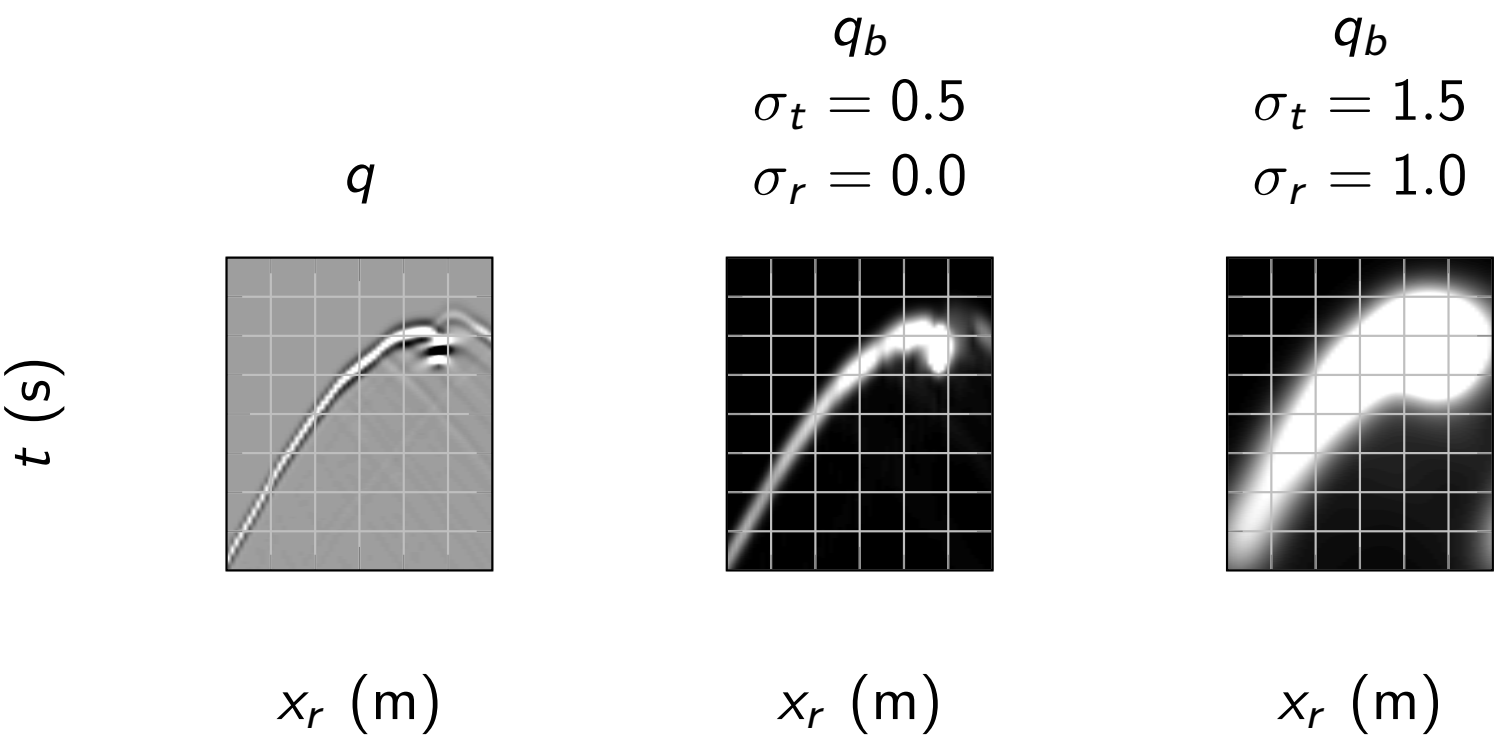


Blurring along a receiver coordinate using σ_r

- squared or absolute-valued data
- Gaussian blurring with $b(x_r, \sigma_r) = \exp \frac{-x_r^2}{\sigma_r^2}$

$$q_b = b(x_r, \sigma_r) *_r b(t, \sigma_t) *_t (q^2)$$

- σ_r controls the width of the bumps

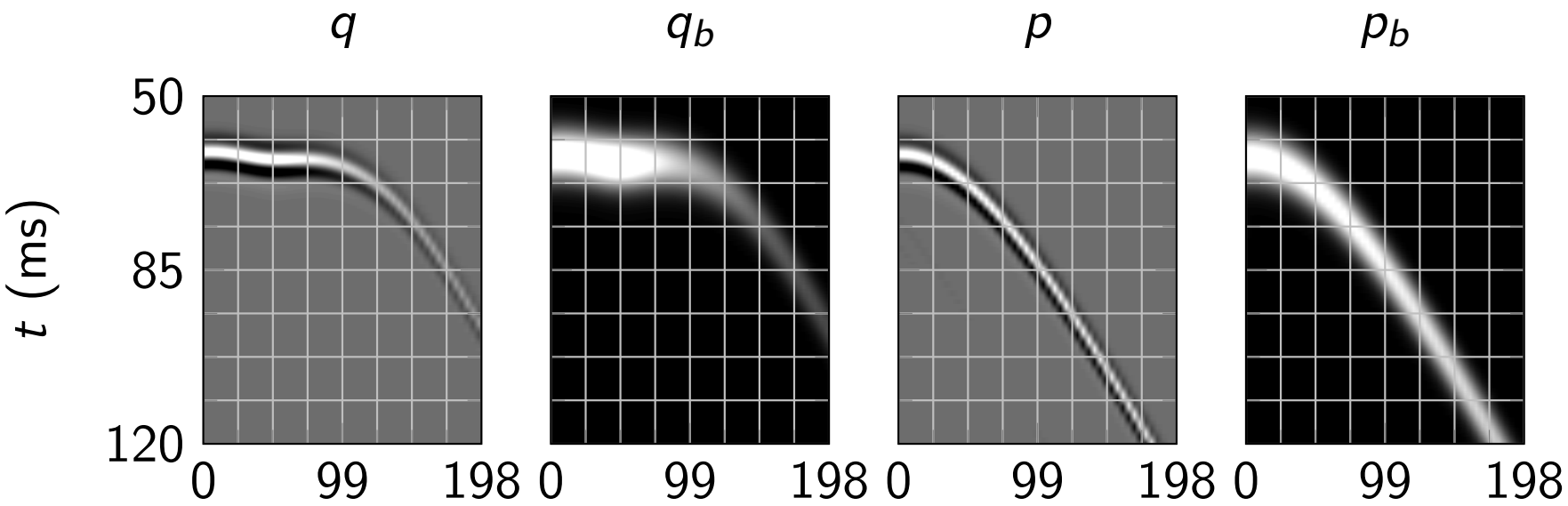


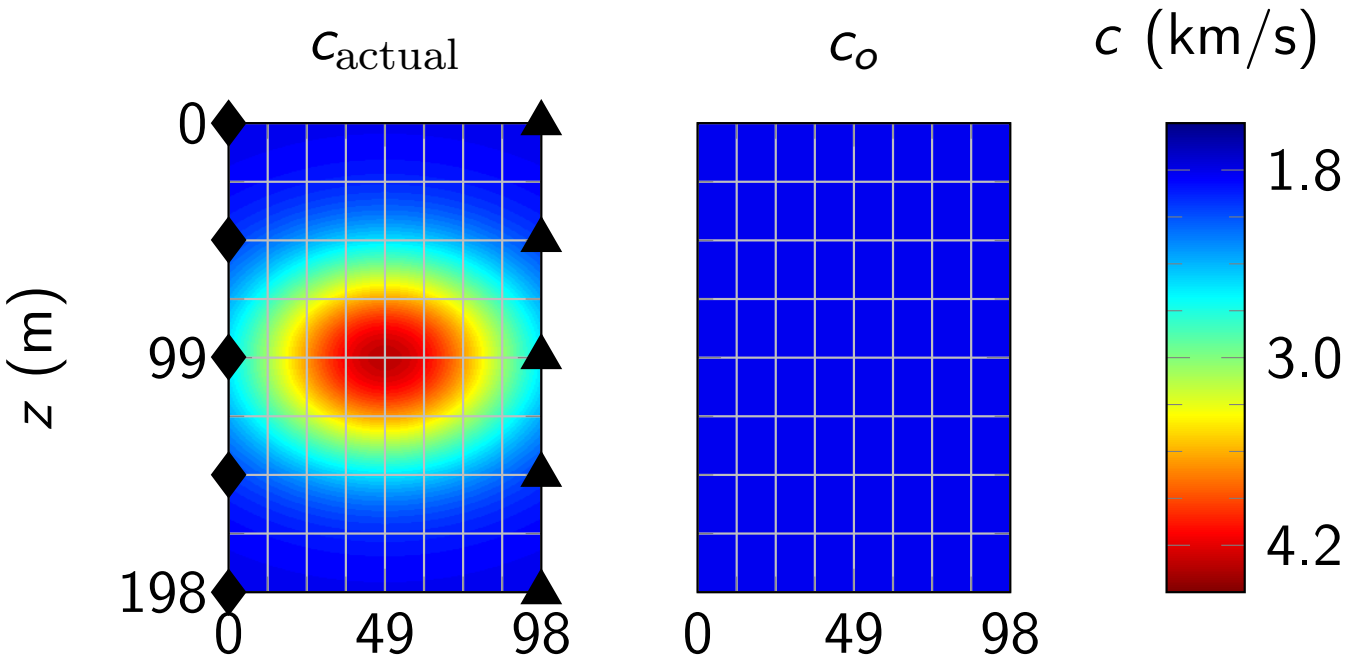
1 Introduction
- Cycle Skipping; Local Minima
- Envelope Functional

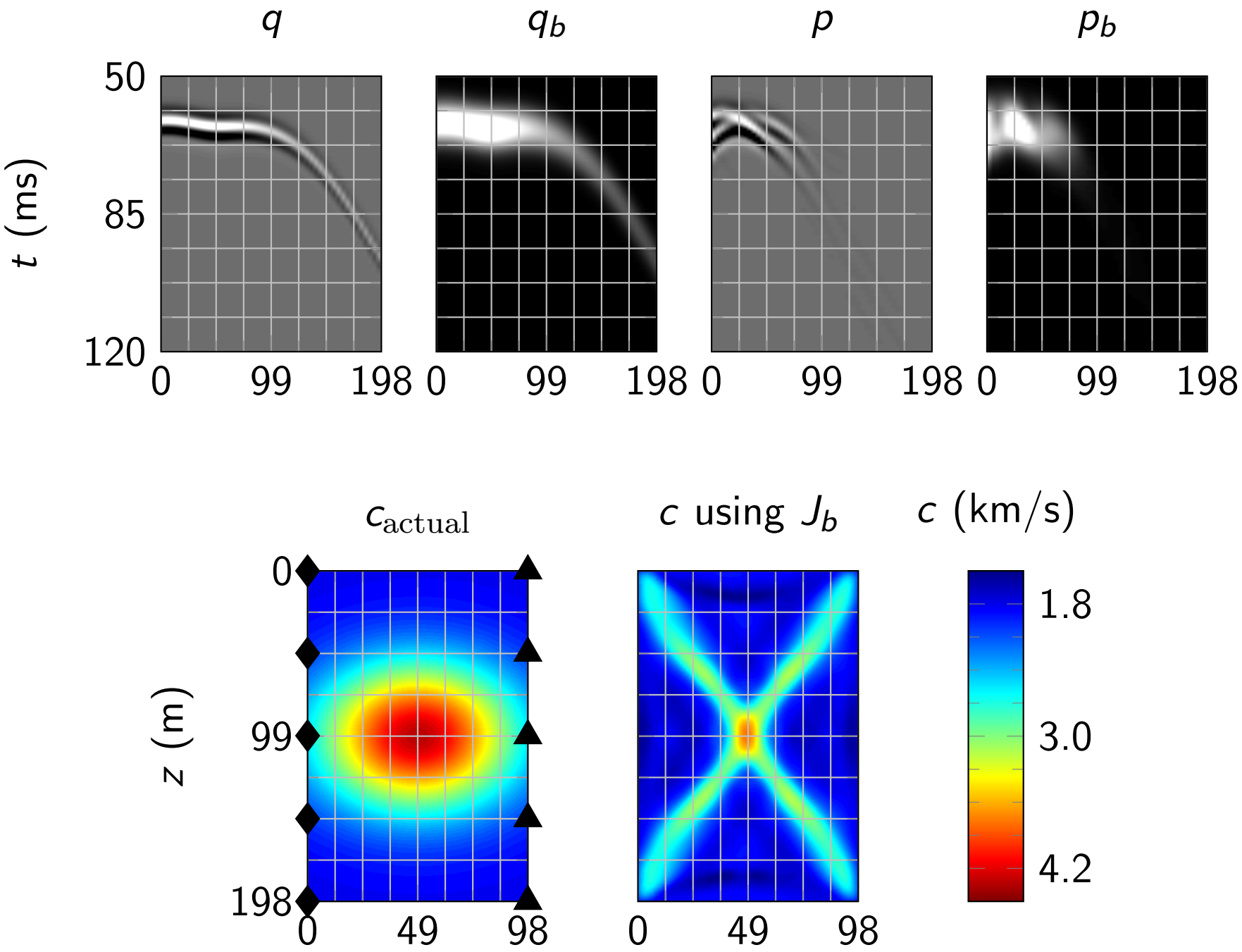
2 The Bump Functional
- Definition
- Characteristics
  - Convergence Robustness
  - Non-uniqueness

3 Multi-objective Inversion
- Motivation
- Illustration
- Realistic Examples

4 Conclusions


q
q_b
p
p_b
t (ms)
50
85
120
0
99
198

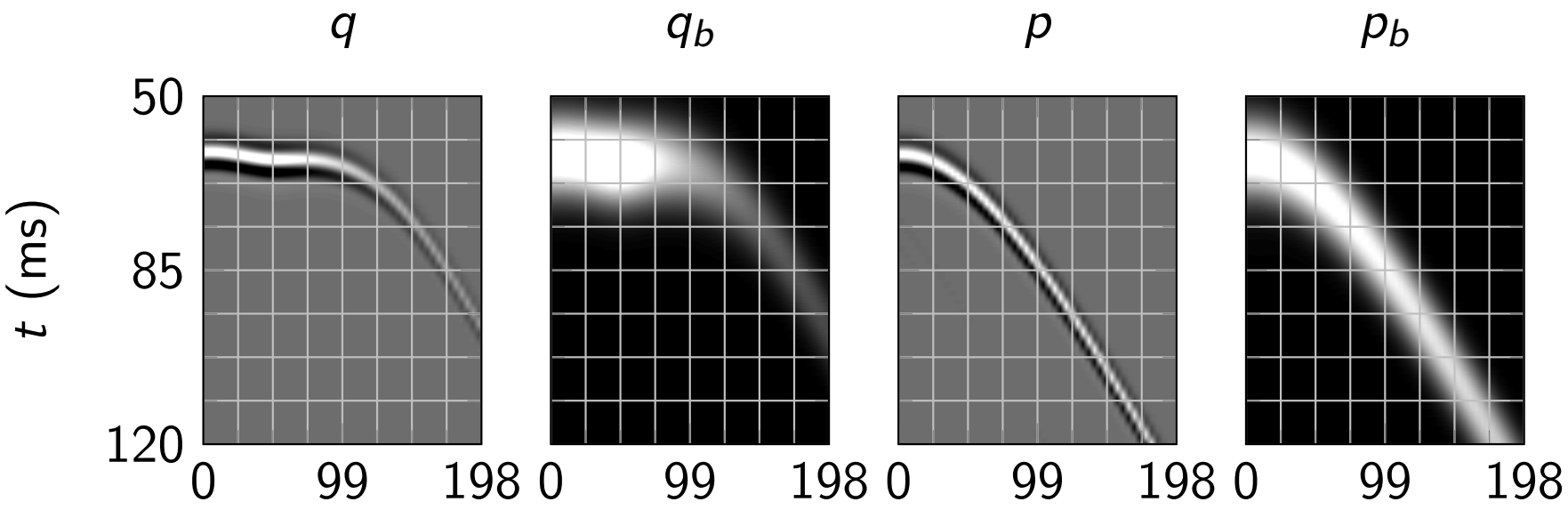

c_{actual}
c_o
c (km/s)
z (m)
0
99
198
0
49
98
1.8
3.0
4.2

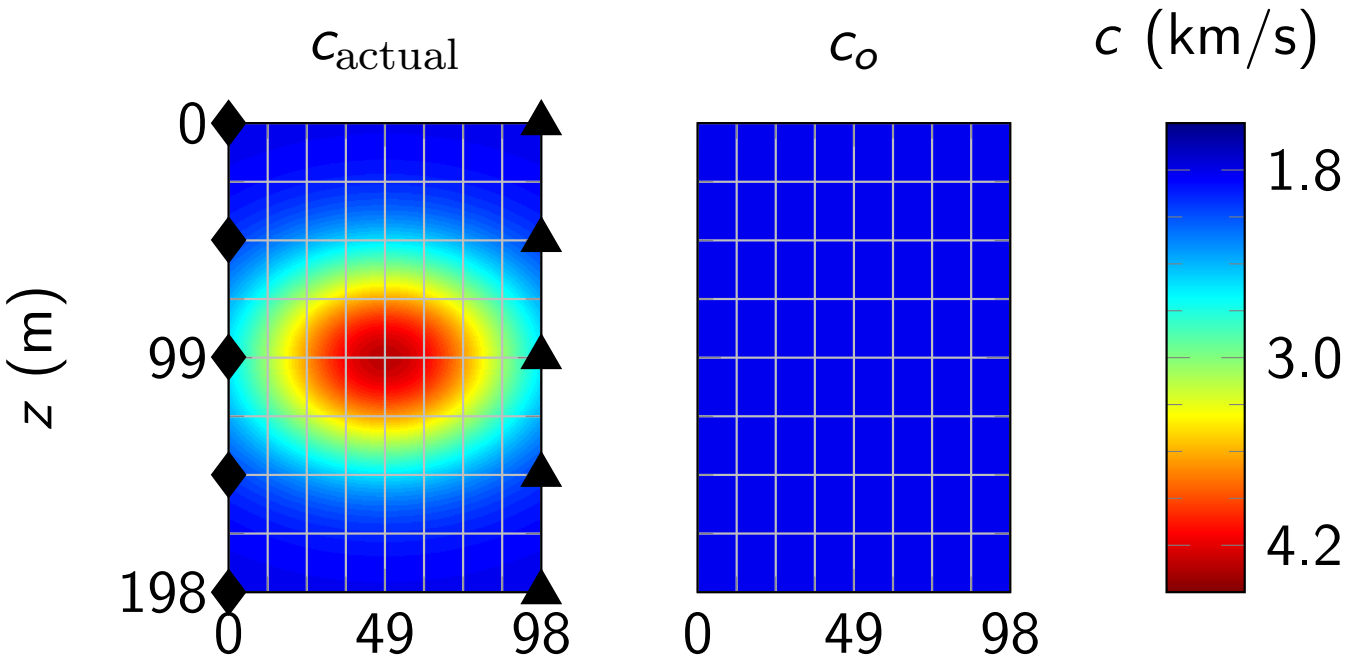

q
q_b
p
p_b
t (ms)
50
85
120
0
99
198
c_{actual}
c using J_b
c (km/s)
z (m)
0
49
98
1.8
3.0
4.2

Cross-well Example

	$p_b == q_b$	$p == q$	**comment**
$\sigma_t = 0.5\tau_d$	no	no	lack of sensitivity
$\sigma_t = \tau_d$	no	no	lack of sensitivity
$\sigma_t = 2\tau_d$	yes	yes	acceptable
$\sigma_t = 4\tau_d$	yes	yes	acceptable
$\sigma_t = 8\tau_d$	yes	no	non-uniqueness

- Sensitivity to arrival time errors $>> \tau_d$ is possible when σ_t is high.
- At the same time, increasing σ_t results in increased non-uniqueness.


q
q_b
p
p_b
t (ms)
50
85
120
0
99
198


c_{actual}
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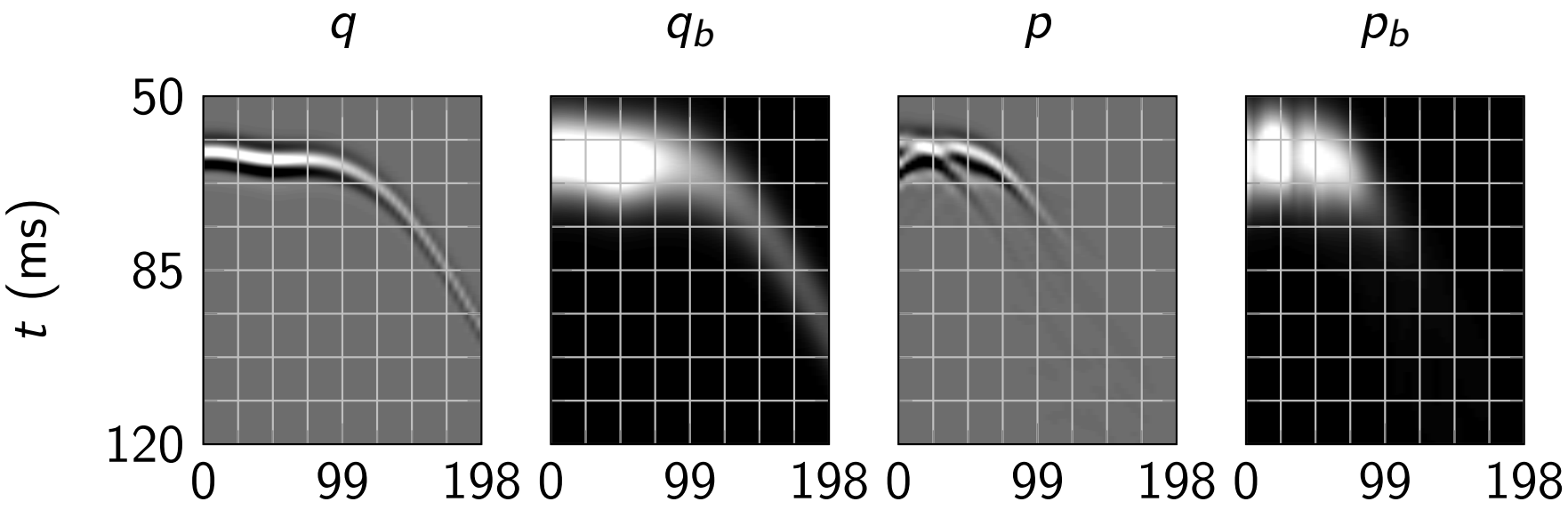
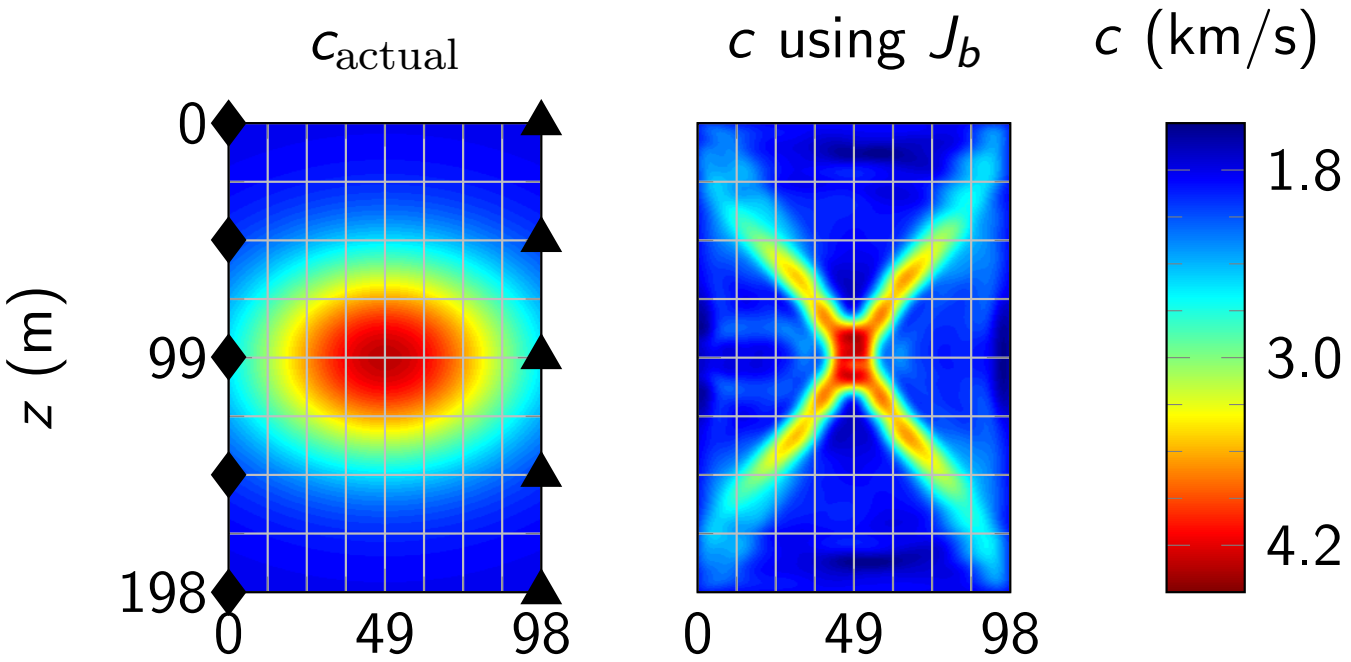

q
q_b
p
p_b
t (ms)
50
85
120
0
99
198


c_{actual}
c using J_b
c (km/s)
z (m)
0
99
198
0
49
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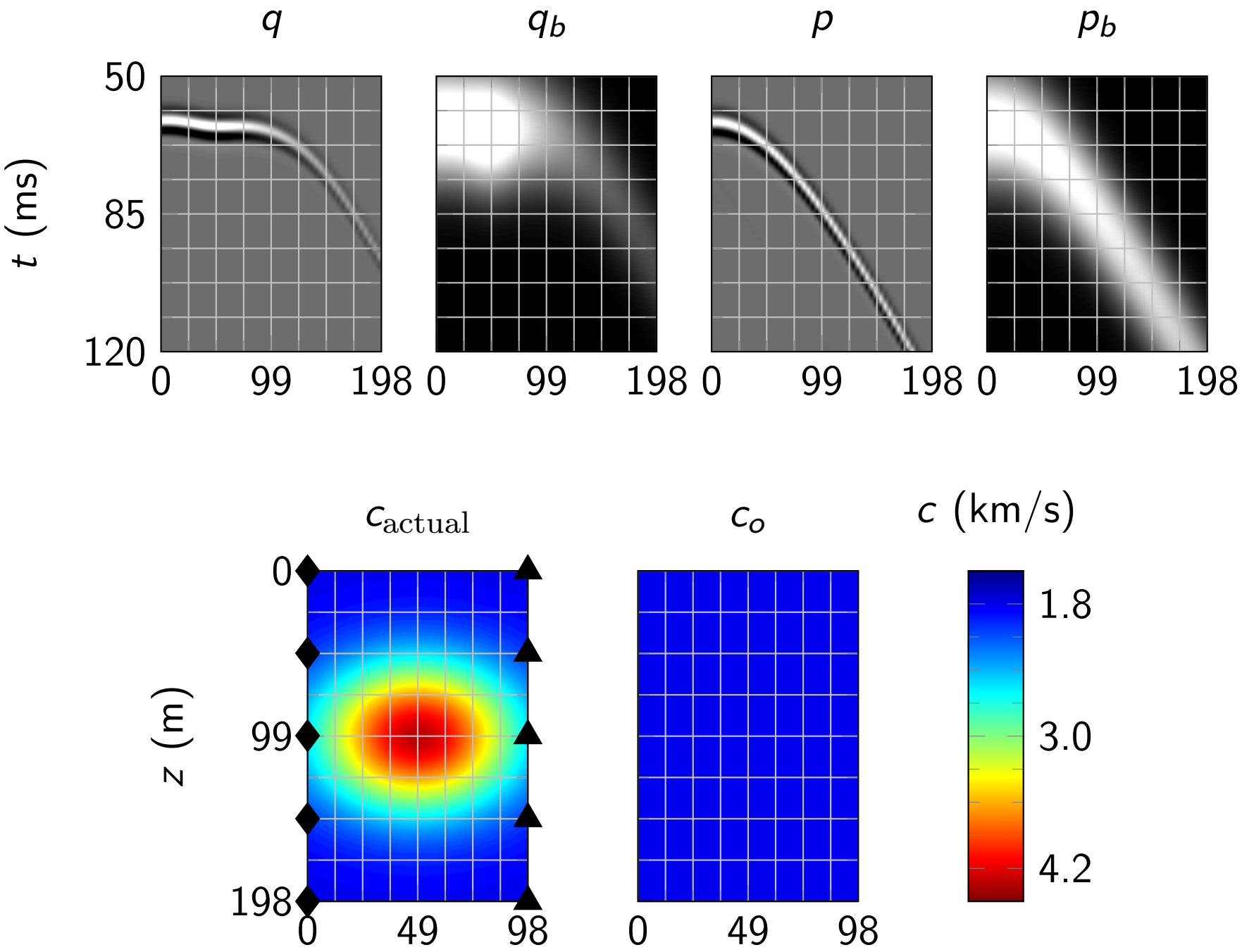
q
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p_b
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50
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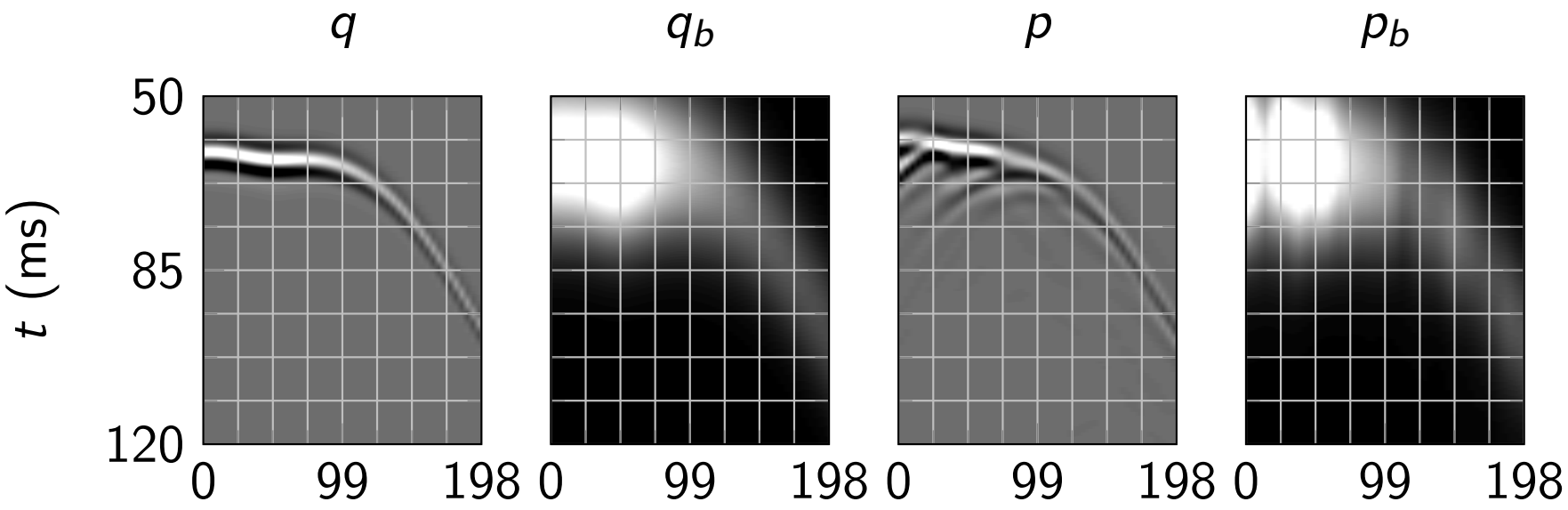
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q_b
p
p_b
t (ms)
50
85
120
0
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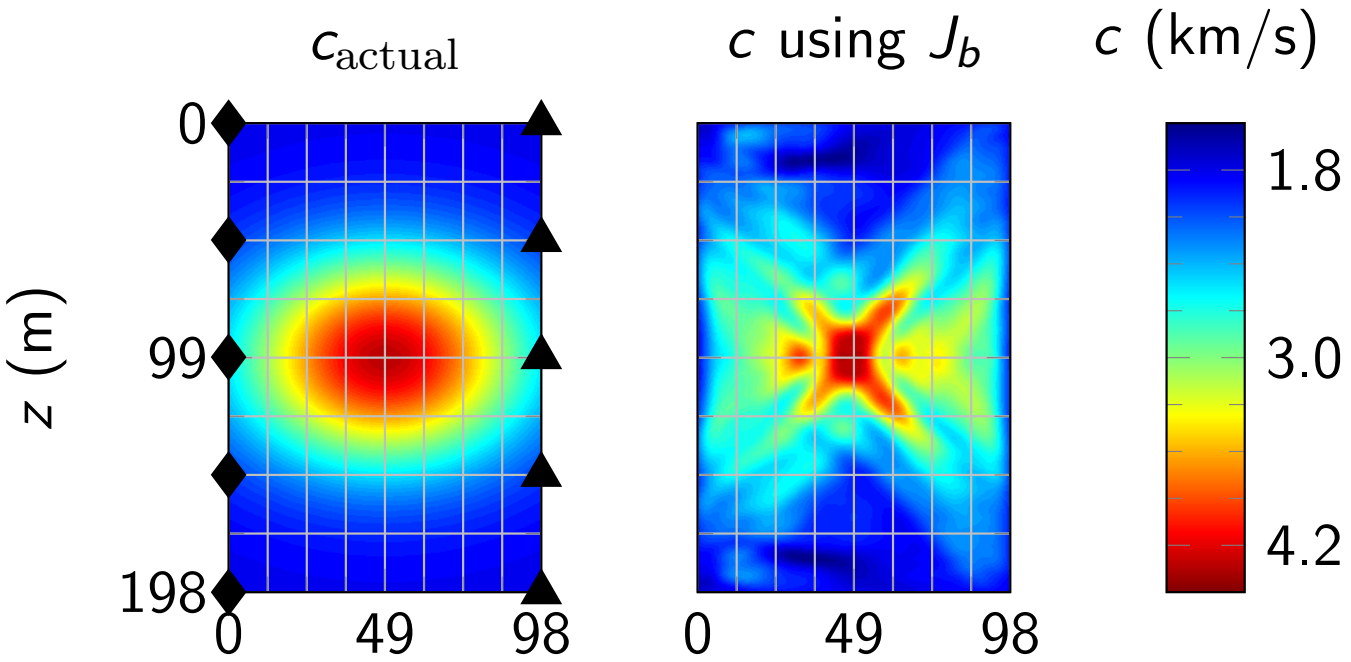
198


c_{actual}
c using J_b
c (km/s)
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1.8
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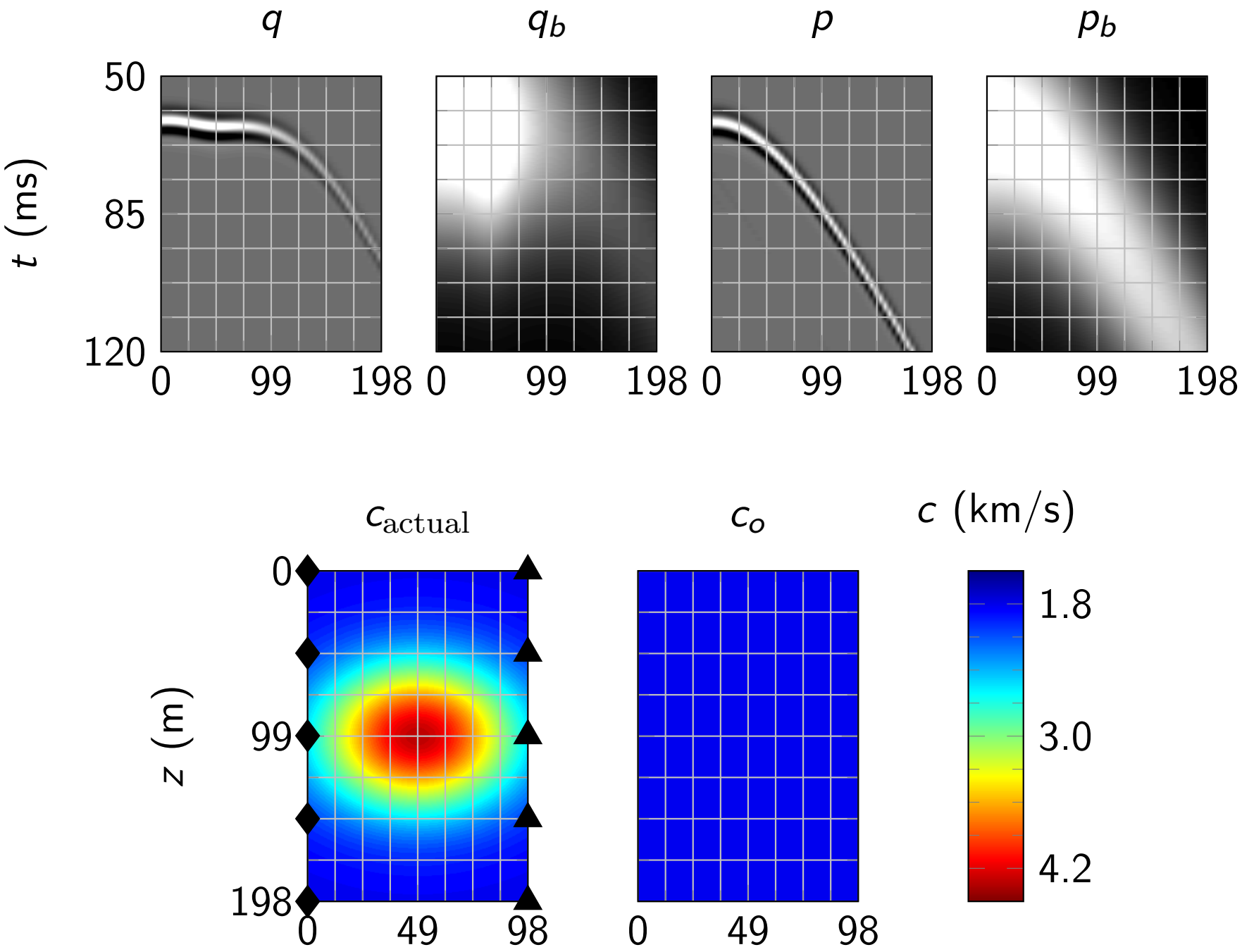
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q_b
p
p_b
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120
0
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c_{actual}
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c (km/s)
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0
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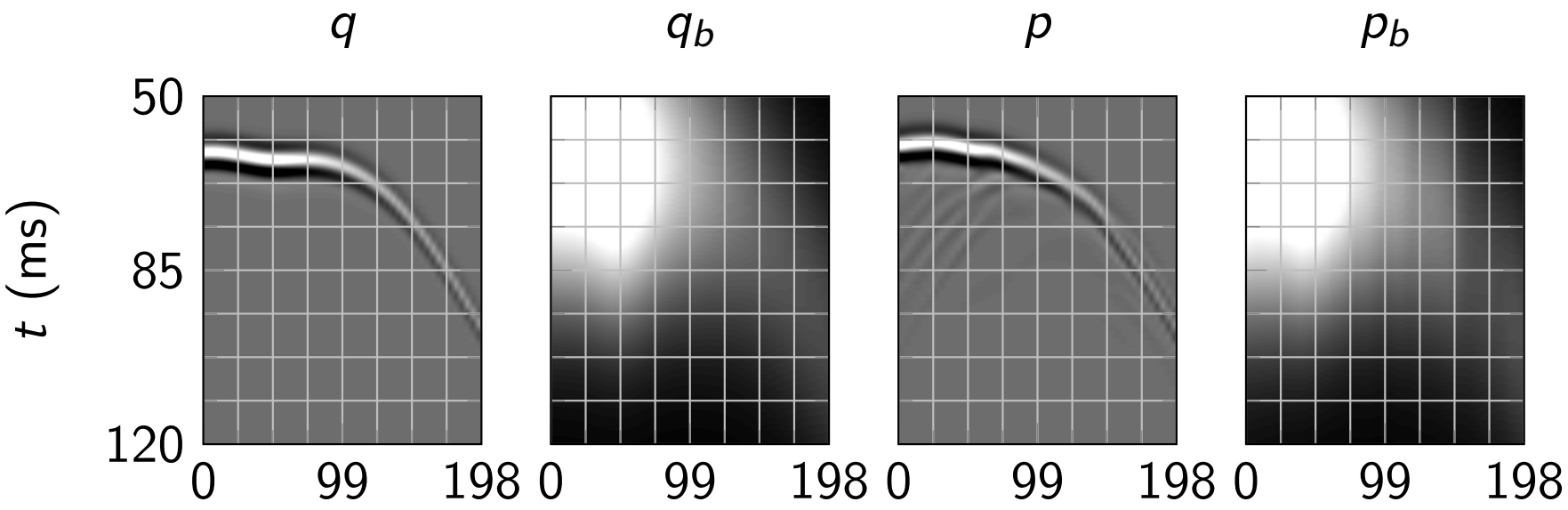
1.8
3.0
4.2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85
120
0
99
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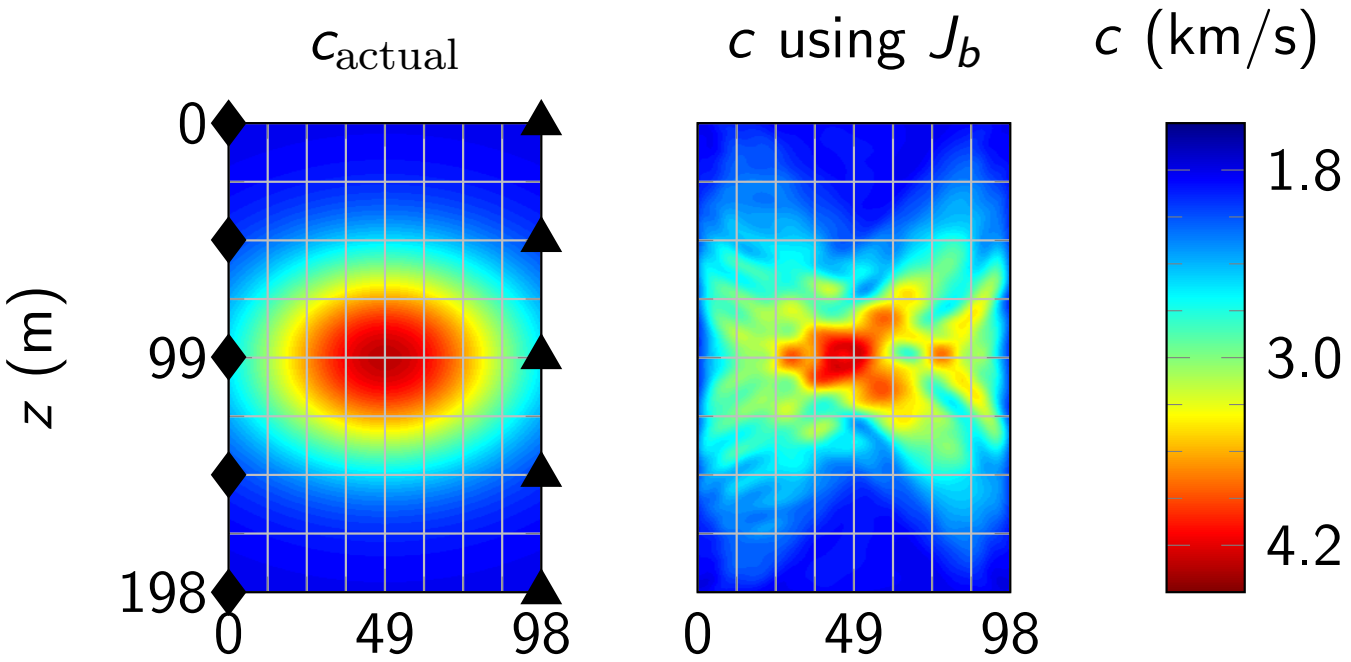
c_{actual}
c using J_b
c (km/s)
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0
99
198
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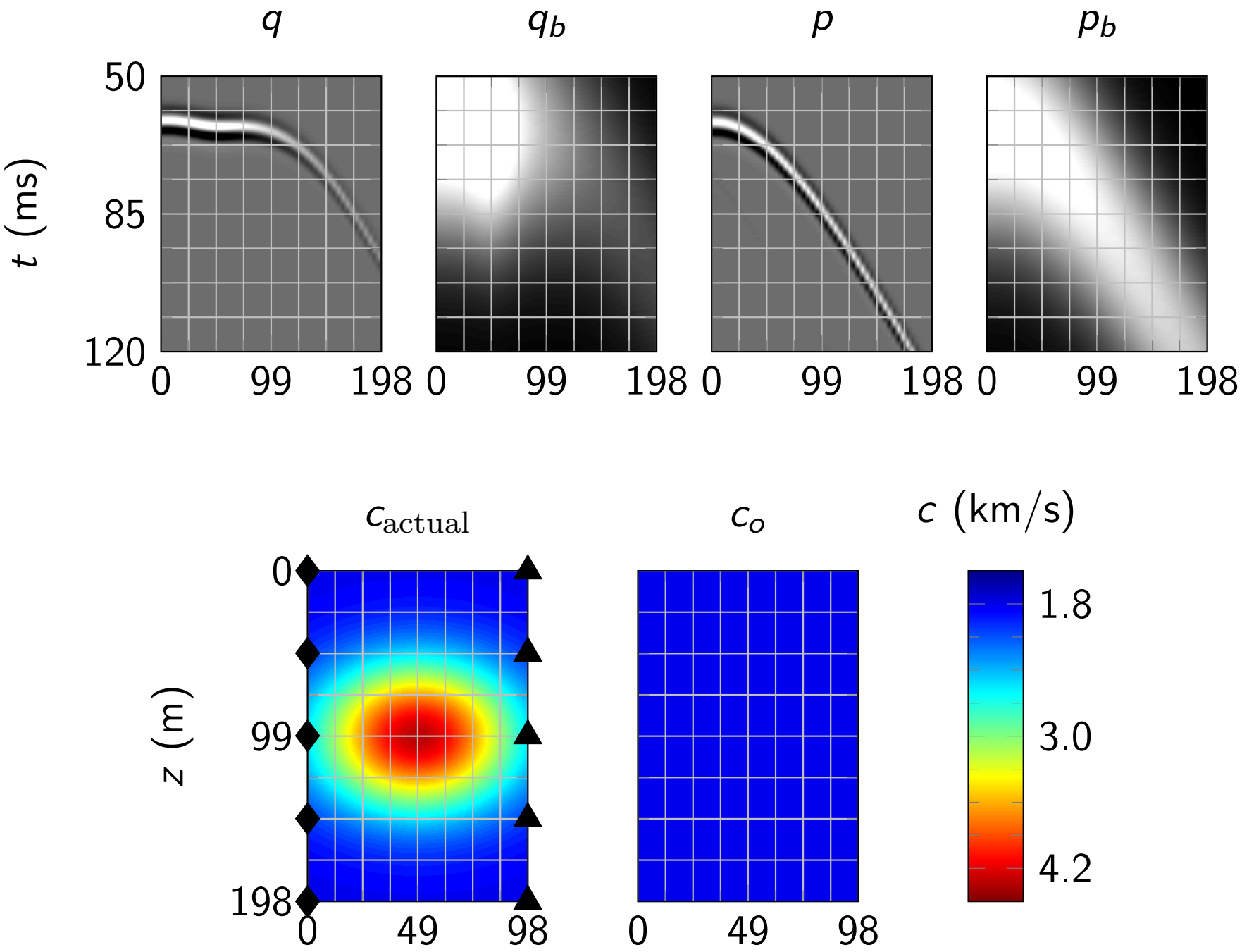
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q_b
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p_b
t (ms)
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120
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c_{actual}
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c (km/s)
z (m)
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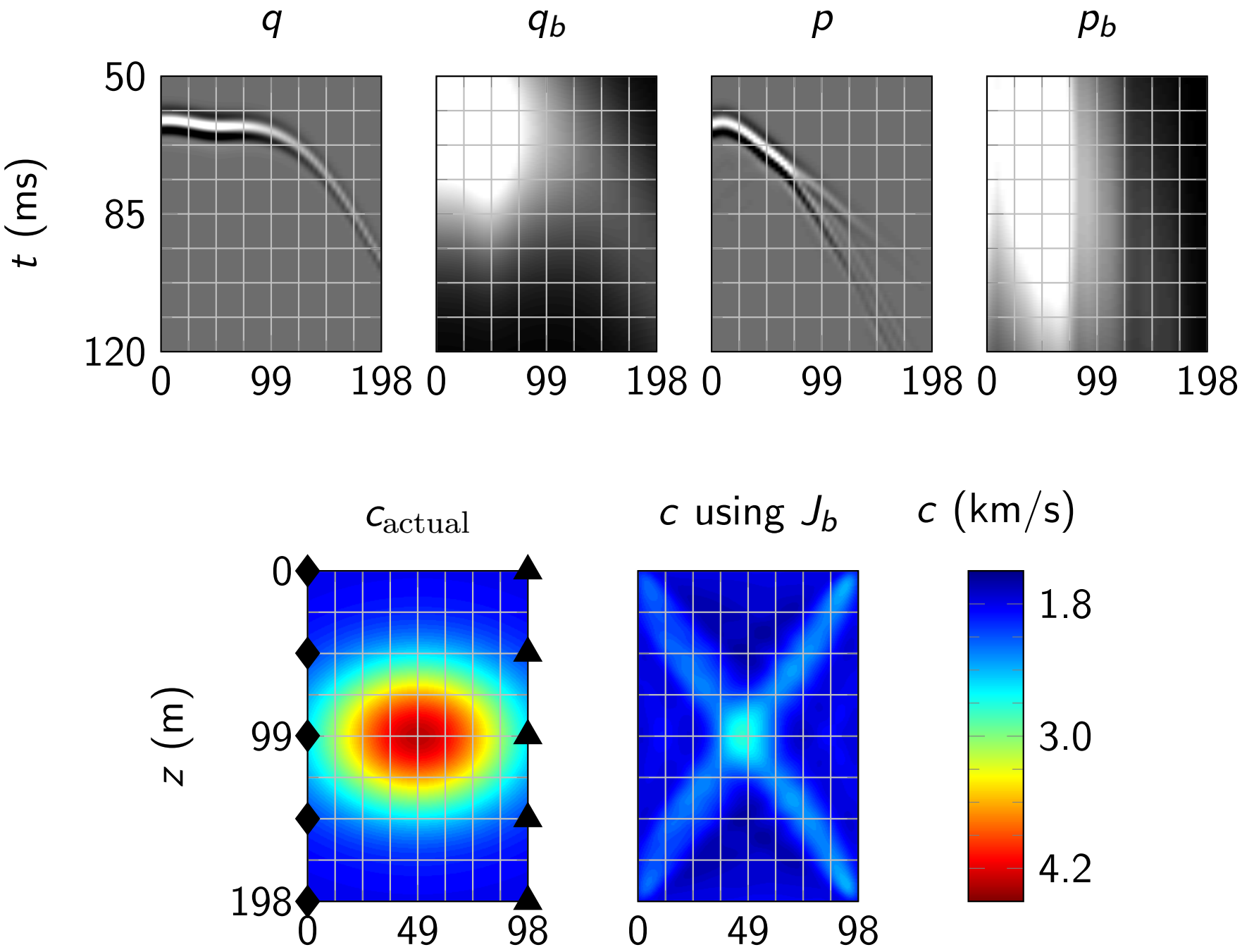
1.8
3.0
4.2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q_b
p
p_b
t (ms)
50
85
120
0
99
198
c_{actual}
c using J_b
c (km/s)
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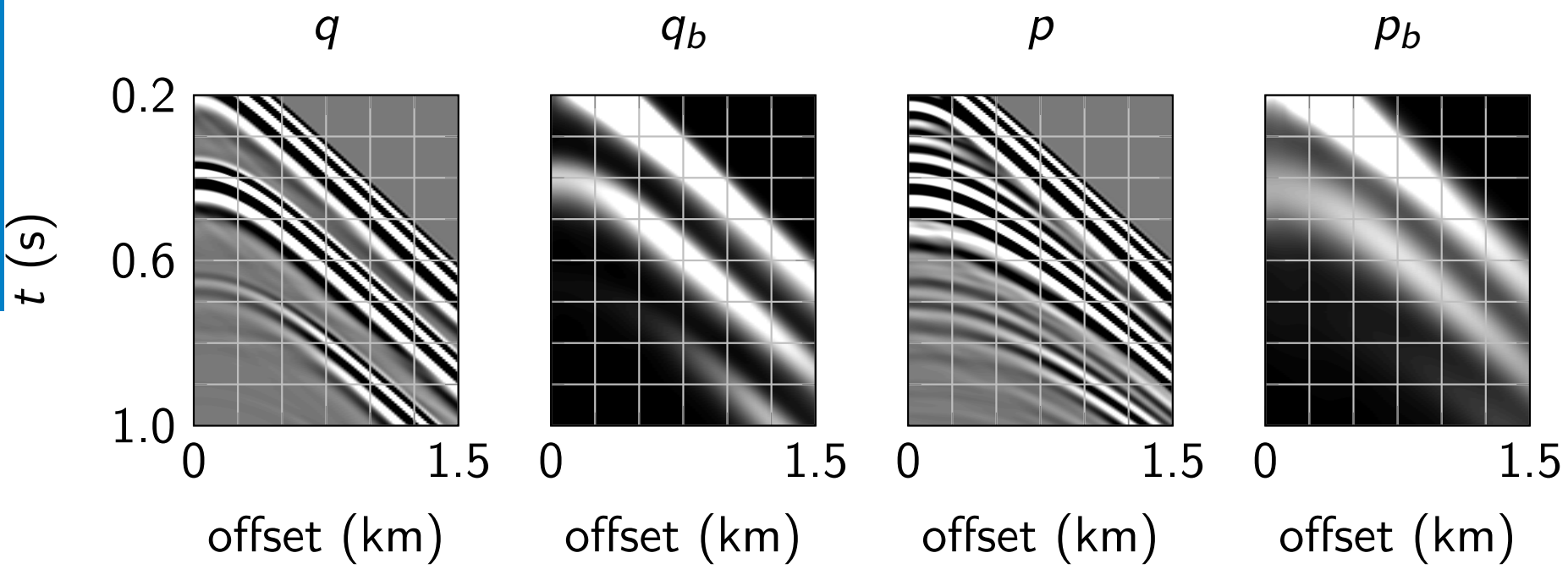
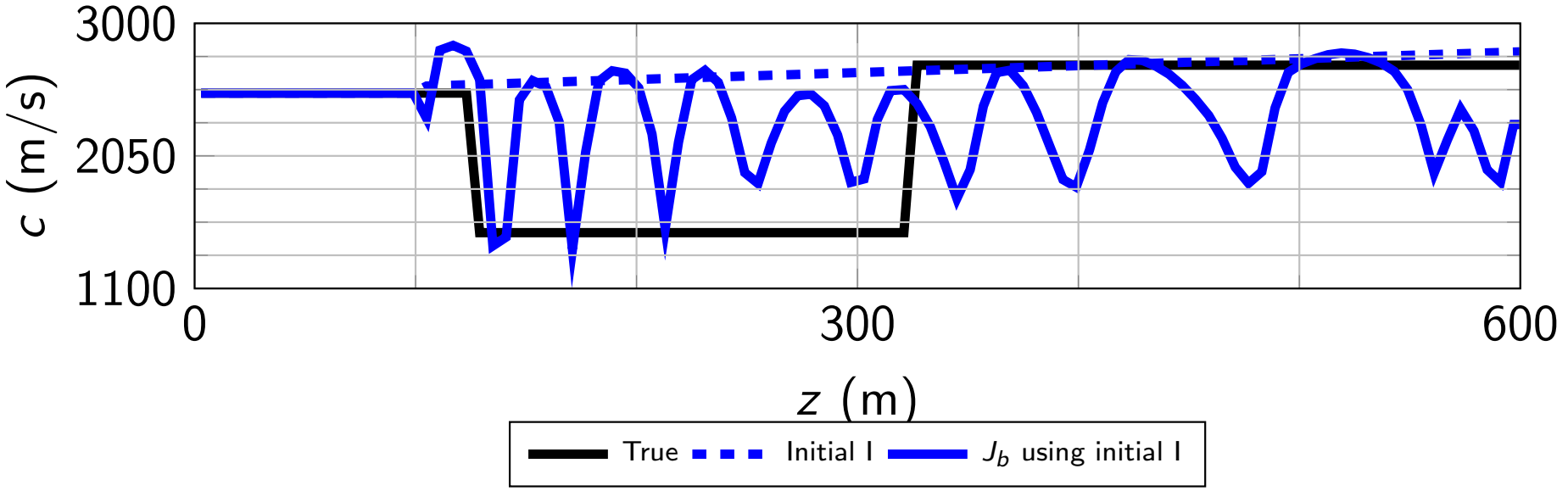
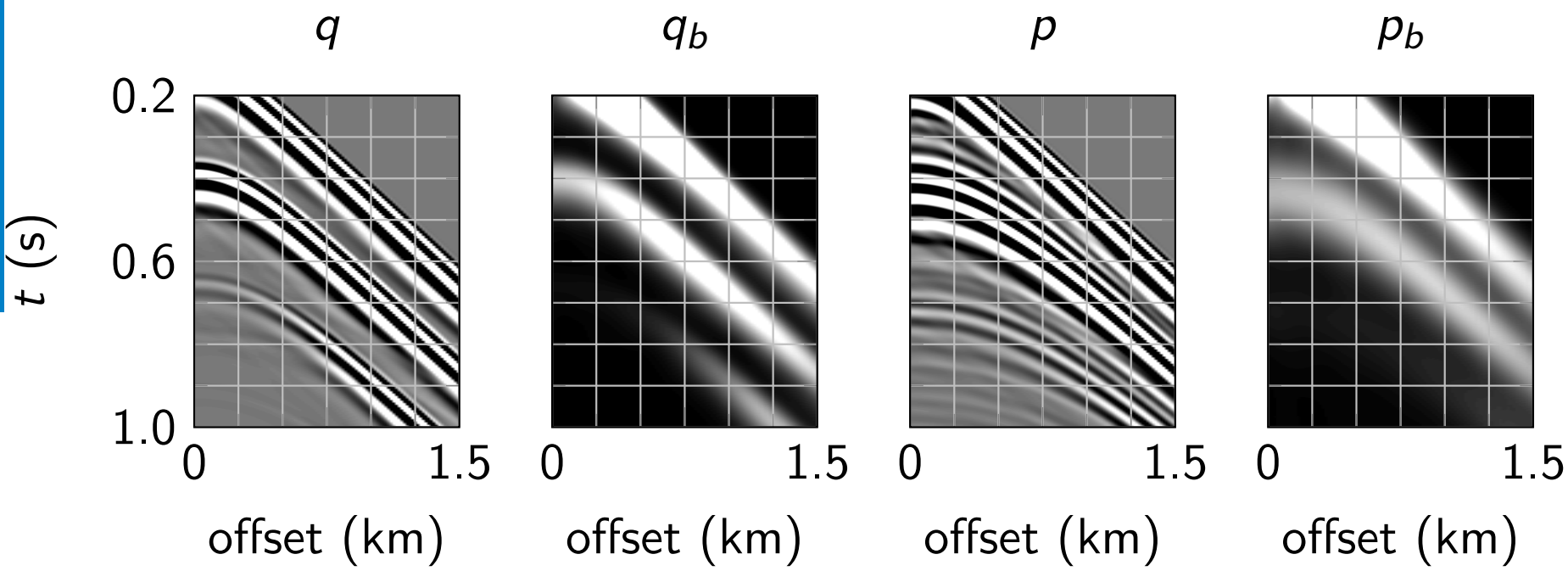

q
q_b
p
p_b
0.2
0.6
1.0
t (s)
0
1.5
offset (km)


3000
2050
1100
c (m/s)
0
300
600
z (m)
True
Initial I
J_b using initial I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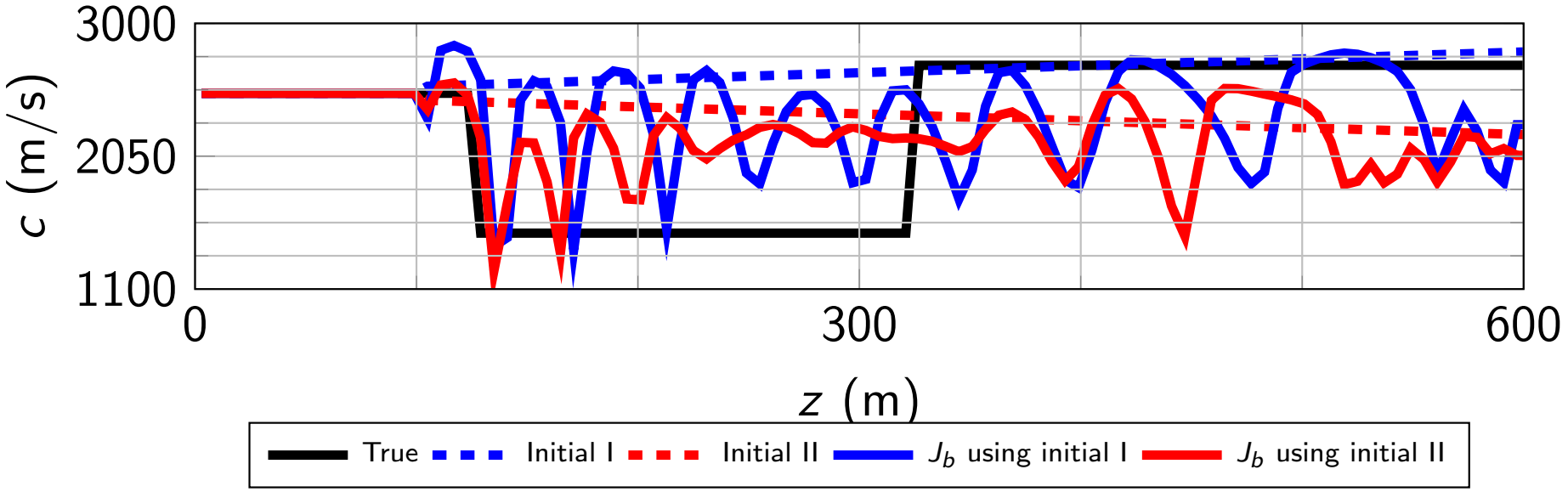
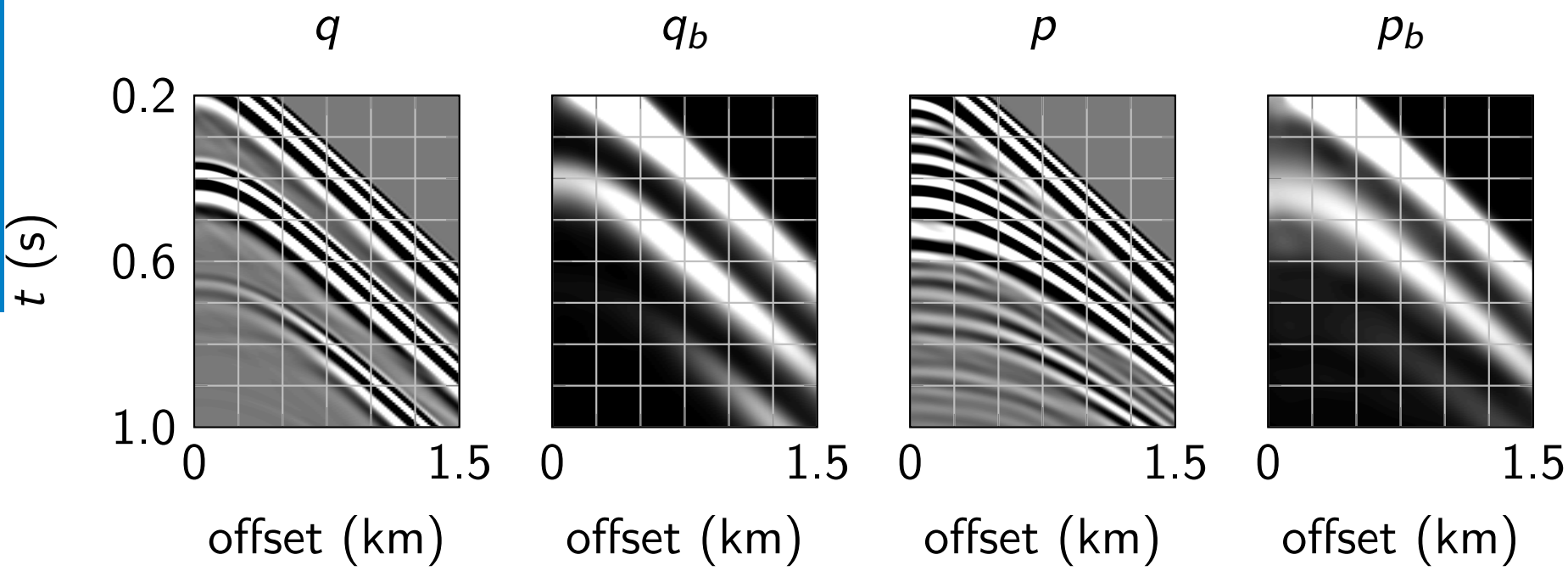
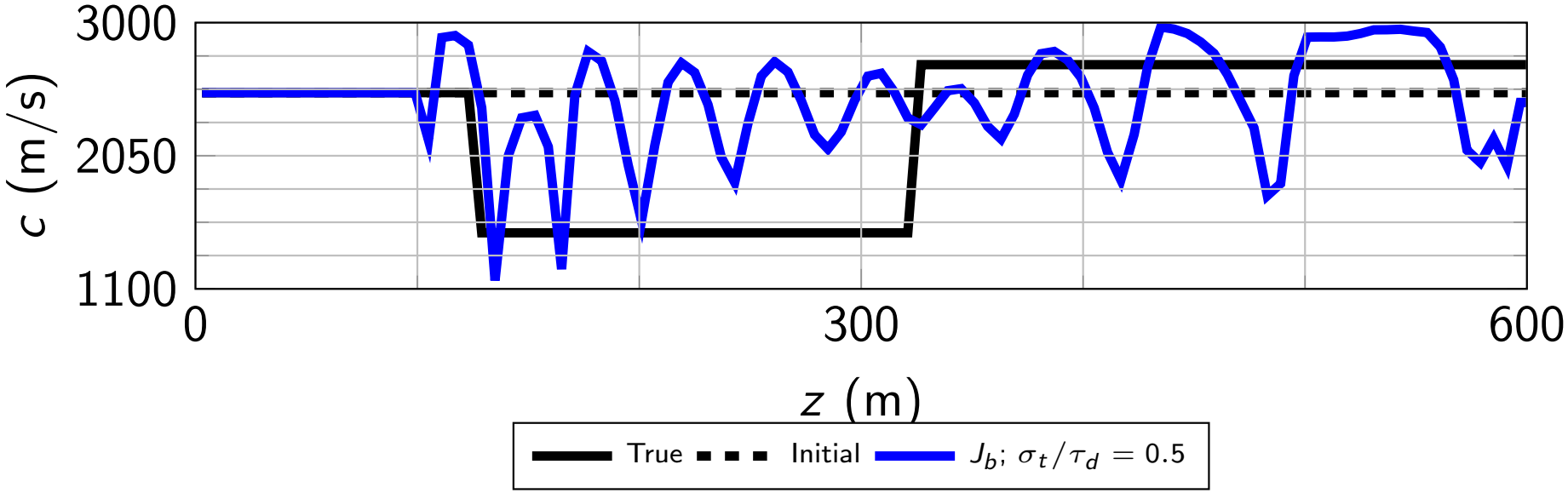
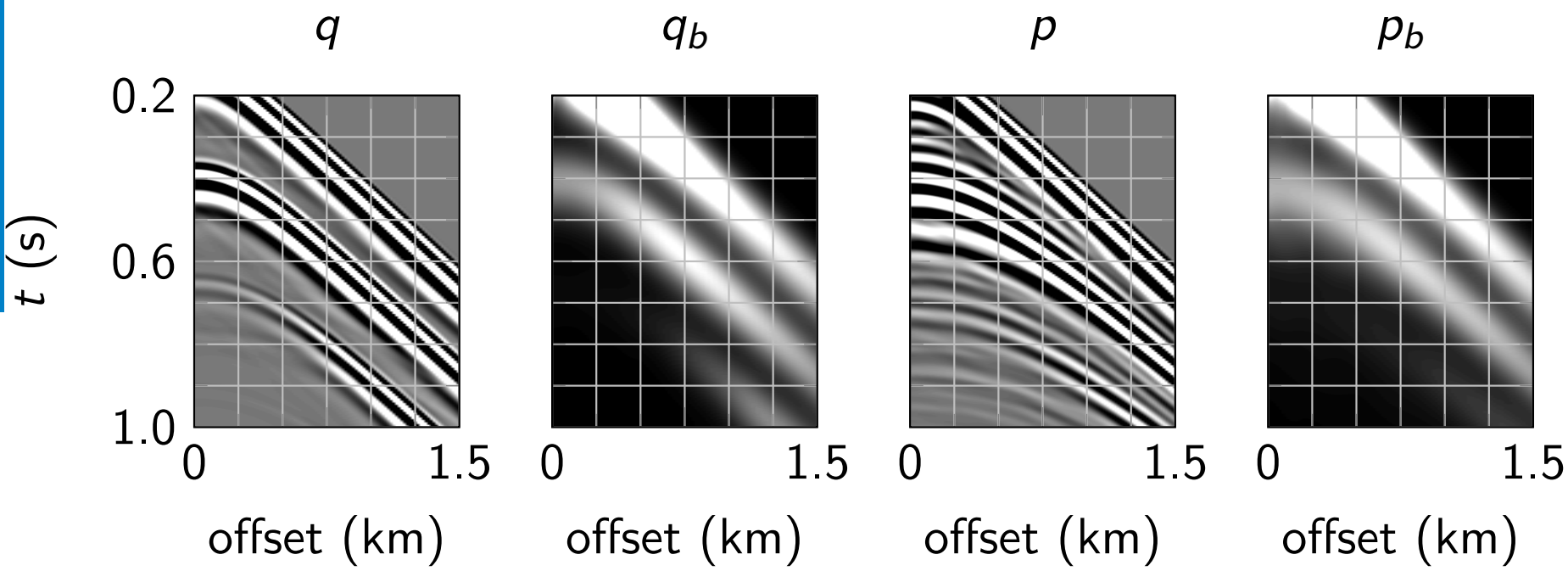
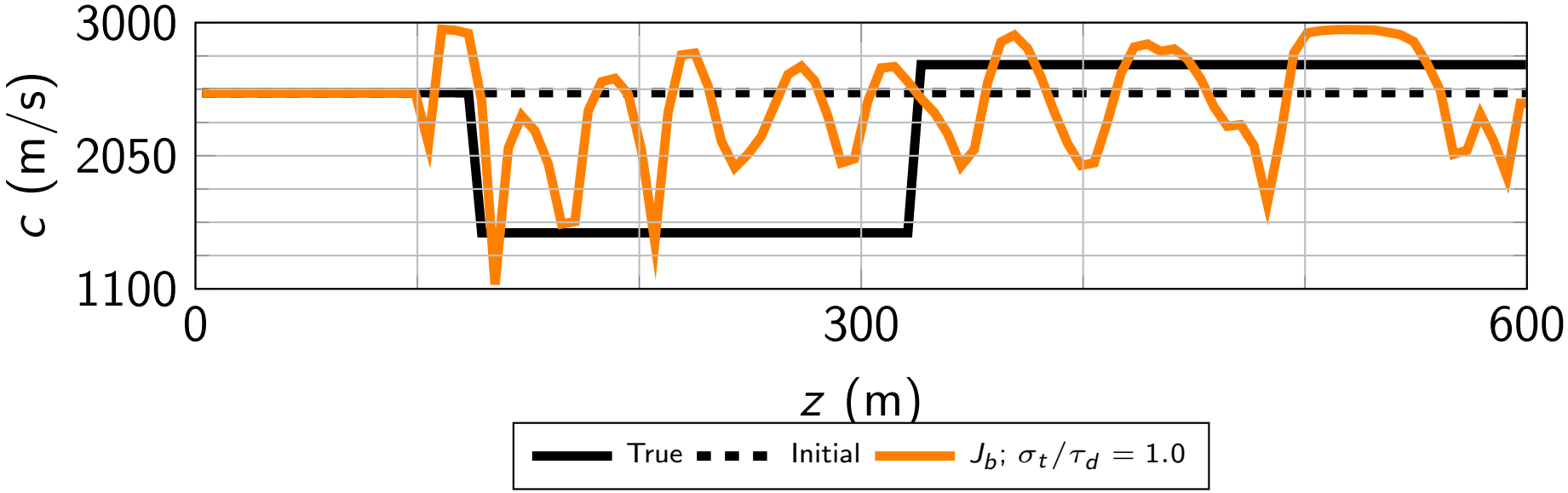
q_b
p
p_b
0.2
0.6
1.0
t (s)
0
1.5
offset (km)

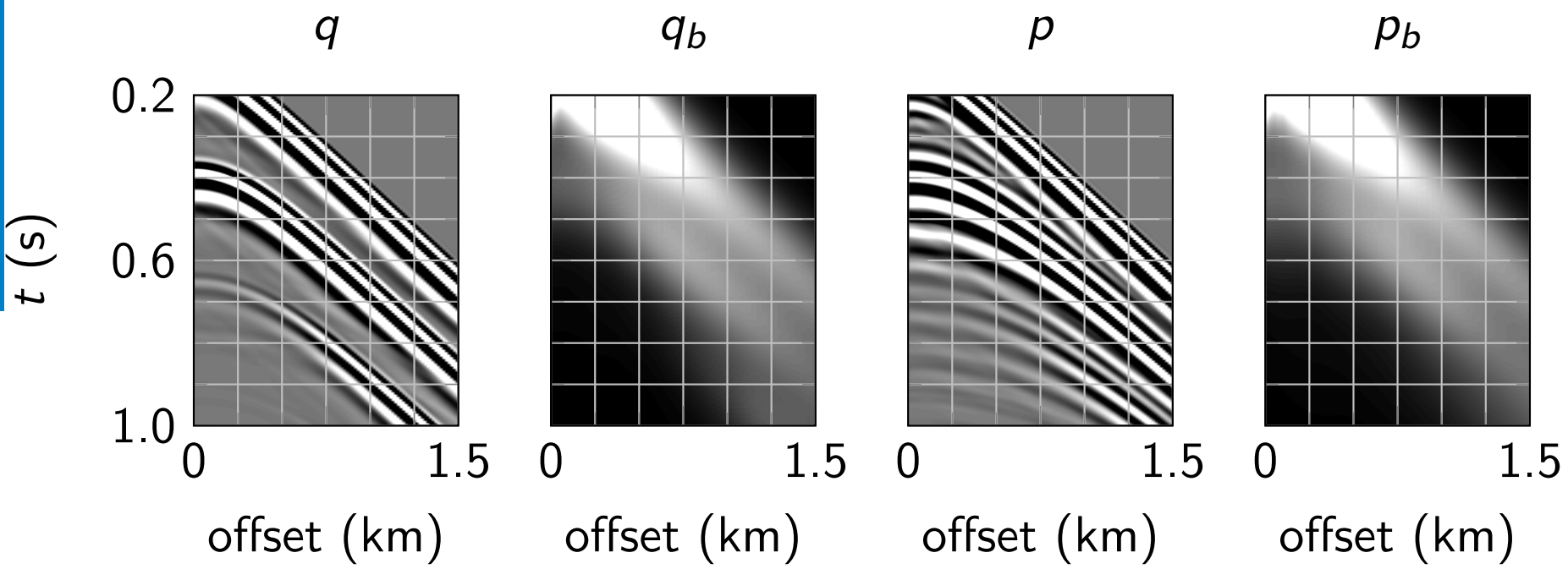

3000
2050
1100
c (m/s)
0
300
600
z (m)
True
Initial I
Initial II
J_b using initial I
J_b using initial II

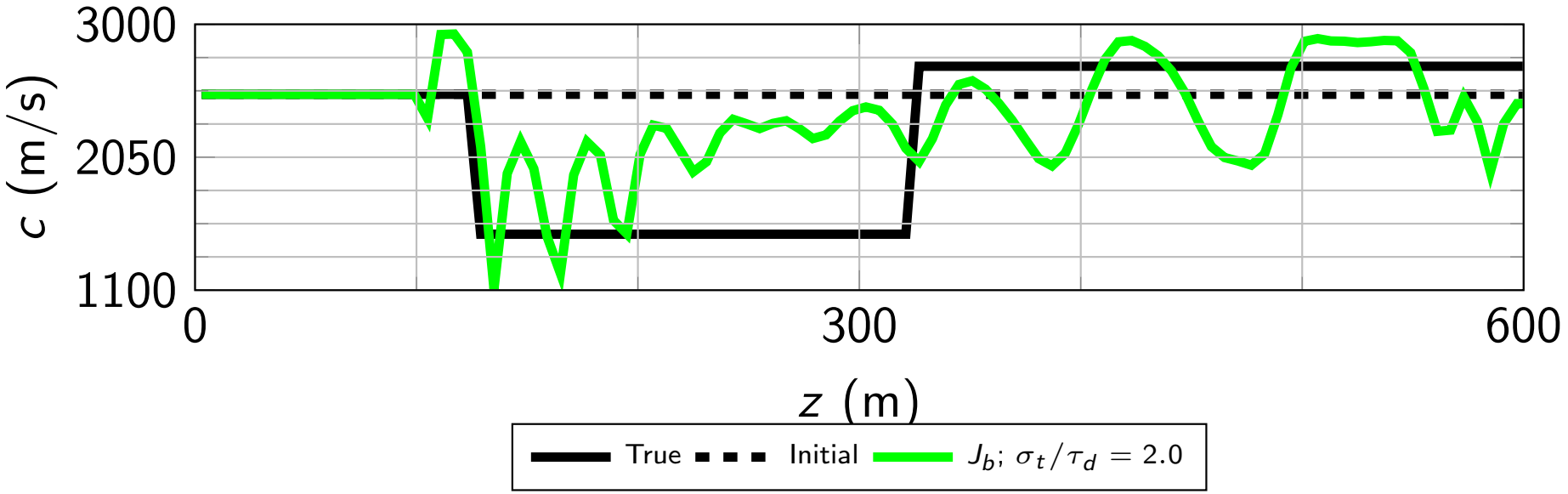

q
q_b
p
p_b
0.2
0.6
1.0
t (s)
0
1.5
offset (km)

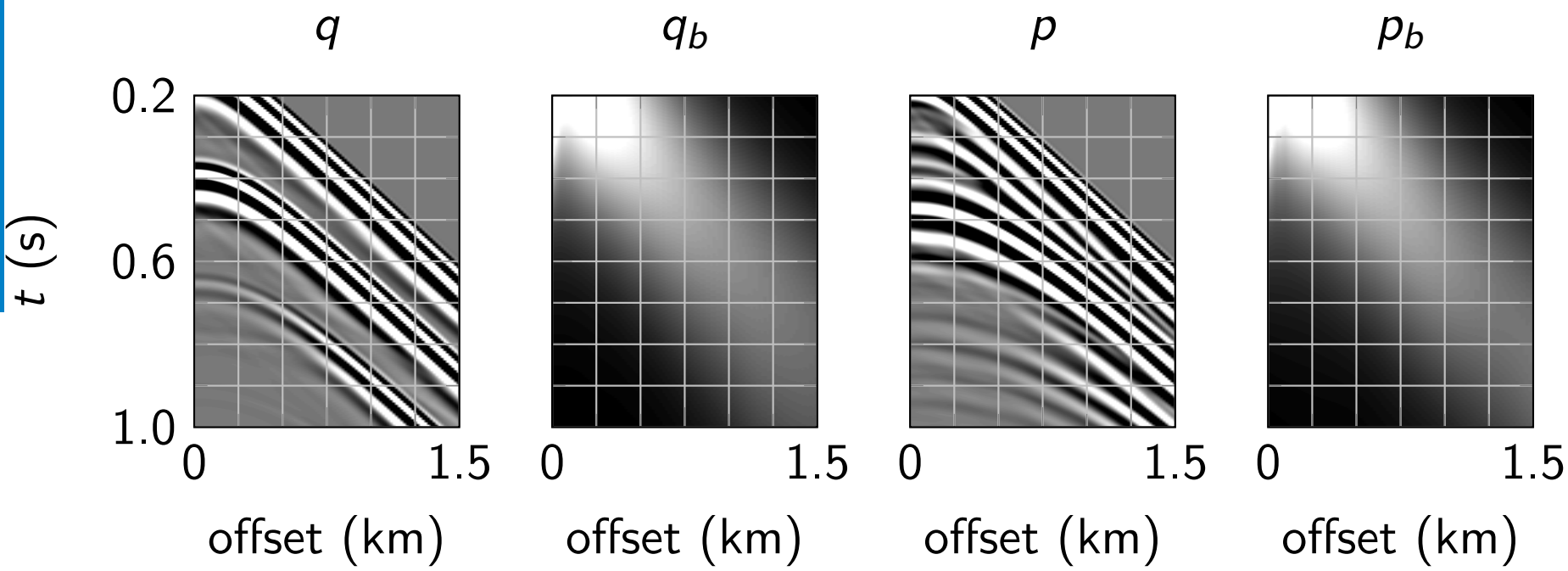

3000
2050
1100
c (m/s)
0
300
600
z (m)
True
Initial
J_b; $\sigma_t/\tau_d = 0.5$

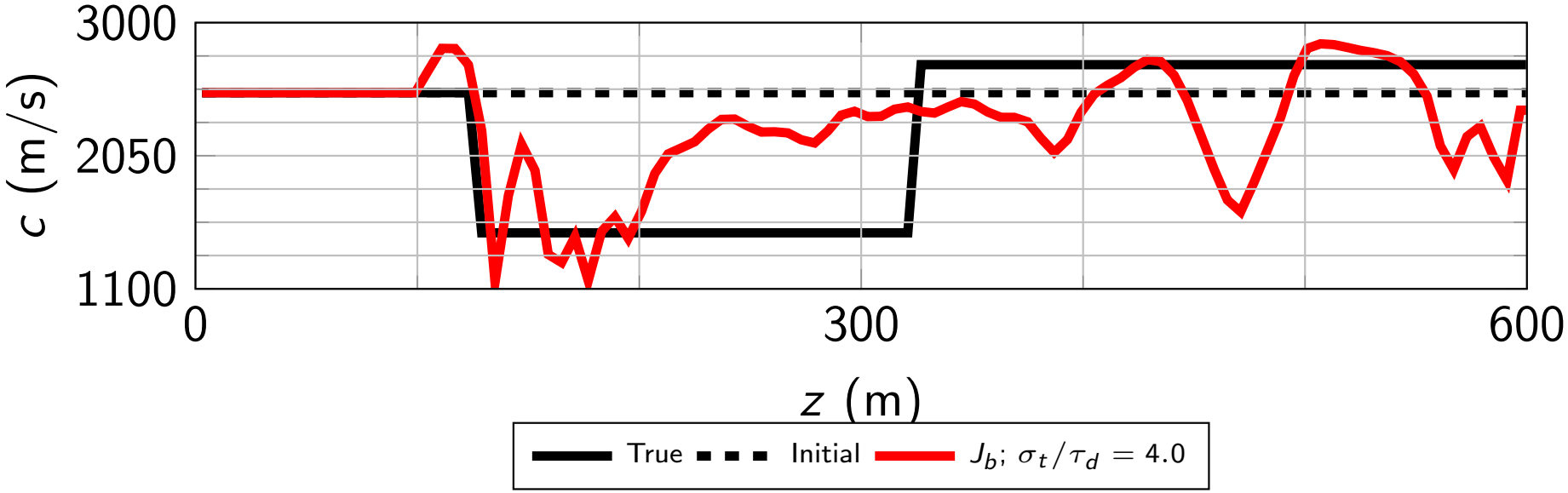

q
q_b
p
p_b
0.2
0.6
1.0
t (s)
0
1.5
offset (km)


3000
2050
1100
c (m/s)
0
300
600
z (m)
True
Initial
J_b; $\sigma_t/\tau_d = 1.0$


q
q_b
p
p_b
0.2
0.6
1.0
t (s)
0
1.5
offset (km)


3000
2050
1100
c (m/s)
0
300
600
z (m)
True
Initial
J_b; $\sigma_t/\tau_d = 2.0$


q
q_b
p
p_b
0.2
0.6
1.0
t (s)
0
1.5
offset (km)


3000
2050
1100
c (m/s)
0
300
600
z (m)
True
Initial
J_b; $\sigma_t/\tau_d = 4.0$

Summary of the Bump Functional

- J_b can be seen as a generalized envelope based misfit with blurring along time and/or receiver coordinate.
- J_b inversion doesn't suffer from cycle skipping.
- J_b is insensitive to the polarity of arrivals and the source signature.
- The size of the basin of attraction of J_b increases with σ_t and σ_r.
- Bump functional minimization suffers from non-uniqueness problems.

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Motivation

- A model that corresponds to the local minima of the least-squares functional doesn't minimize the bump functional.
- A model that minimizes the bump functional can contain false arrivals and these false arrivals increase the least-squares functional.
- The model that corresponds to the global minimum of the least-squares objective simultaneously minimizes both J_b and J_{ls}.

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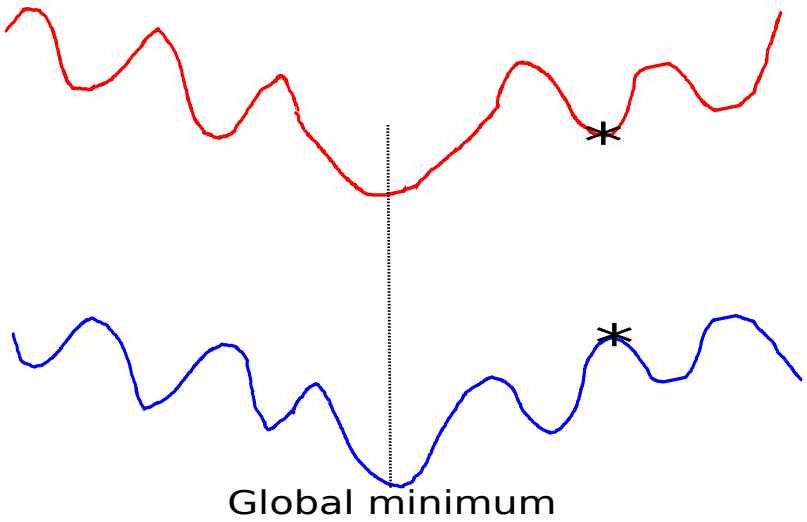

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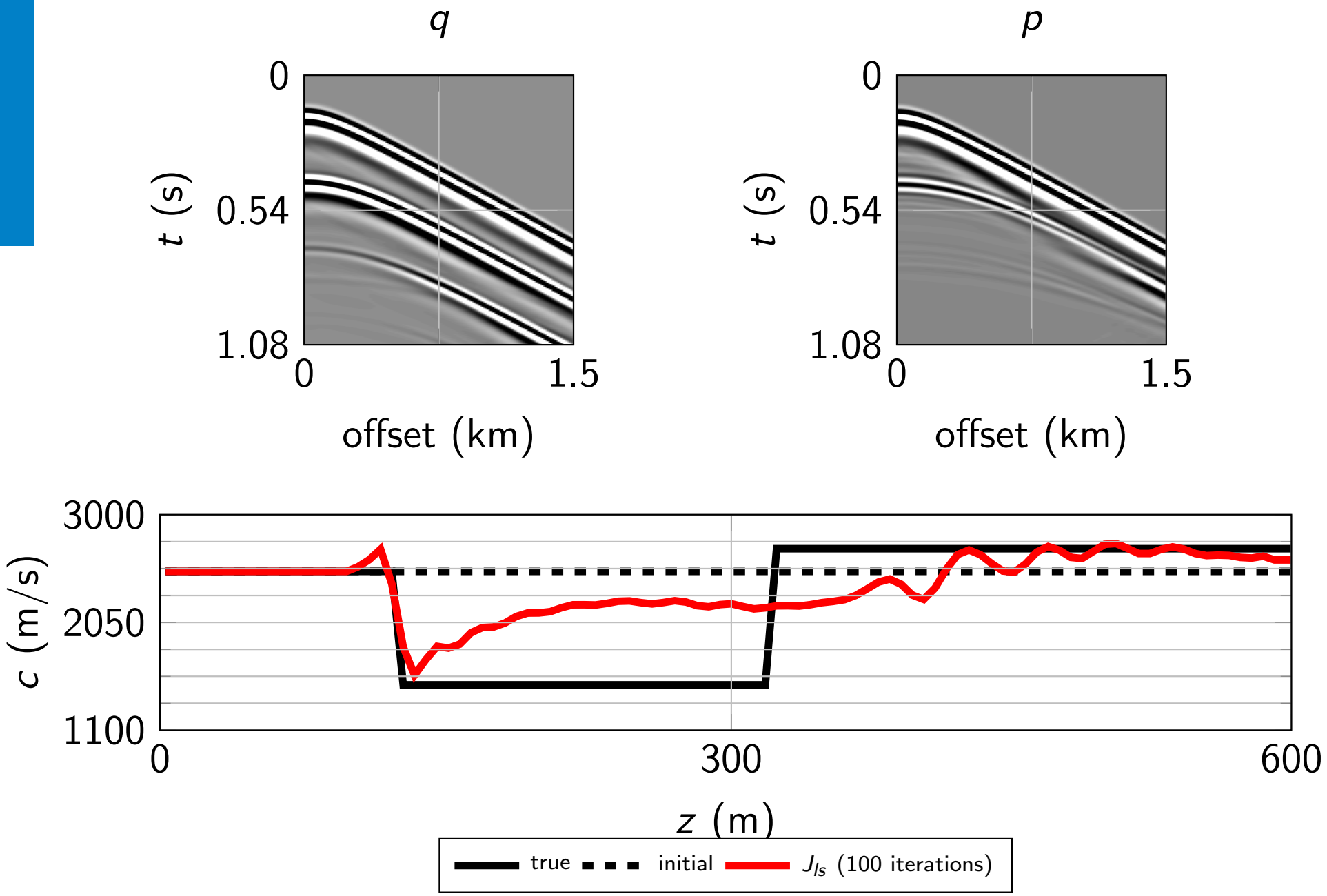
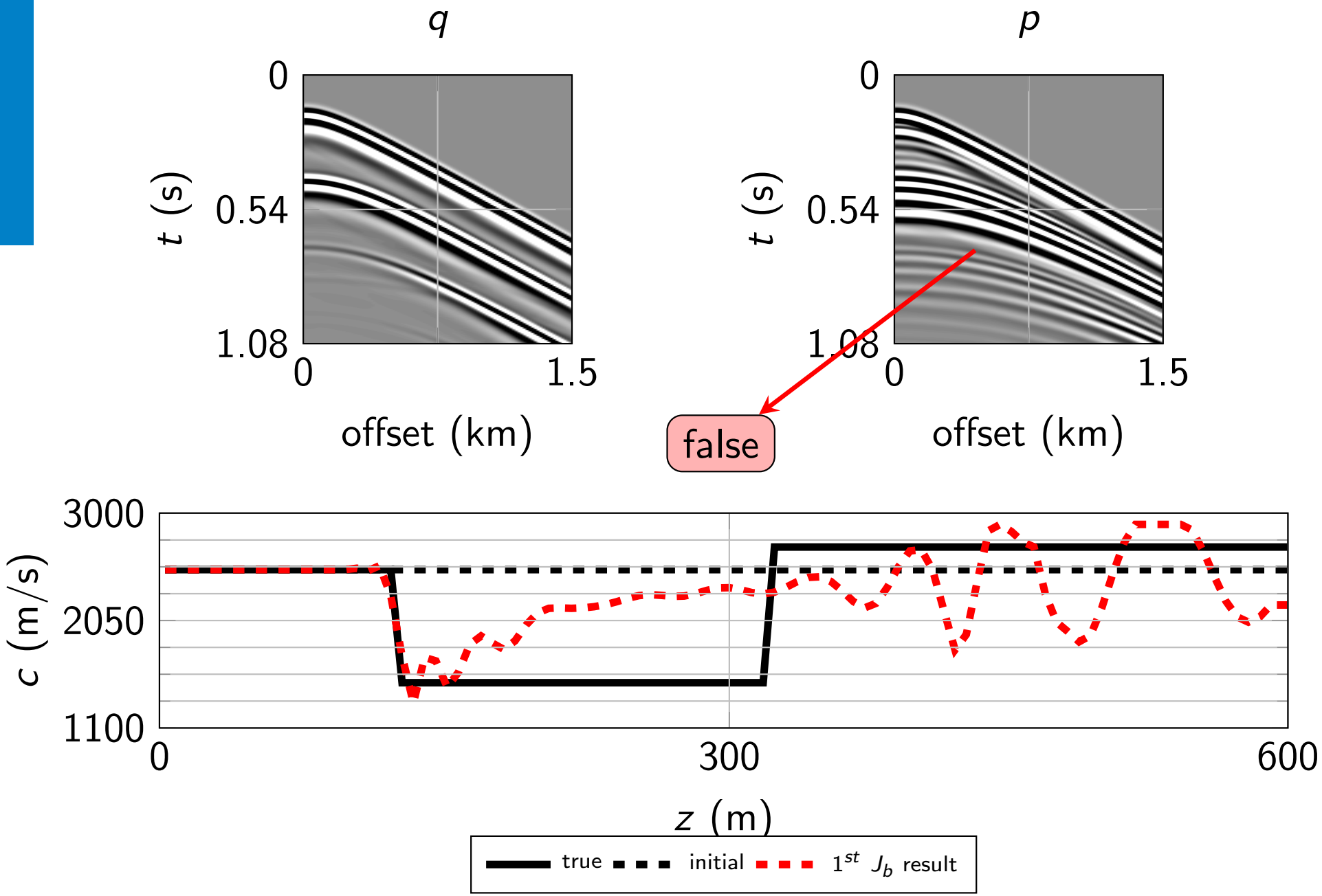
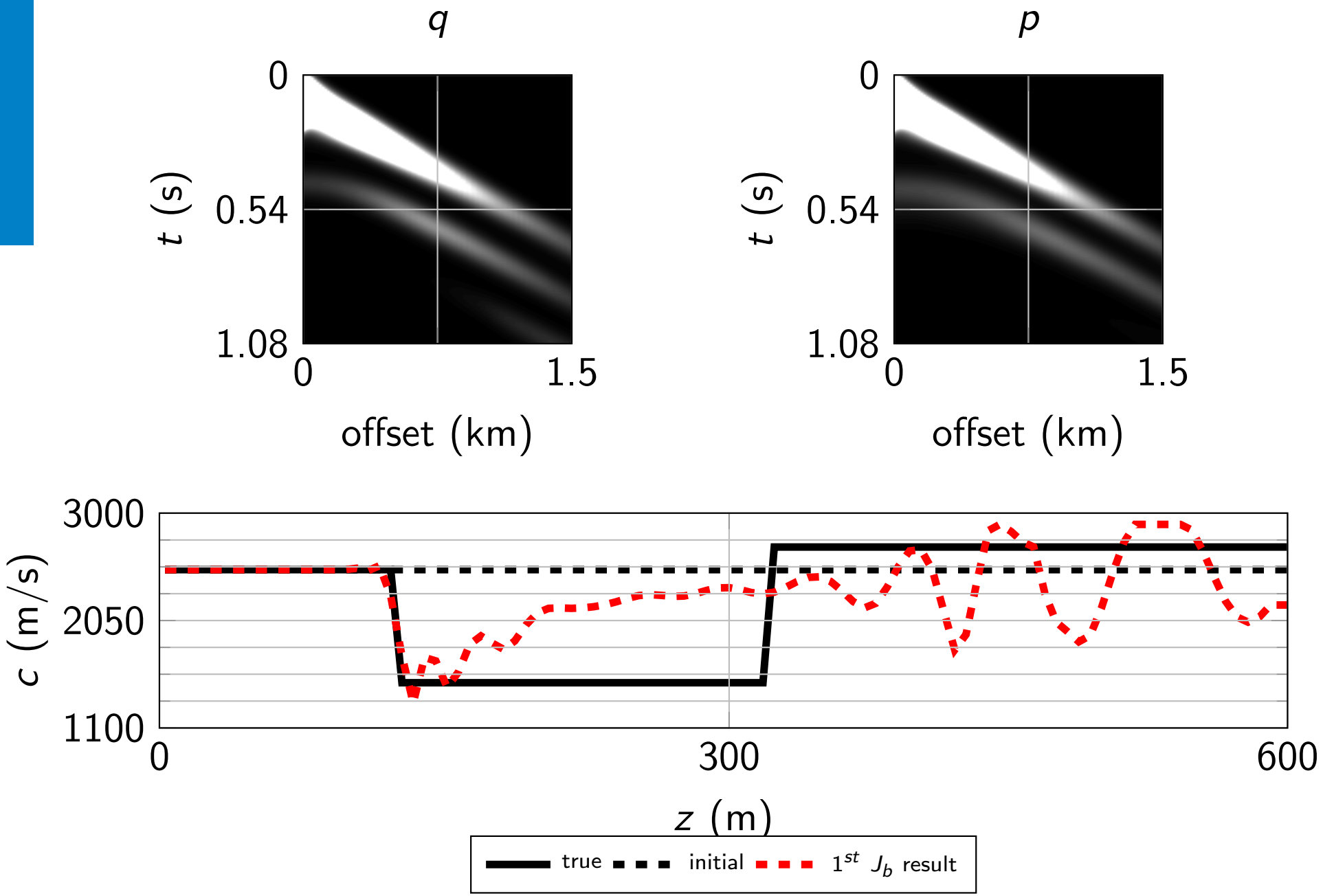
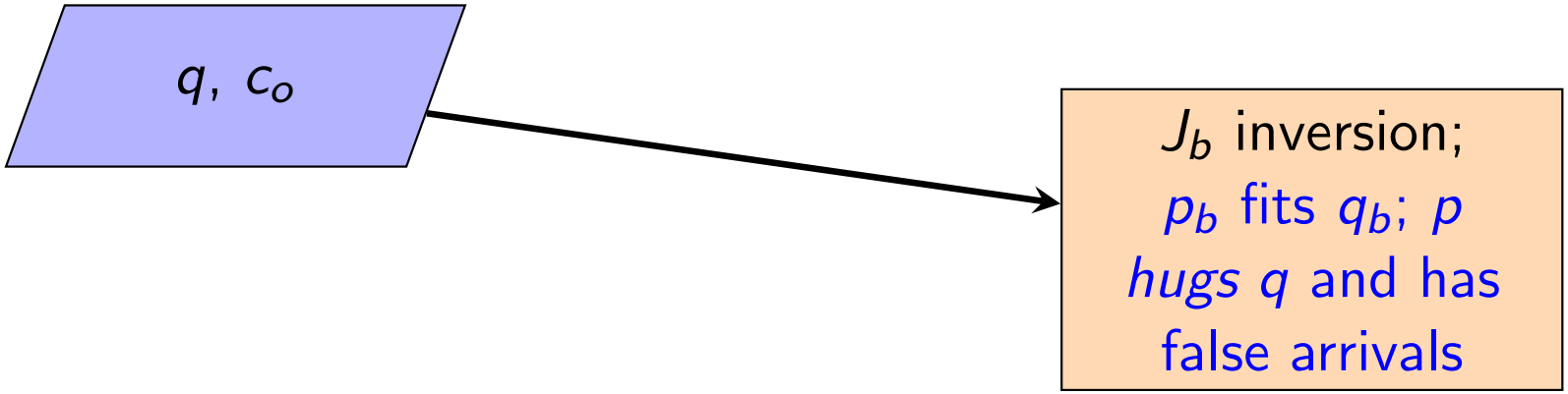
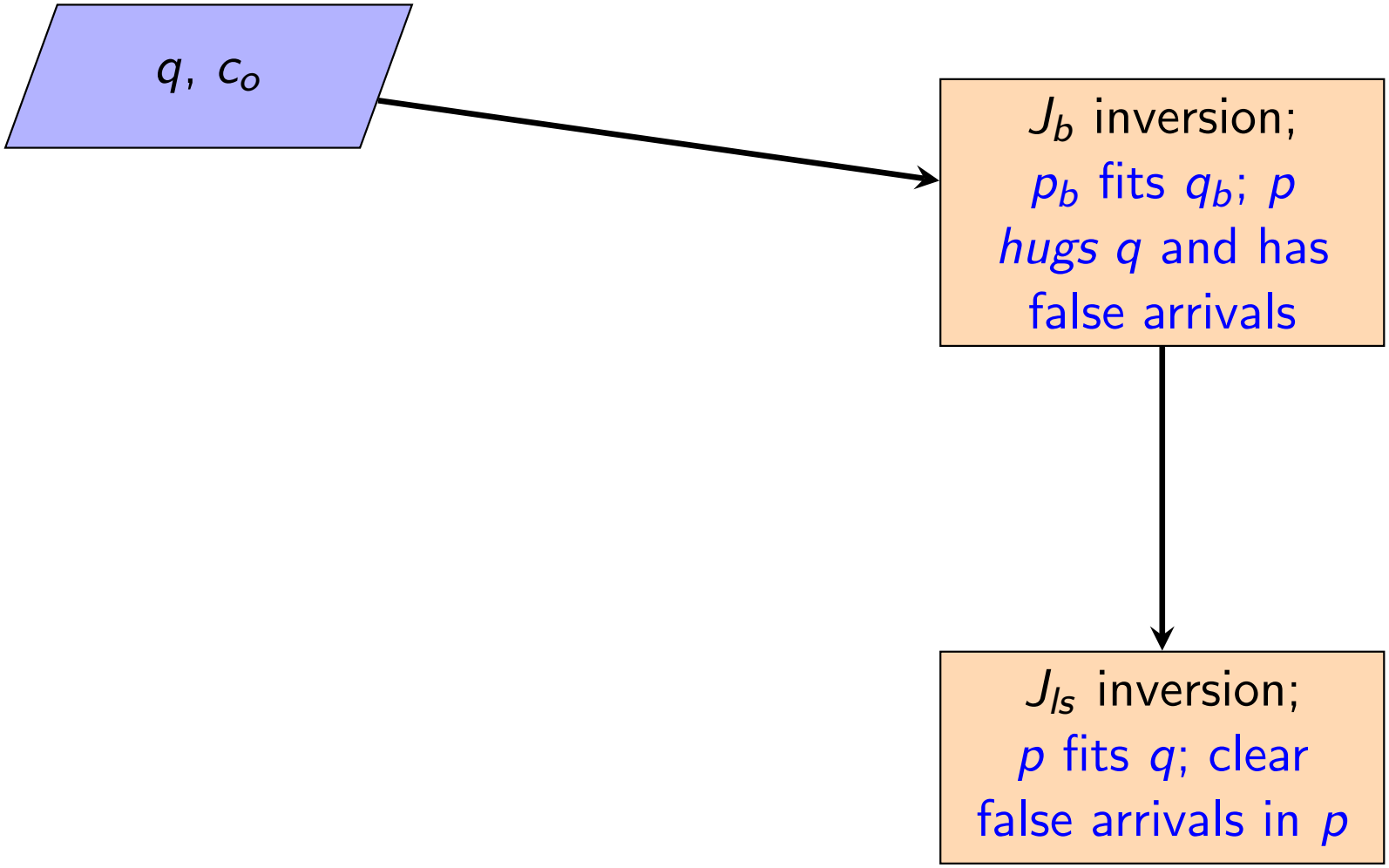
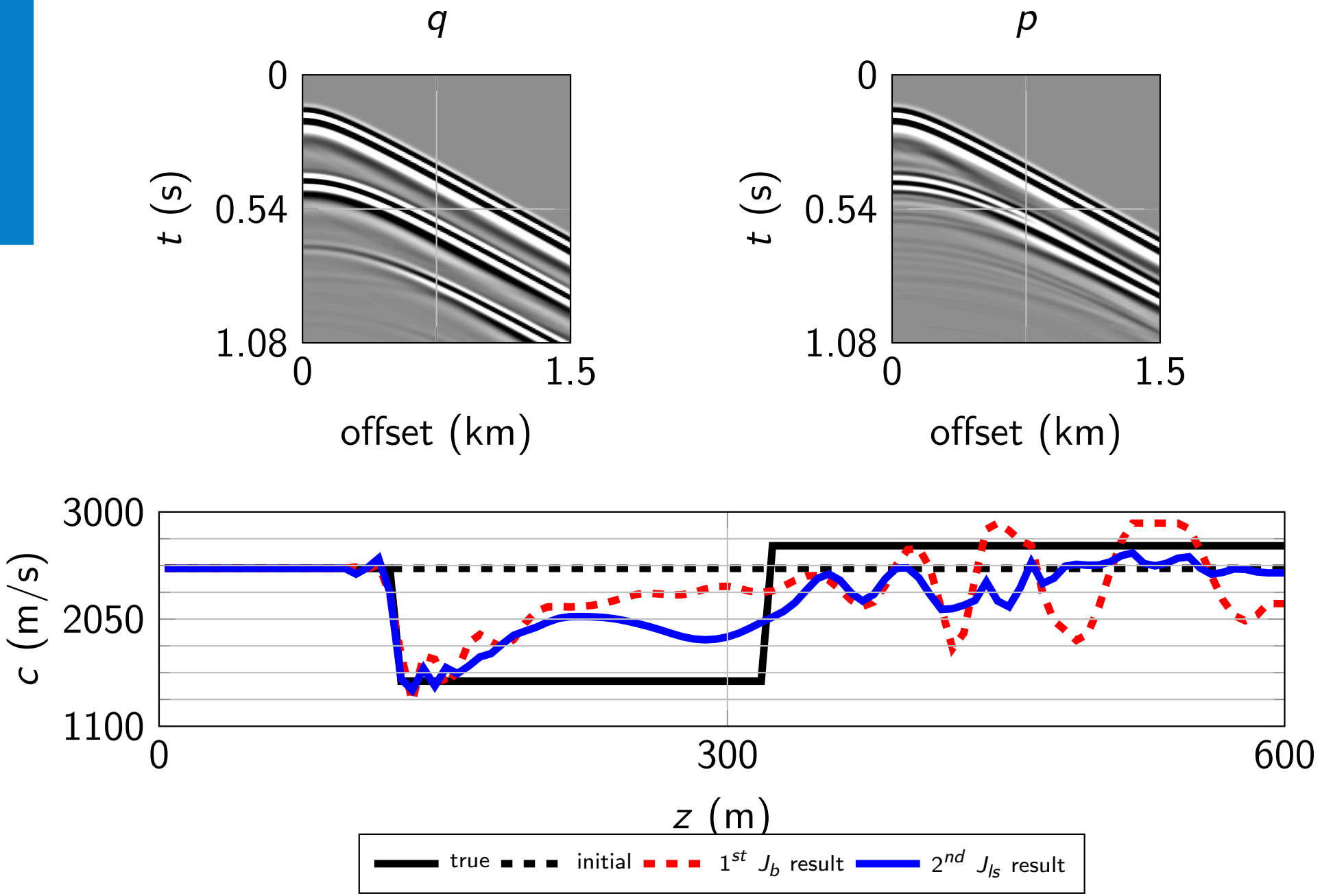

q
p
0
0.54
1.08
t (s)
0
1.5
offset (km)
3000
2050
1100
c (m/s)
300
600
z (m)
true
initial
J_{ls} (100 iterations)

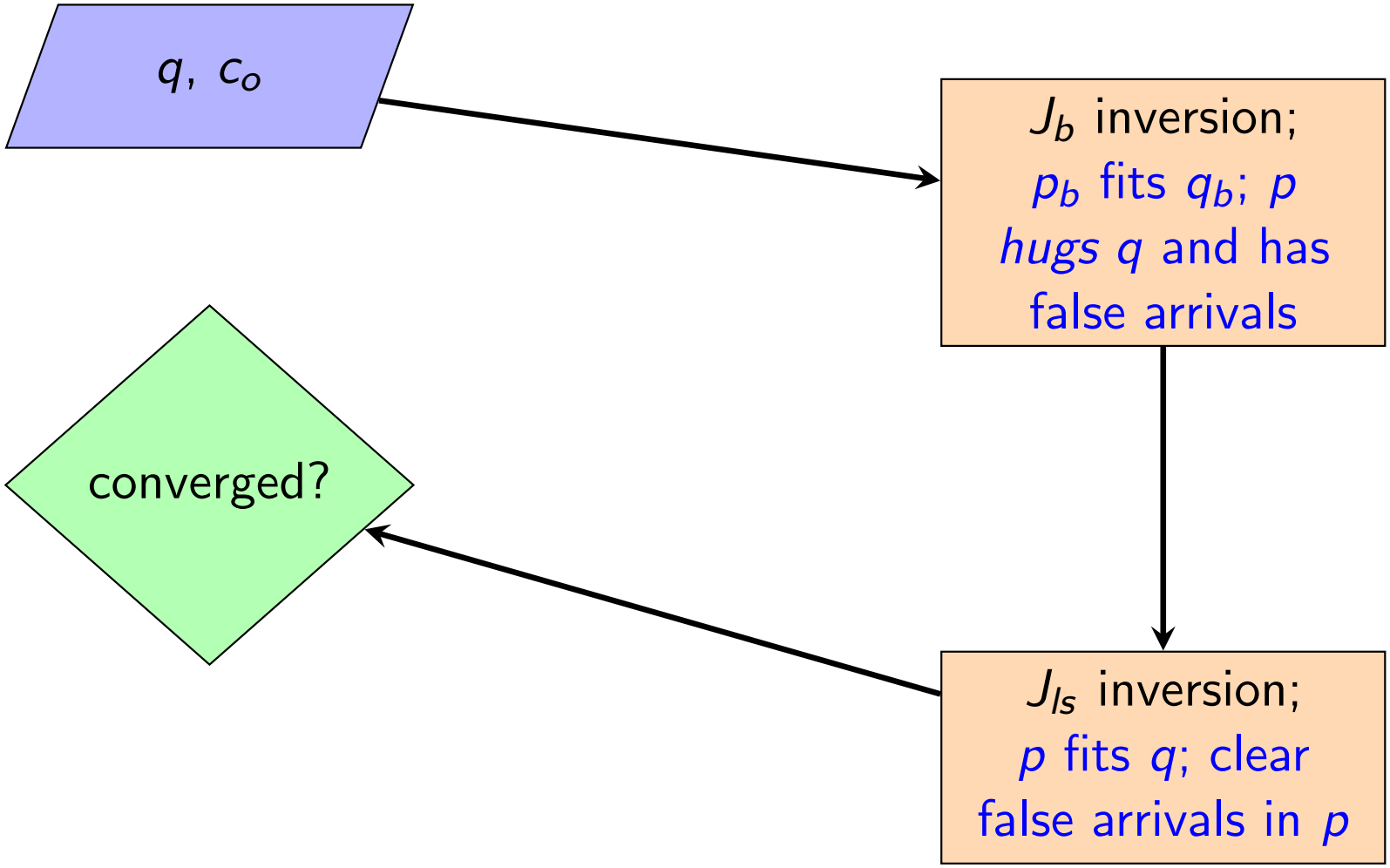

q
p
0
0.54
1.08
t (s)
0
1.5
offset (km)
false
3000
2050
1100
c (m/s)
300
600
z (m)
true
initial
1st J_b result

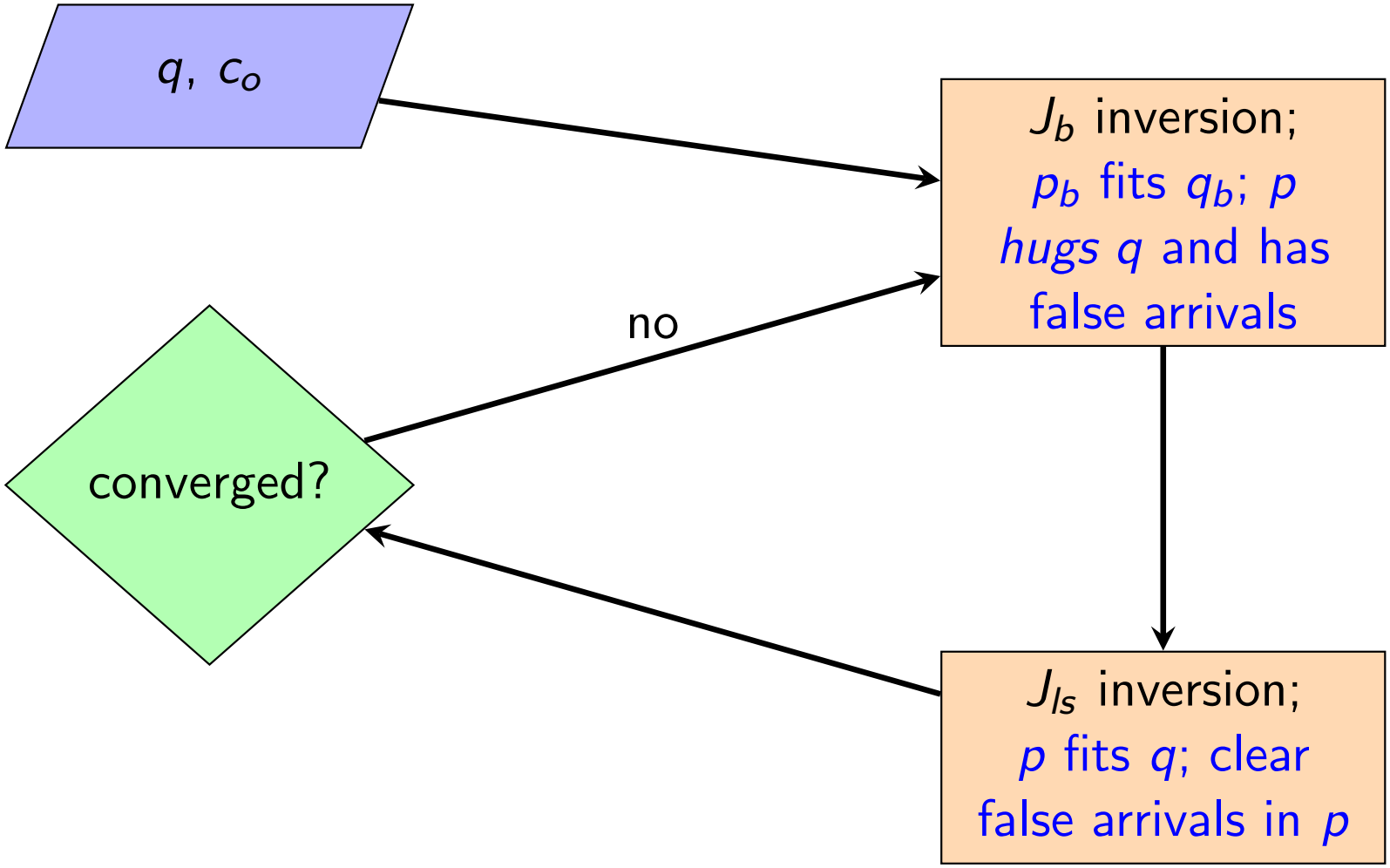

q
p
0
0.54
1.08
t (s)
0
1.5
offset (km)
3000
2050
1100
c (m/s)
300
600
z (m)
true
initial
1st J_b result

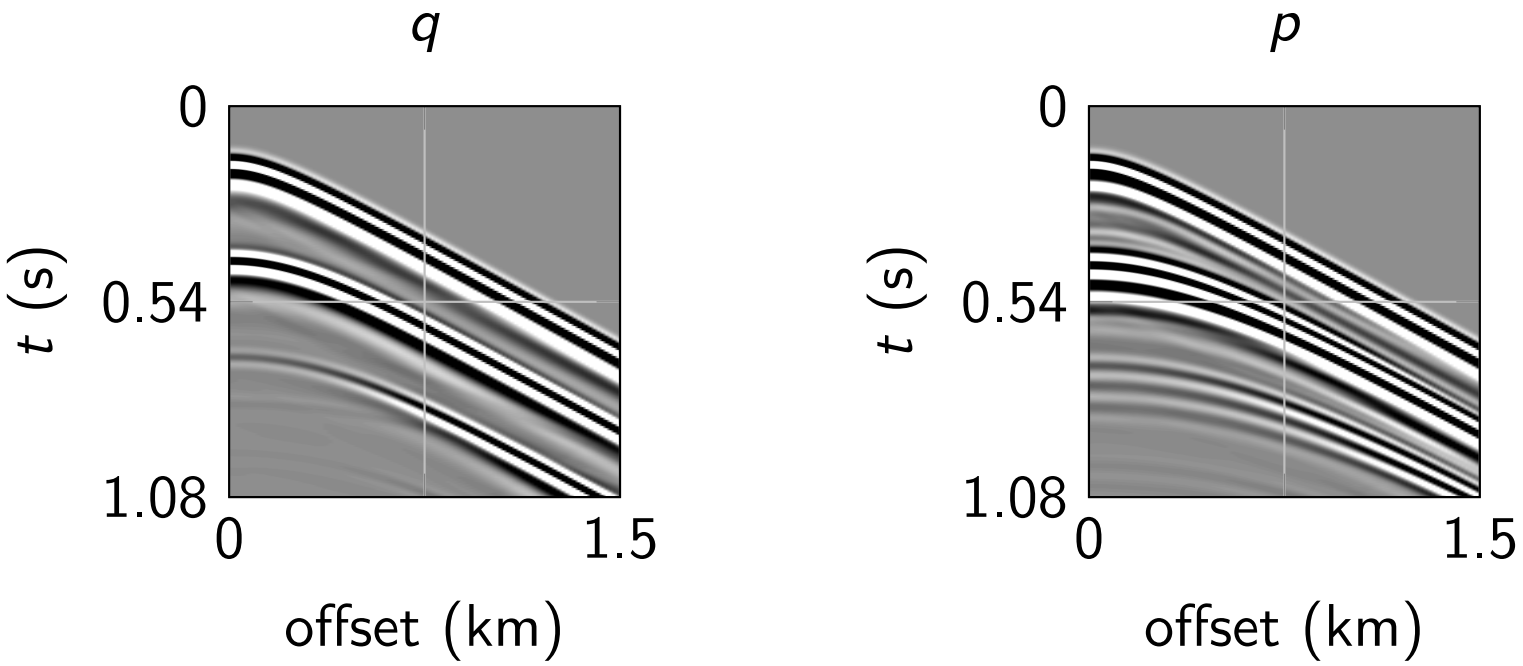

q, c_o
J_b inversion;
p_b fits q_b; p
hugs q and has
false arrivals

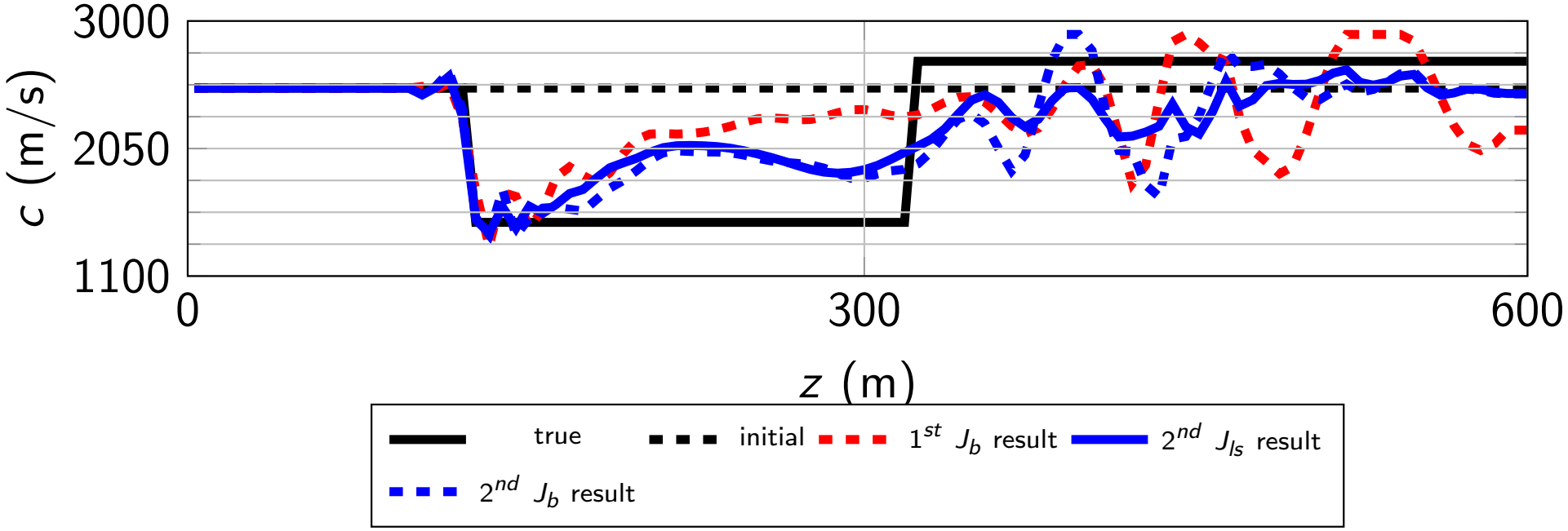

q, c_o
J_b inversion;
p_b fits q_b; p
hugs q and has
false arrivals
J_{ls} inversion;
p fits q; clear
false arrivals in p

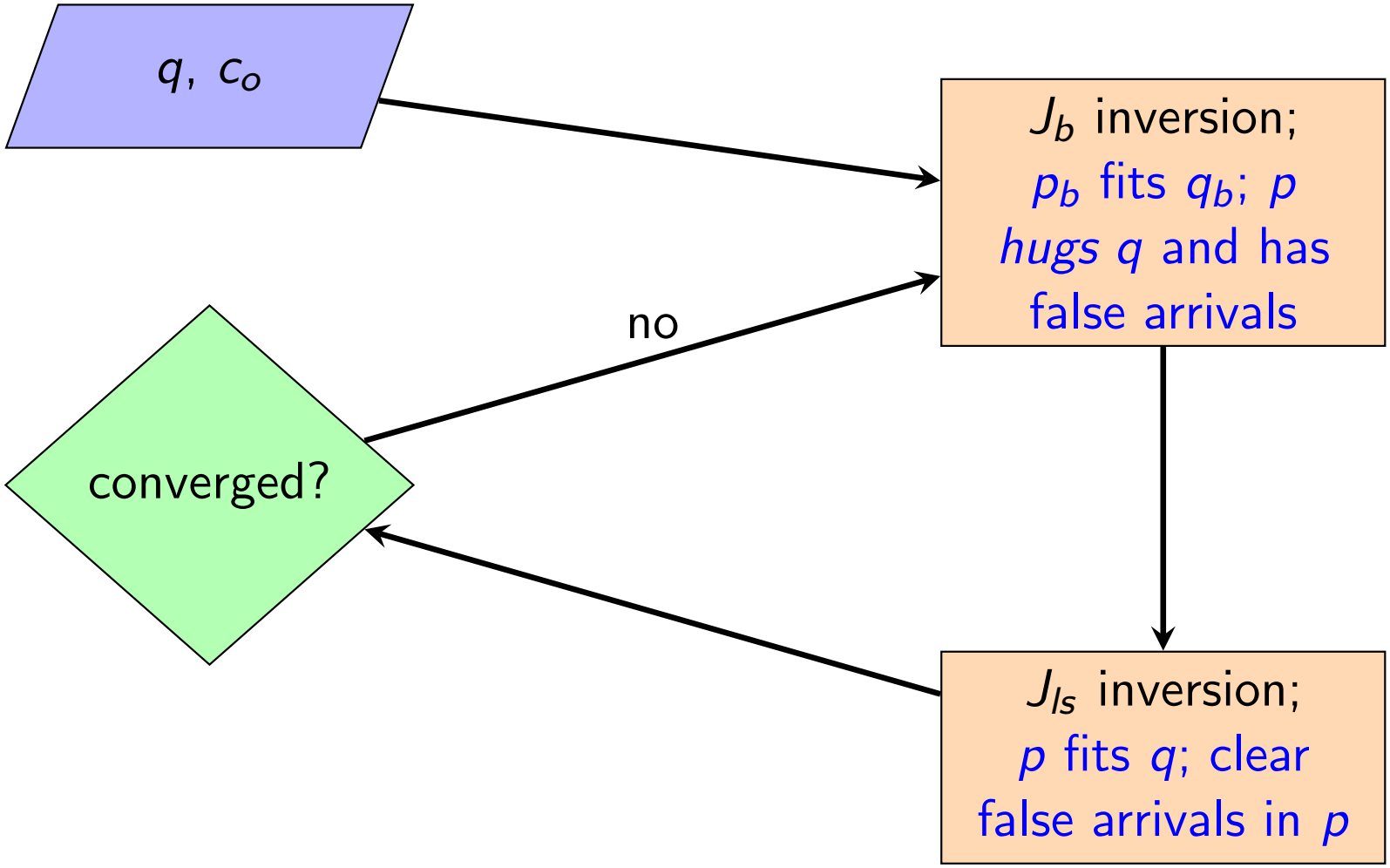

q
p
0
0.54
1.08
t (s)
0
1.5
offset (km)
3000
2050
1100
c (m/s)
300
600
z (m)
true
initial
1st Jb result
2nd Jls result

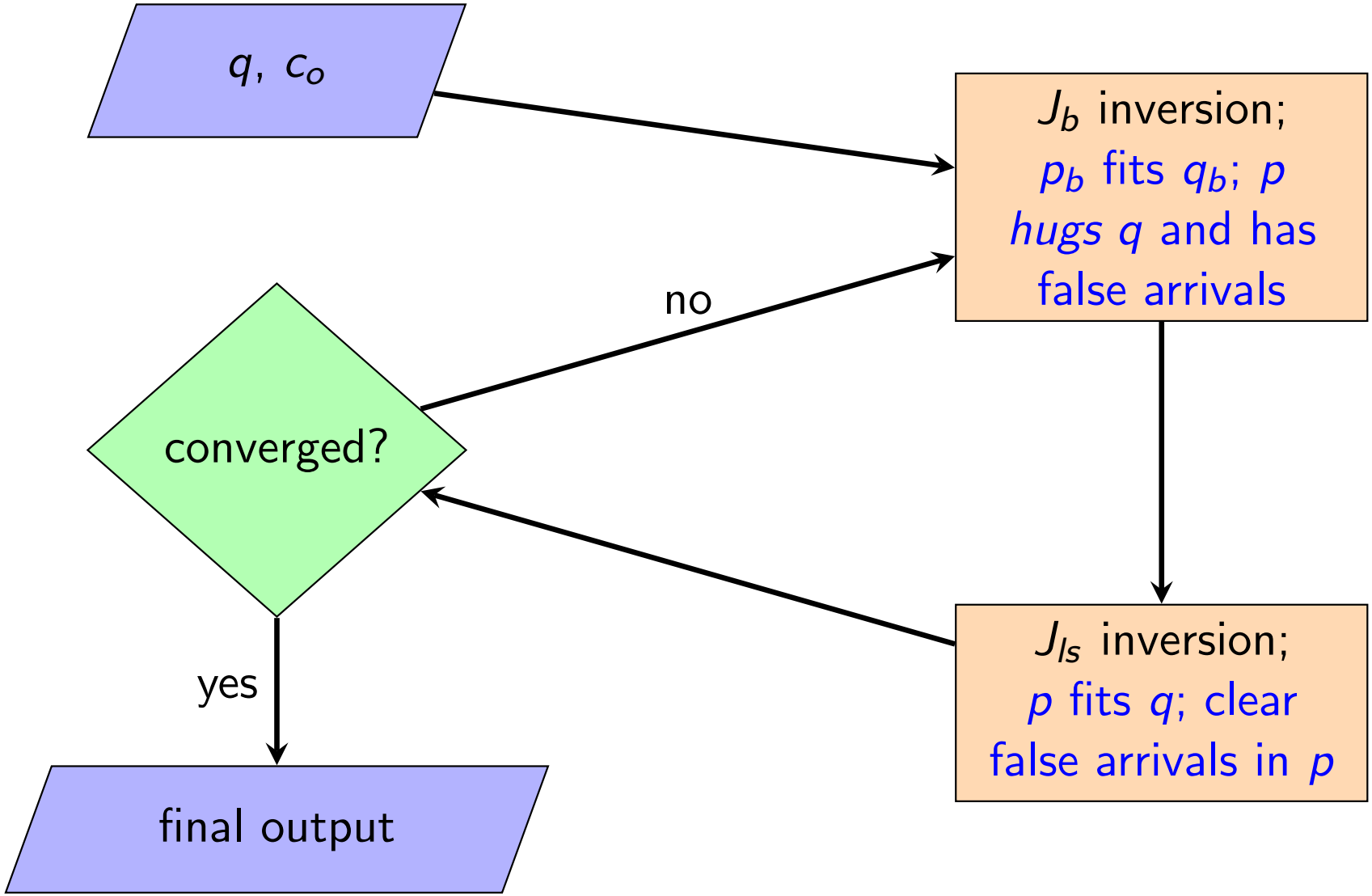

q, c_o
J_b inversion; p_b fits q_b; p *hugs* q and has false arrivals
J_{ls} inversion; p fits q; clear false arrivals in p
converged?

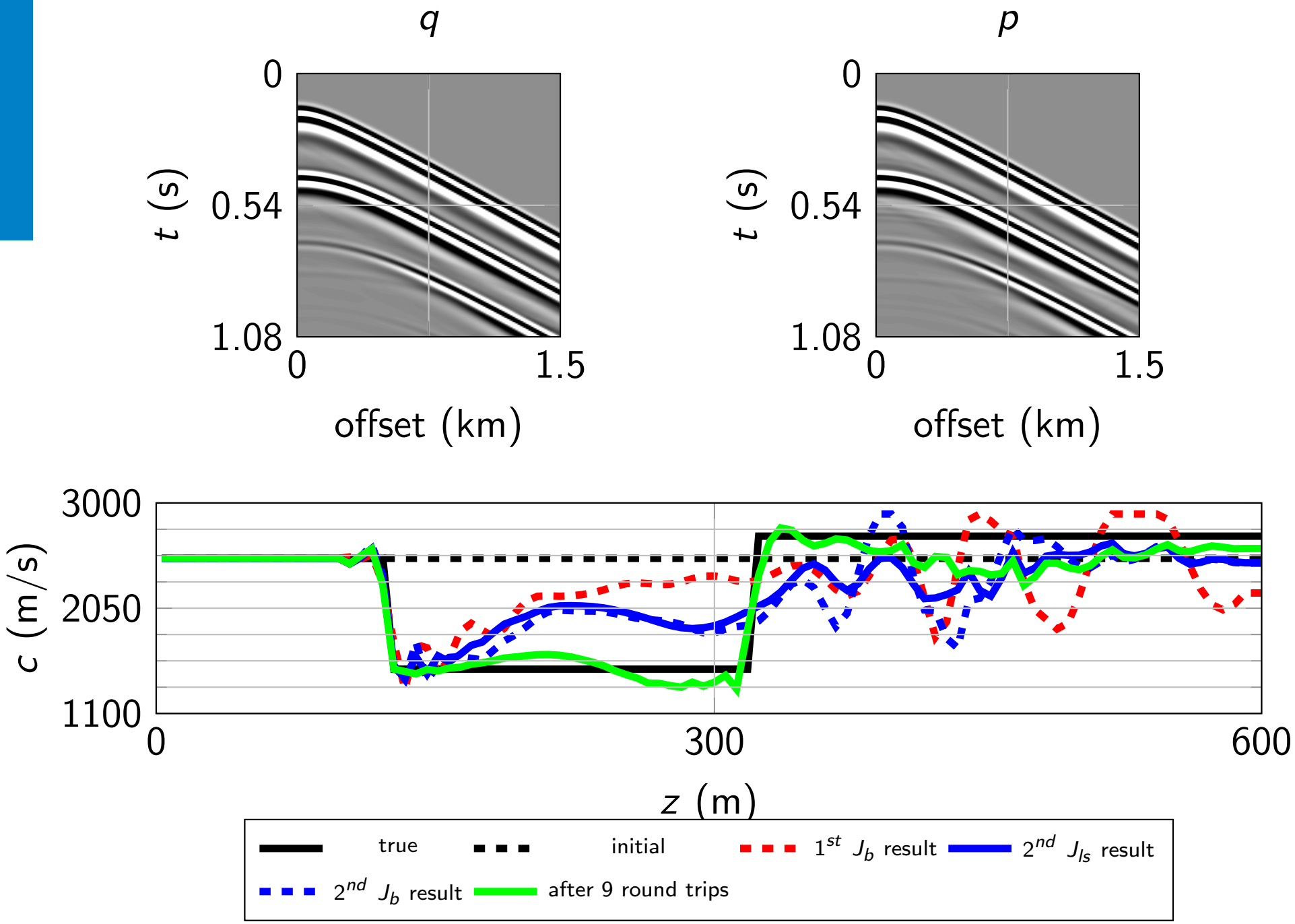

q, c_o
J_b inversion;
p_b fits q_b; p
hugs q and has
false arrivals
no
converged?
J_{ls} inversion;
p fits q; clear
false arrivals in p


q
p
0
0.54
1.08
t (s)
0
1.5
offset (km)


3000
2050
1100
c (m/s)
0
300
600
z (m)
true
initial
1st Jb result
2nd Jls result
2nd Jb result


q, c_o
J_b inversion;
p_b fits q_b; p
hugs q and has
false arrivals
no
converged?
J_{ls} inversion;
p fits q; clear
false arrivals in p


q, c_o
J_b inversion;
p_b fits q_b; p
hugs q and has
false arrivals
no
converged?
yes
J_{ls} inversion;
p fits q; clear
false arrivals in p
final output


q
p
0
0.54
1.08
t (s)
0
1.5
offset (km)
3000
2050
1100
c (m/s)
300
600
z (m)
true
initial
1st J_b result
2nd J_{ls} result
2nd J_b result
after 9 round trips

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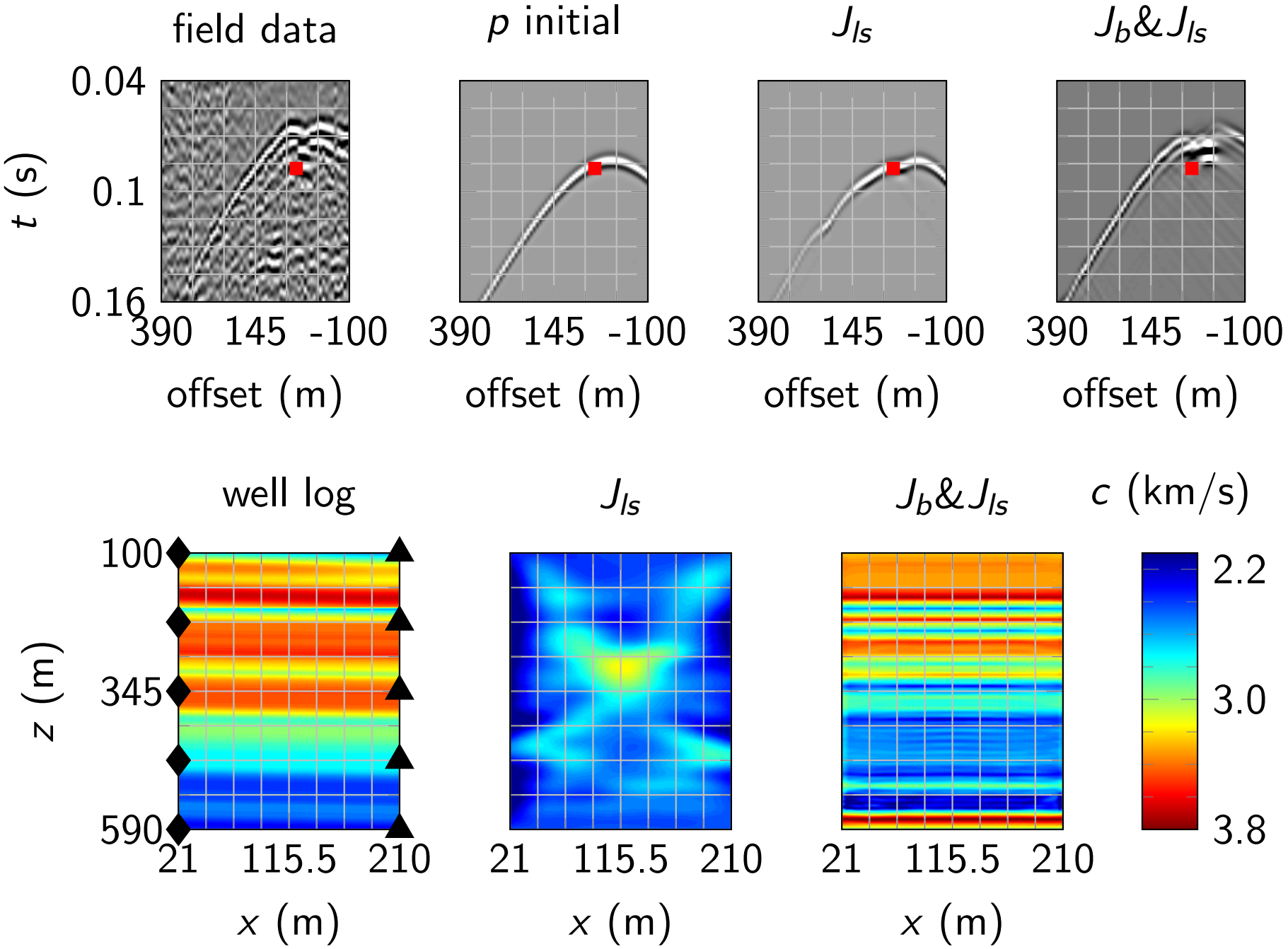
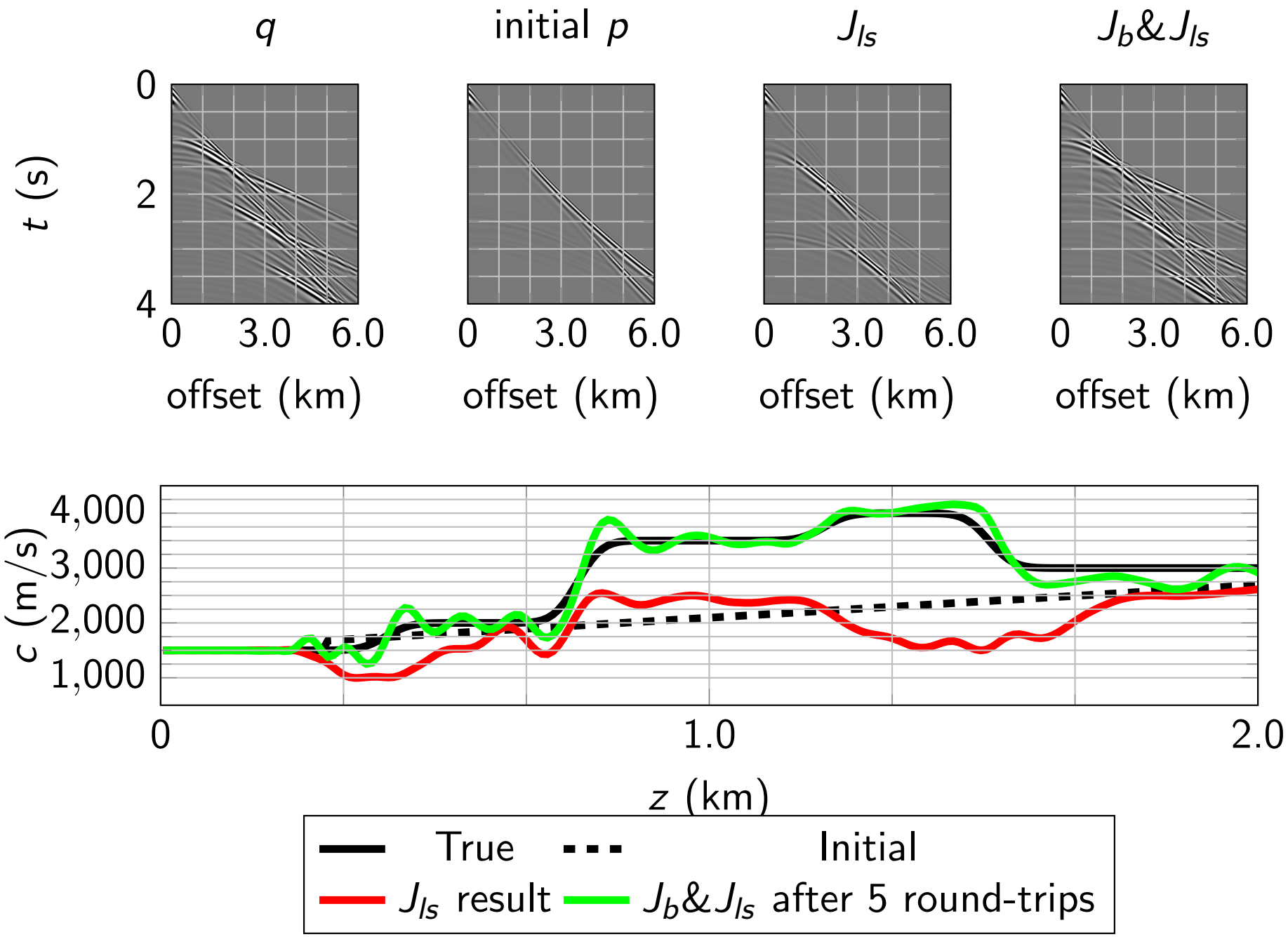
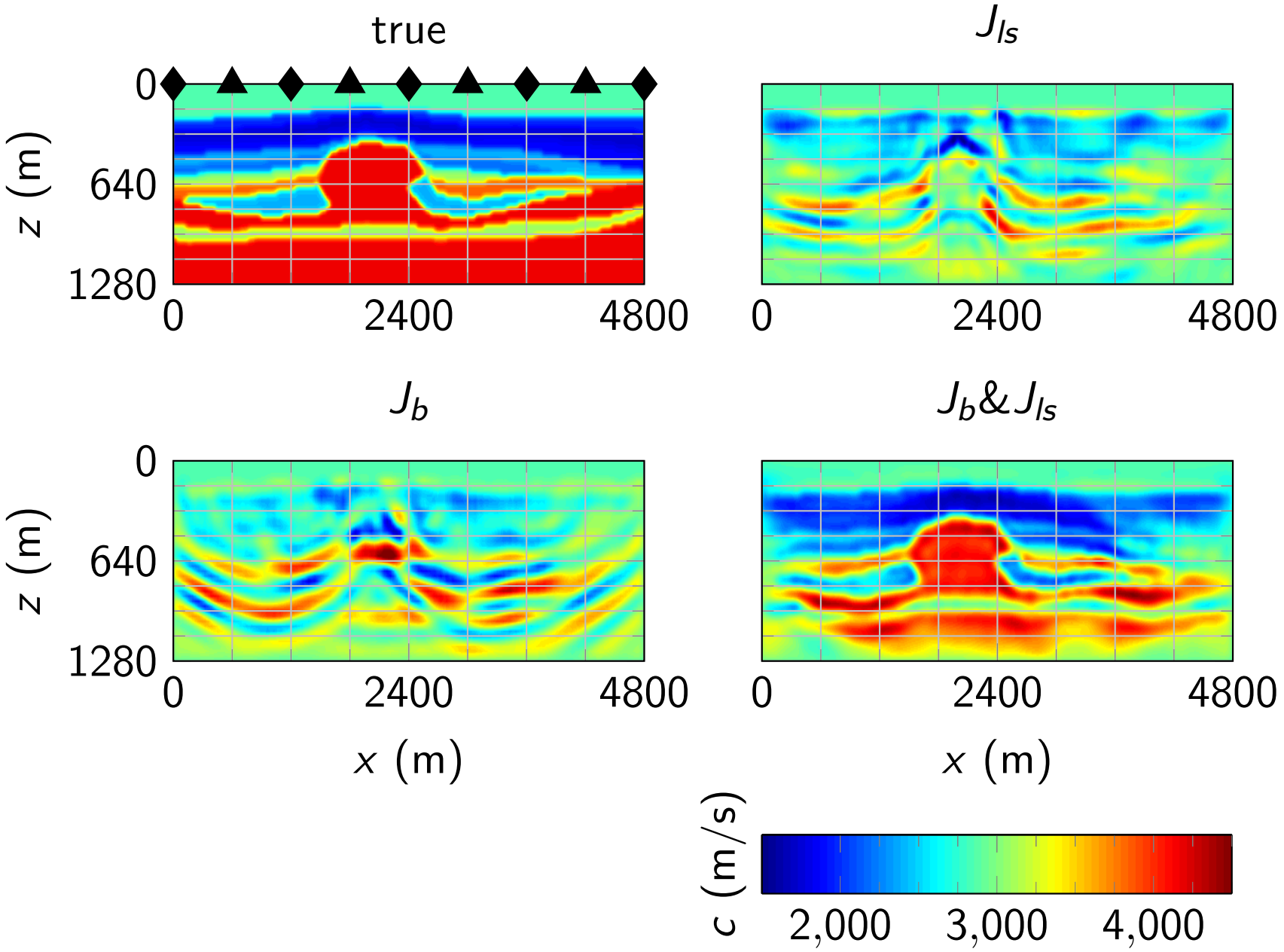

field data
p initial
J_{ls}
J_b&J_{ls}
0.04
0.1
0.16
t (s)
390 145 -100
offset (m)
well log
J_{ls}
J_b&J_{ls}
c (km/s)
100
345
590
z (m)
21 115.5 210
x (m)
2.2
3.0
3.8


q
initial p
J_{ls}
J_b&J_{ls}
t (s)
0
2
4
0
3.0
6.0
offset (km)
c (m/s)
4,000
3,000
2,000
1,000
z (km)
1.0
2.0
True
Initial
J_{ls} result
J_b&J_{ls} after 5 round-trips



homogeneous initial model with $c_o = 3000\,\mathrm{m/s}$

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Bump functional

- Takes multiple arrivals in the data into account.
- It has a larger basin of attraction compared to envelope misfit.
- The size of its attraction basin depends on σ_t and σ_r.
- A model that minimizes it can be non-unique.

Bump functional optimization

- J_b inversion is robust while fitting transmitted arrivals.
- J_b inversion cannot update the background velocity using reflections due to non-uniqueness.

Multi-objective inversion using J_b and J_{ls}

- J_b can be used as an auxiliary objective during conventional full waveform inversion to prevent local minima convergence.

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Acknowledgements

- Wim Mulder and Guy Drijkoningen, my PhD supervisors.
- This work was carried out on the Dutch national e-infrastructure with the support of SURF Foundation.
- NeTTUN project, funded from the European Commission's Seventh Framework Programme for Research, Technological Development and Demonstration (FP7 2007-2013) under Grant Agreement 280712.
- Jan Thorbecke, my HPC guru at TU Delft.

Thank you! Any questions?